

Introduction

Using a cloud computing platform

What is Cloud Computing

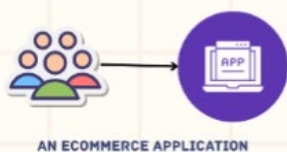
Delivery of computing services over the Internet.

You get access to services that allow you to deploy compute machines, host databases, build and deploy applications.

There are several benefits to using cloud computing platforms

1. Cost - You don't need to pay anything upfront. You only pay based on the usage.
2. Speed - You can deploy applications faster because you are able to provision infrastructure on-demand.
3. Latest - You get access to the latest hardware and software services.

APPLICATIONS



AN ECOMMERCE APPLICATION



WRITE THE RELEVANT CODE AND BUILD THE APPLICATION



SERVERS



STORAGE



NETWORK

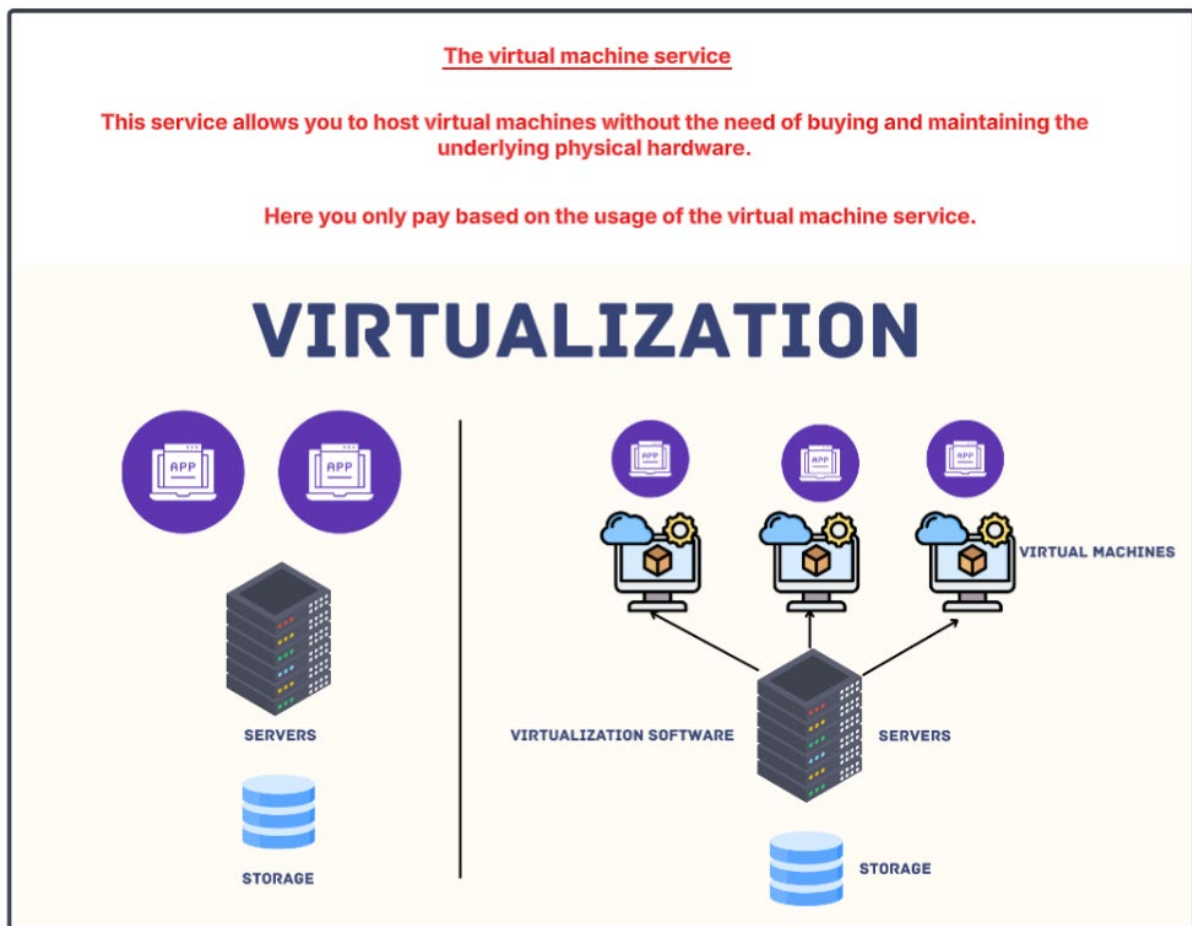


DATA CENTERS

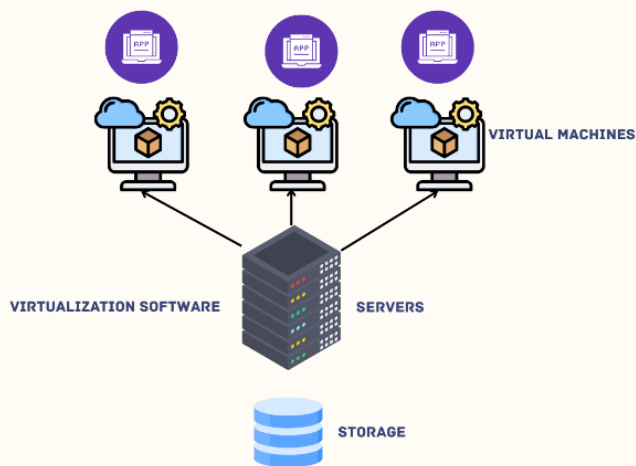


Describe Azure architecture and services - Azure compute

Virtual Machines in Azure



VIRTUALIZATION

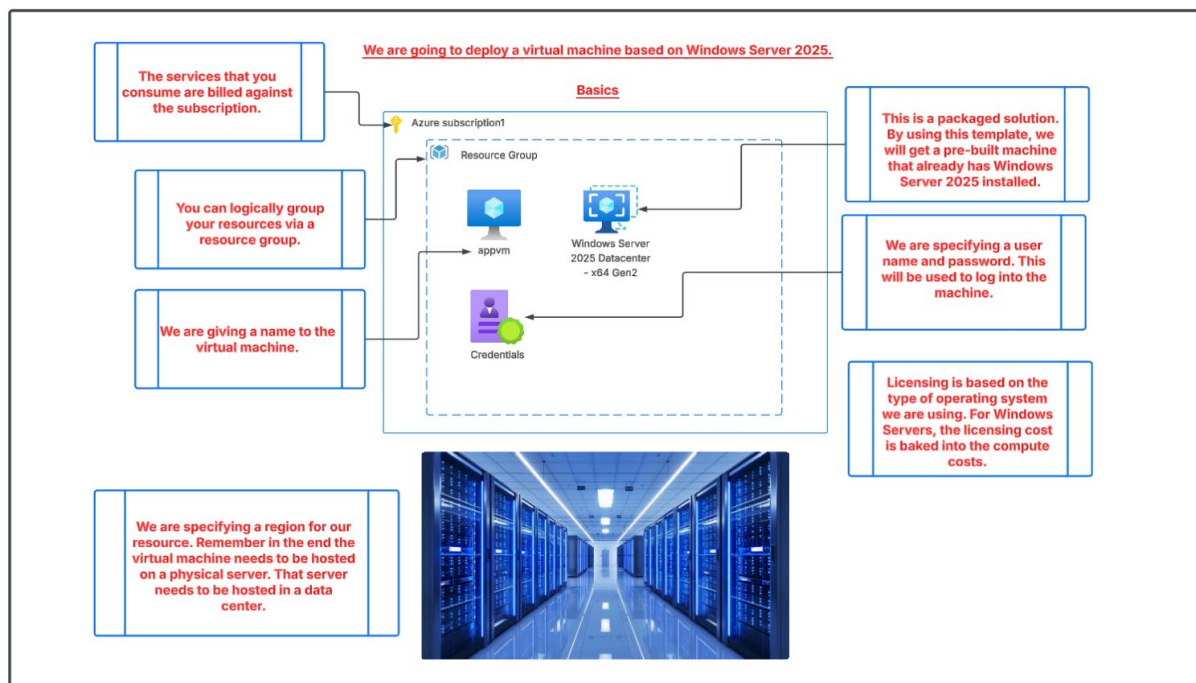


- Buy physical servers
- Buy physical storage
- Deploy the virtualization software
- Create virtual machines

When it comes to the Azure platform, you don't need to buy or maintain the physical infrastructure.

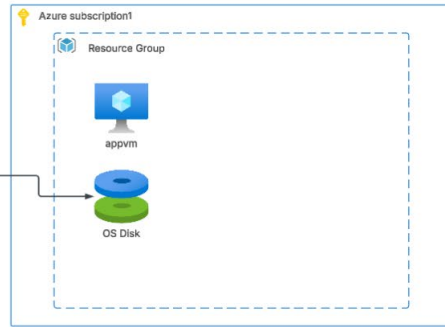
This is managed by Azure. You can just use the service to start deploying virtual machines.

Lab - Deploying a Windows-based virtual machine - Initial configuration



Disks

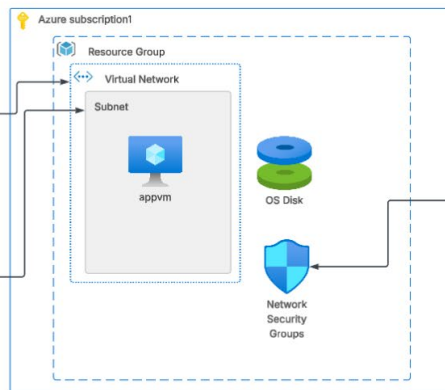
Just like disks that are part of a laptop or computer system, we get disks that can be attached to a virtual machine. By default there will be an OS level disk which will have the operating system files. You can also create and attach multiple data disks.



Networking

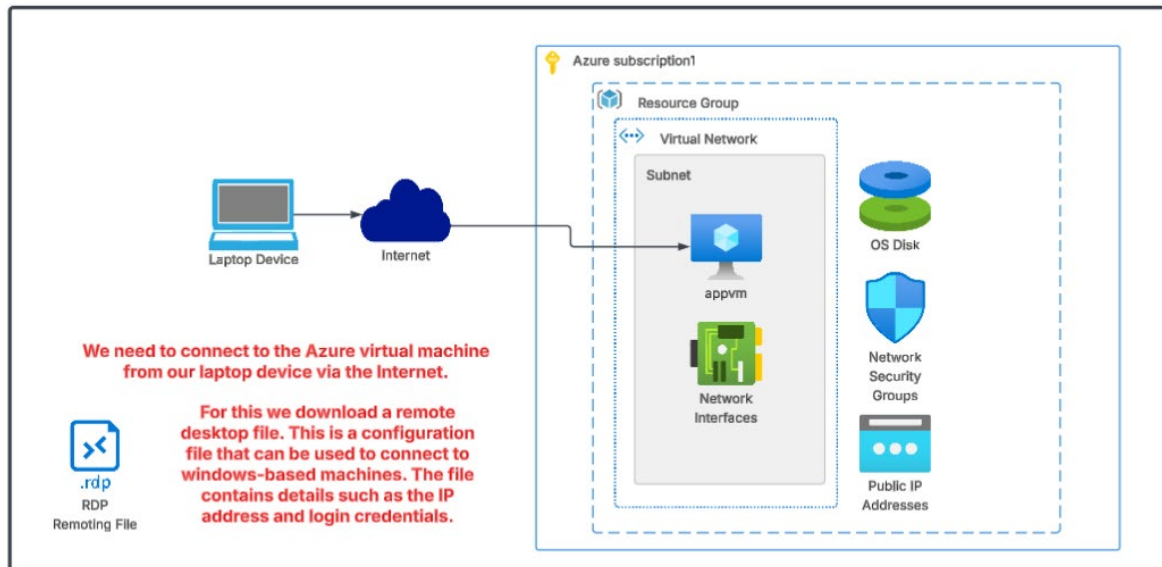
The virtual network is a logically isolated network on the cloud. This is similar to companies having physical networks in their on-premises data centers.

Subnets are just used to break down the network into further logical segments.



The network security group is a light-weight firewall that is used to limit the inbound and outbound connections to the virtual machine.

Lab - Connecting to the virtual machine

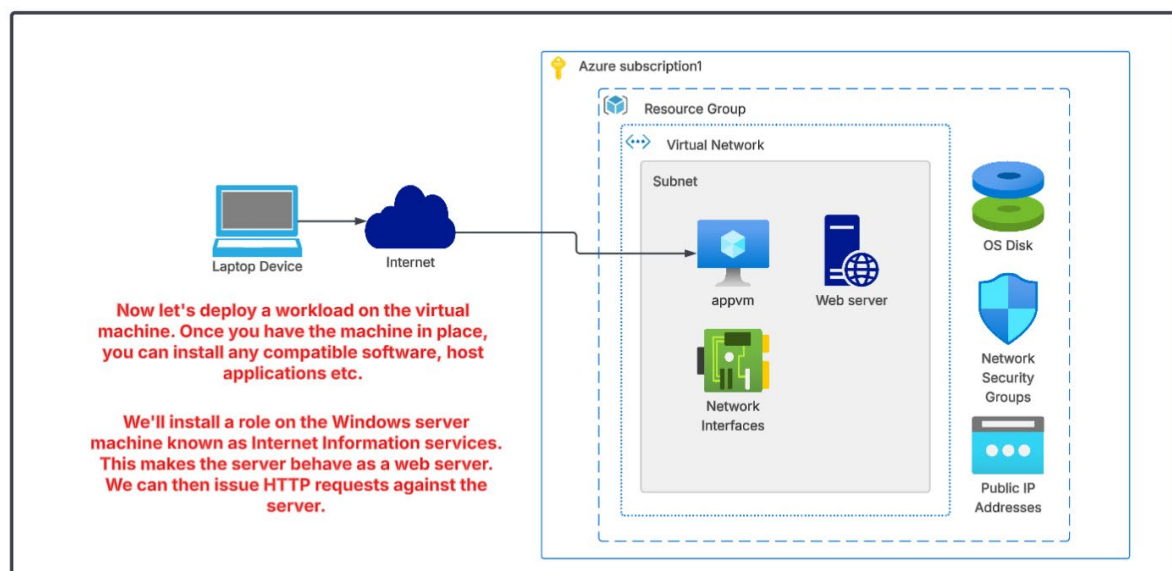


When we connect to the Azure virtual machine, we are connecting using a Public IP address that is assigned to the machine.

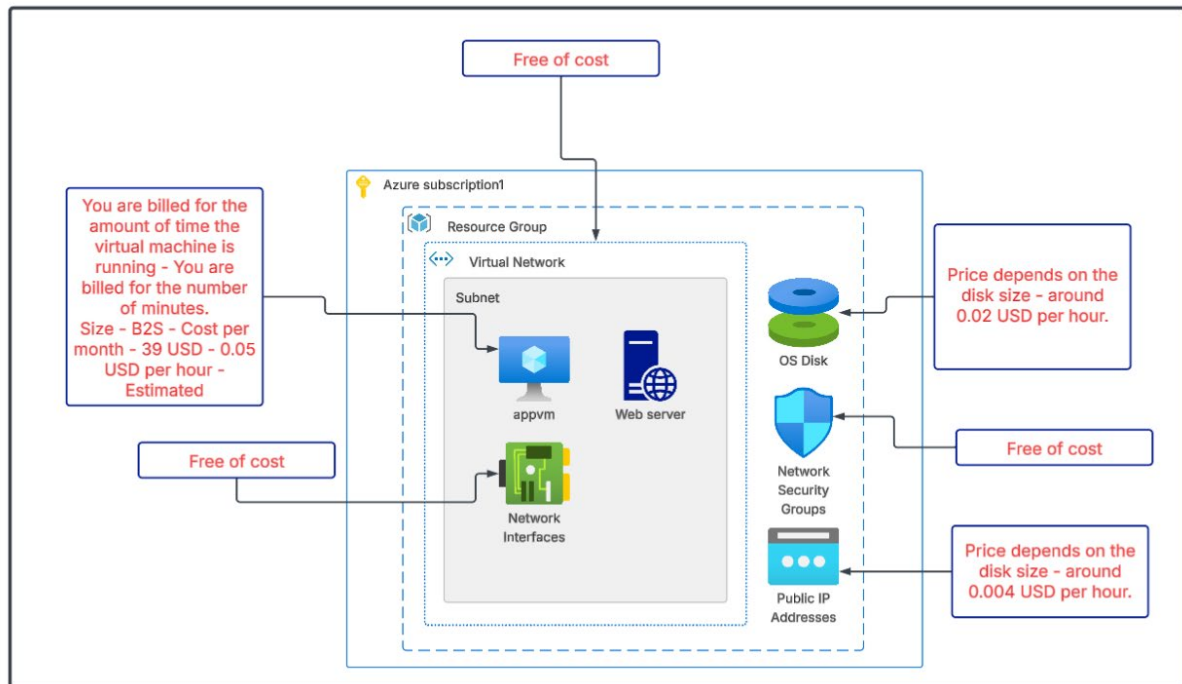
When we build the Azure virtual machine, Azure will create an additional resource known as a Public IP address.

Azure also creates a virtual network interface that is attached to the virtual machine. The public IP address is then attached to the network interface.

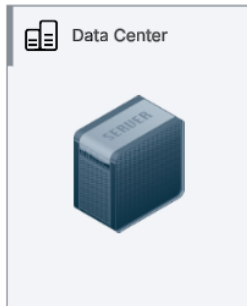
Lab - Deploying a web server on the Windows machine



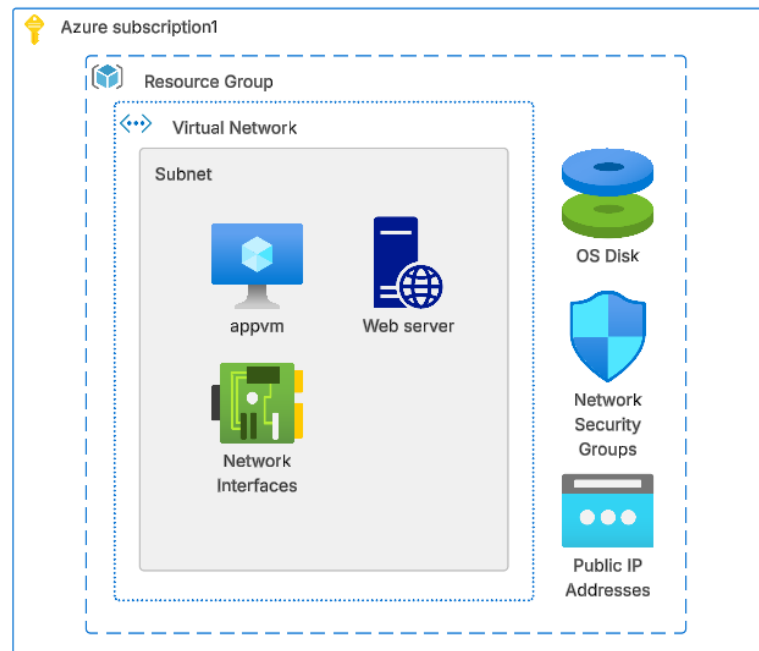
Costing for Azure services



Shutting down the virtual machine



Your virtual machines are hosted on a physical server in an Azure data center.



We can shutdown the virtual machine at the operating system level.

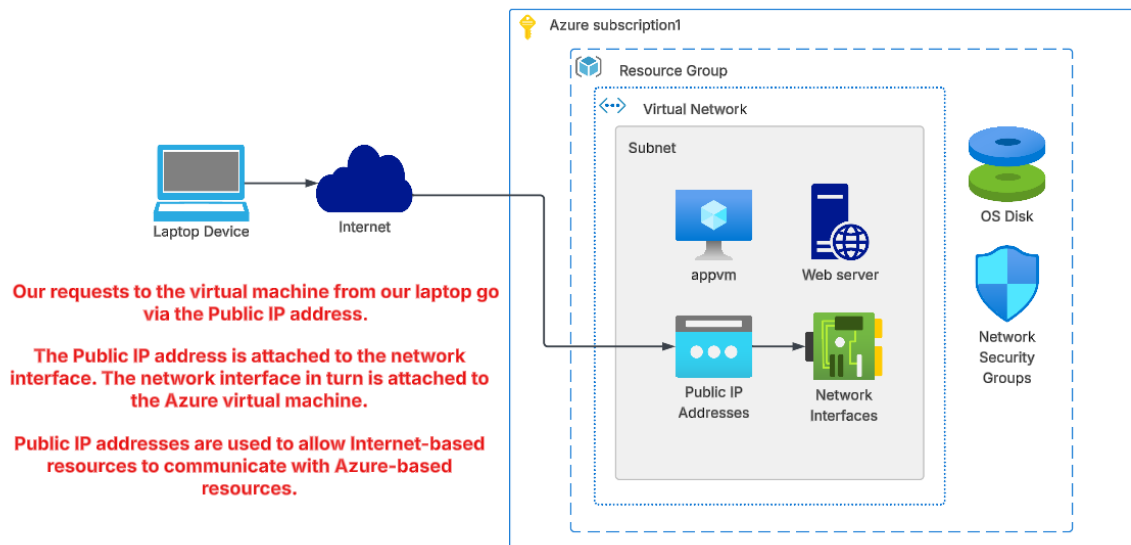
Or we can Stop(Deallocate) the machine from the Azure portal.

When you stop the machine at the operating system level, the machine will still be retained on the physical host.

When you deallocate the machine, then the virtual machine will be removed from the physical host. When you start the virtual machine, it could be launched on the same or different physical host.

When you deallocate the machine, the data on the temporary drive is lost.

About Public IP addresses for virtual machines

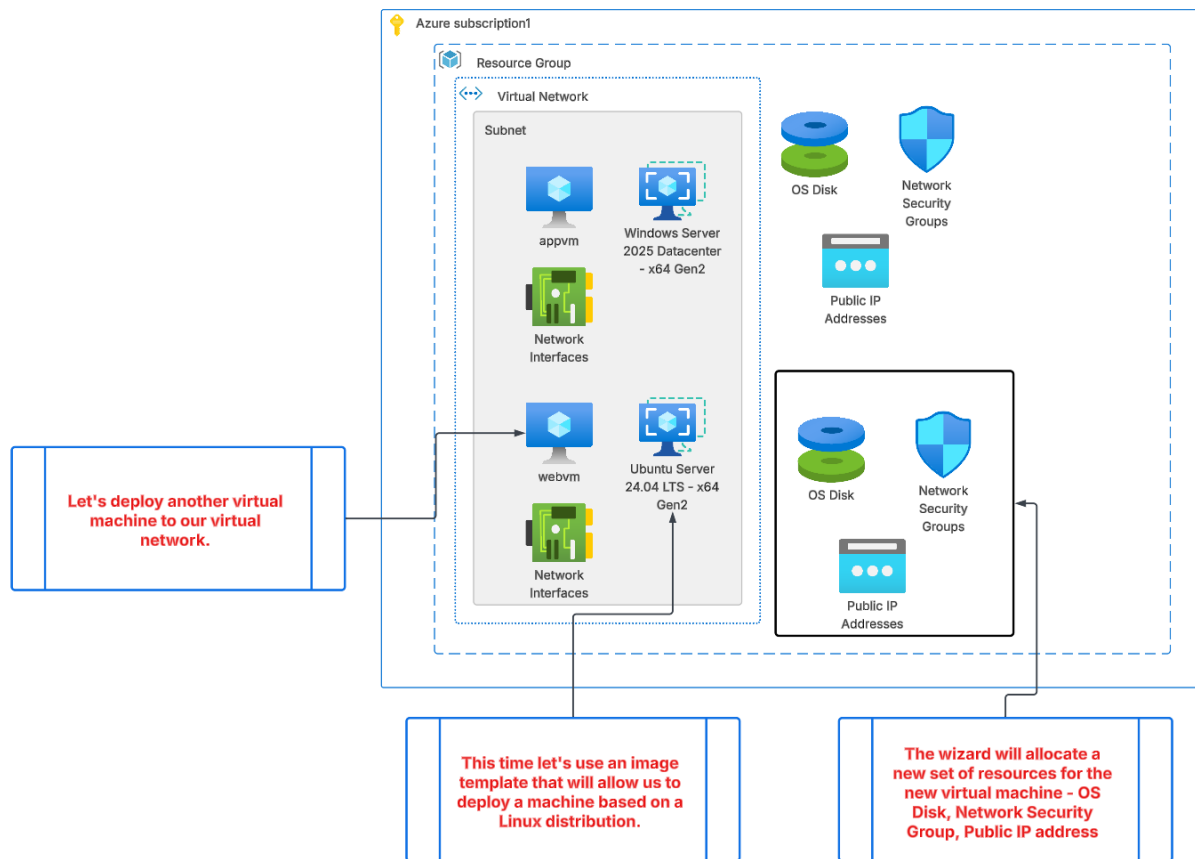


There are two types of assignments when it comes to Public IP's

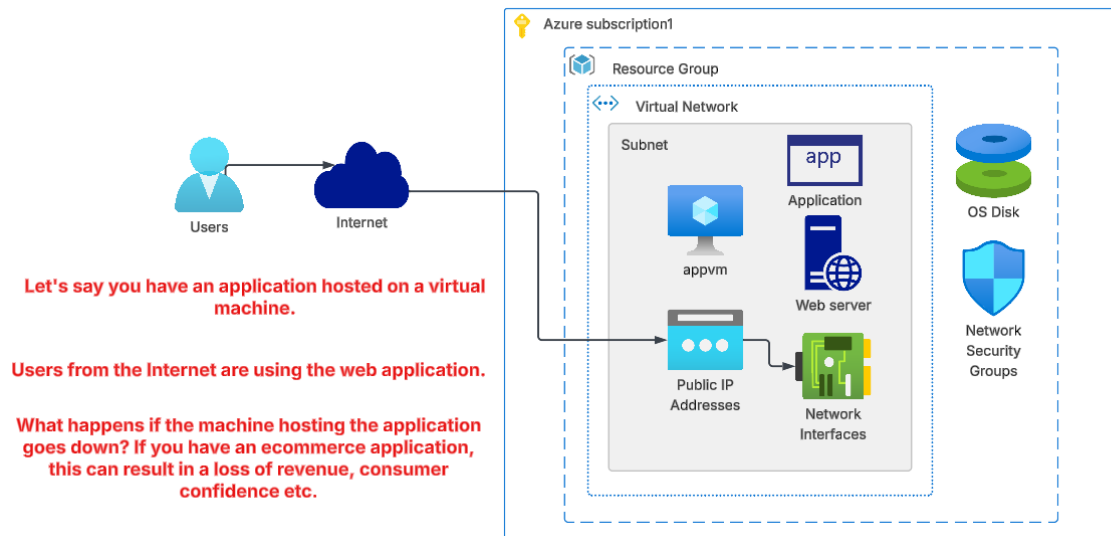
Dynamic - Here the IP address is released when you stop or delete the resource.

Static - Here the IP address is released when the resource is deleted.

Lab - Creating a Linux-based virtual machine



Adding stability for your systems



Azure as a cloud computing platform ensures high availability for your services.

But no one can ever guarantee a complete 100% availability for online services.

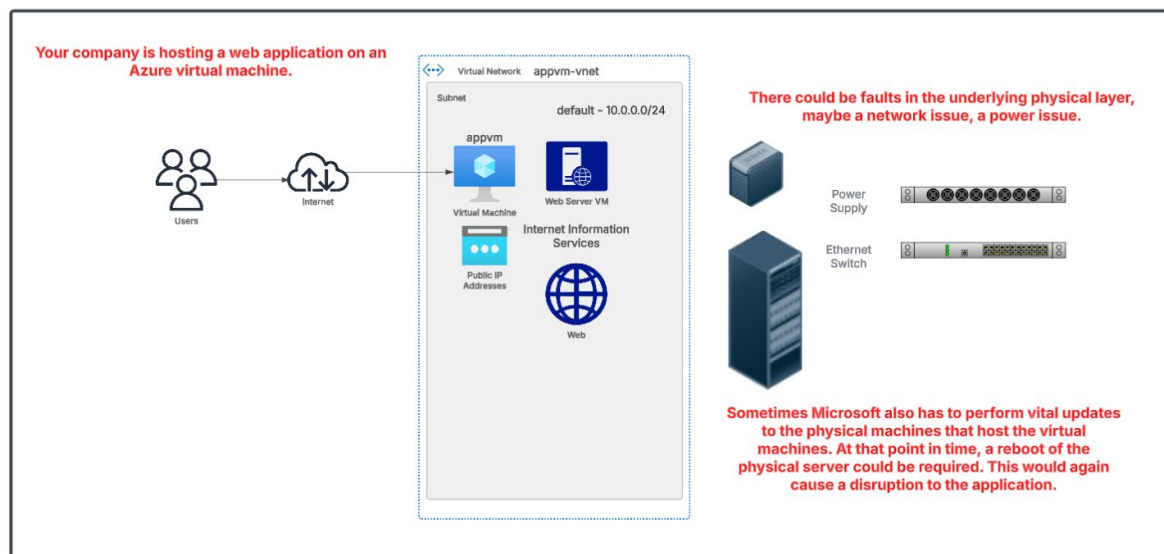
Businesses that have critical and revenue generating applications on the cloud need to look into options into making their application tolerant to failures.

You will hear terms like High Availability, Fault tolerance etc.

How can we make our system tolerant to faults at the infrastructure level.

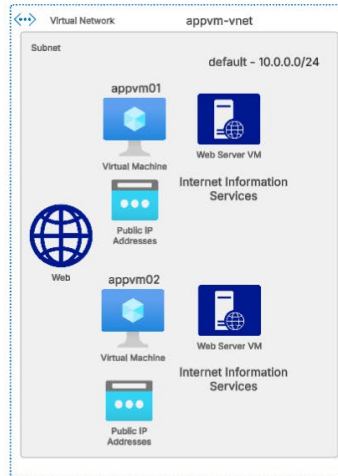
How can we increase the overall availability of our entire system.

Availability Sets



We need to look at increasing the availability of our system.

You can look towards hosting your application on multiple virtual machines.



But what happens if both virtual machines are launched on the same physical server. By default we can't specify the underlying physical servers for the virtual machines.

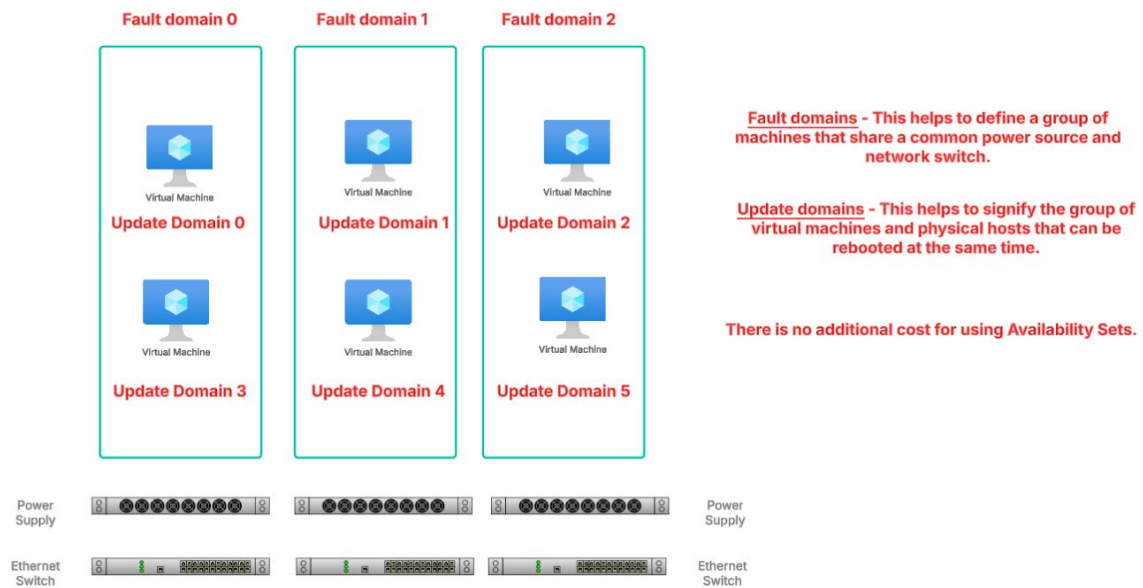
Availability sets

This is a logical grouping of machines that helps to reduce the chances of multiple VM's going down because of hardware issues.

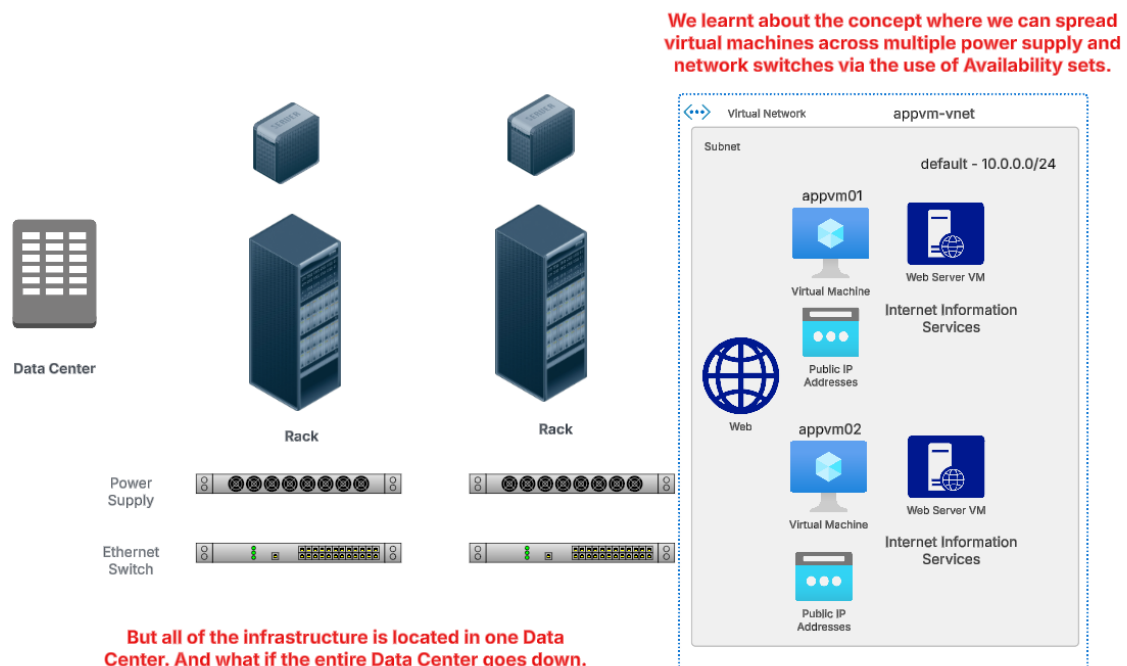
To make use of Availability sets, you need to deploy a Virtual Machine to an Availability Set. You can just create an Availability Set and deploy the machine to the set. The machine can only be part of a set when the machine is created.

The virtual machine is placed as part of a fault and update domain in the Availability set.





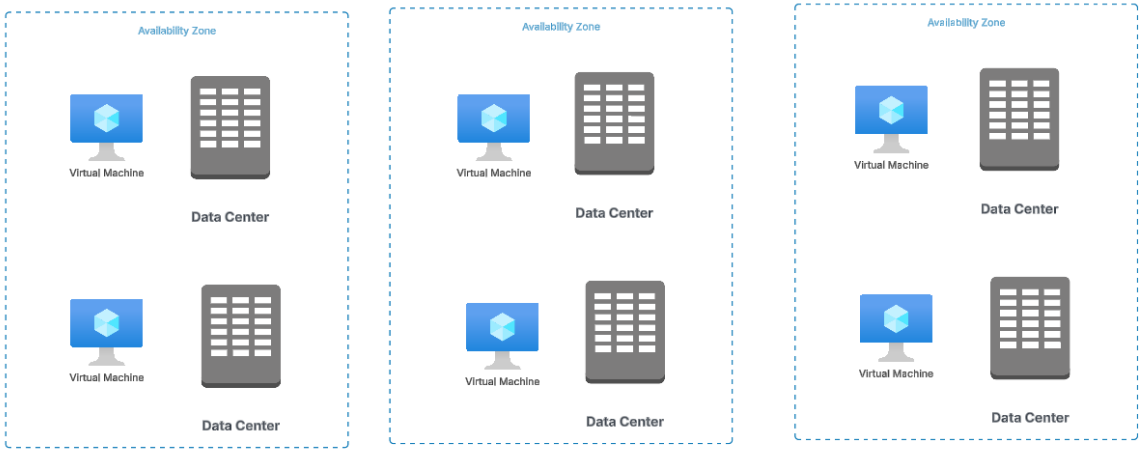
Availability Zones



You can spread your infrastructure across multiple Availability Zones.

An Availability zone is a group of data centers. There are fast links across Availability Zones to ensure low latency.

An Azure region has multiple Availability zones.



Data Transfer	Price
Data Transfer In	Free
Data transfer between Availability Zones(Egress and Ingress)*	\$0.01 per GB
Data transfer within same Availability Zone	Free

There is no cost for using an Availability zone, but there is a bandwidth cost.

Azure Dedicated Host



This can be done with the help of Azure Dedicated Hosts. Here the physical servers are in the Azure Data Center. But they are allocated as a hardware resource to the company.

Multiple virtual machines can be launched on the same physical server. You can also control the maintenance events on the physical server.



When you launch a virtual machine, its located on a physical server in an Azure Data Center.

Other Azure customers with their own accounts can also have their virtual machines located on the same physical server.

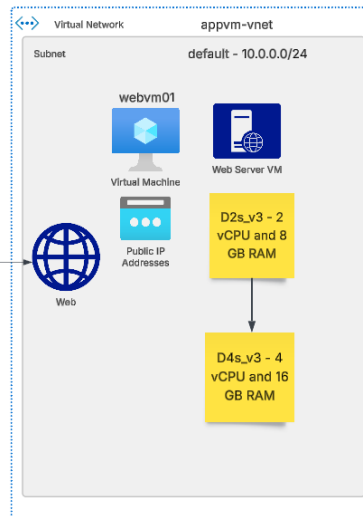
But a large company might want to have a dedicated server for launching their virtual machines.

Virtual Machine Scale Set service

Consider that you have an application hosted on a virtual machine. Users from the Internet are using the application.



Let's say that the application usage has increased and the current hardware is not capable of handling the application load.



We can resize the virtual machine to a higher hardware configuration to manage the extra application load.

This is an example of vertical scaling wherein we add more CPU and RAM to the existing machine.

This is easy to implement. But at some point, increasing the hardware may not solve application performance issues.

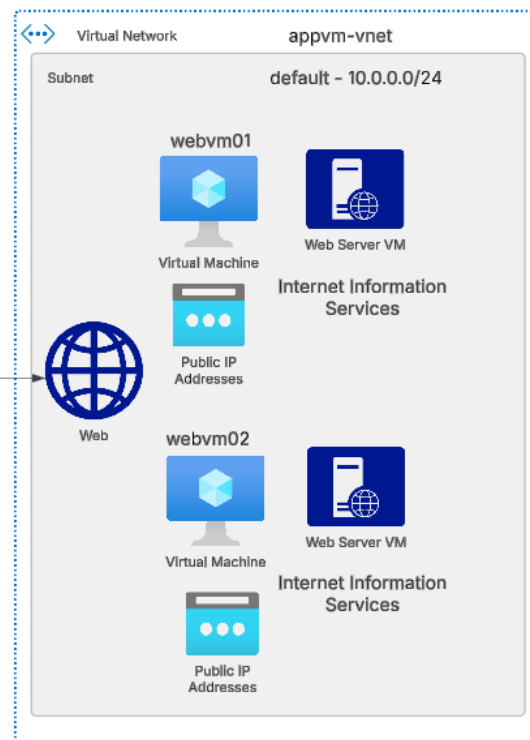
Also this type of scaling might require downtime of the machine.

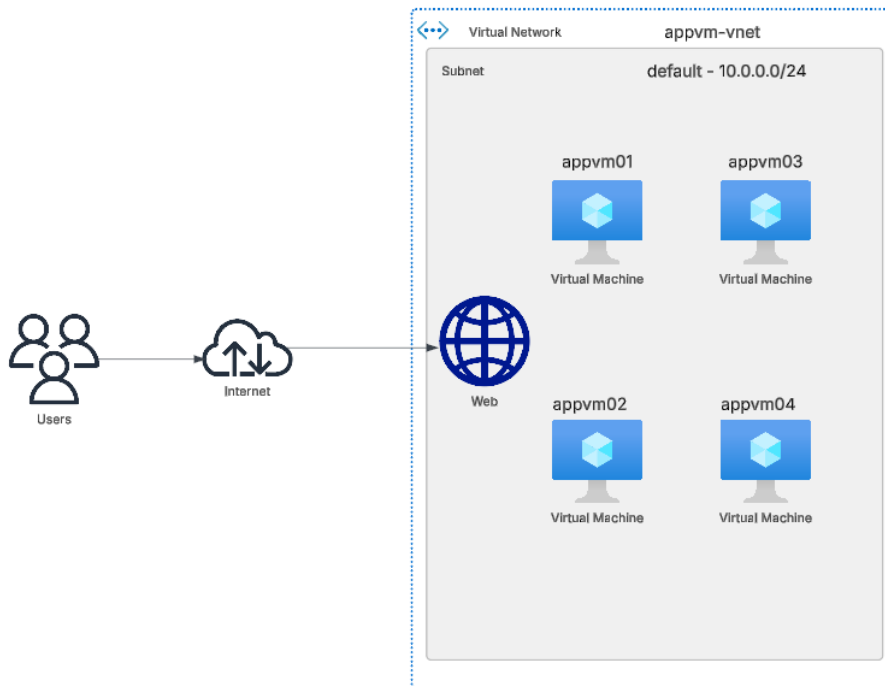
Another option for scalability is horizontal scaling.



Here we add more virtual machines. The application load can then get distributed across the machines.

You can achieve better performance overall with this architecture. You can minimize downtime because you just need to add machines to the existing architecture.





How do we manage the adding of machines to our architecture?

Maybe the traffic is high during the day time and light during the night time and we don't need so many machines during the night. Why does this concern us? Well the cost of having machines in place.

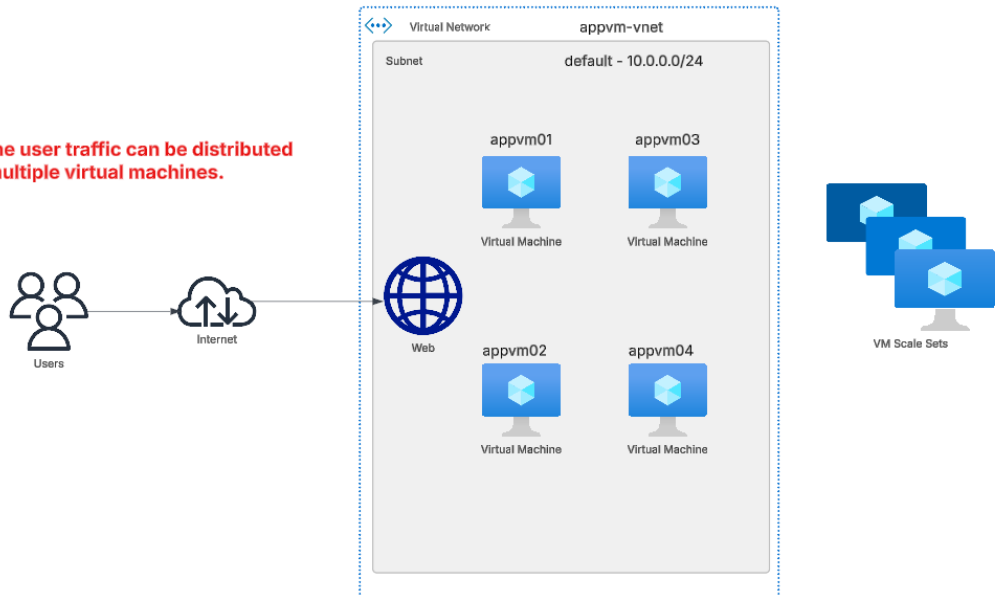
Are you manually going to monitor whether the traffic is high or low, and create and delete machines whenever required.

Azure virtual machine scale sets

This service helps you to create and manage a group of load balanced virtual machines.

The number of virtual machines can then grow based on demand or on a schedule.

By Scalability, the user traffic can be distributed across multiple virtual machines.

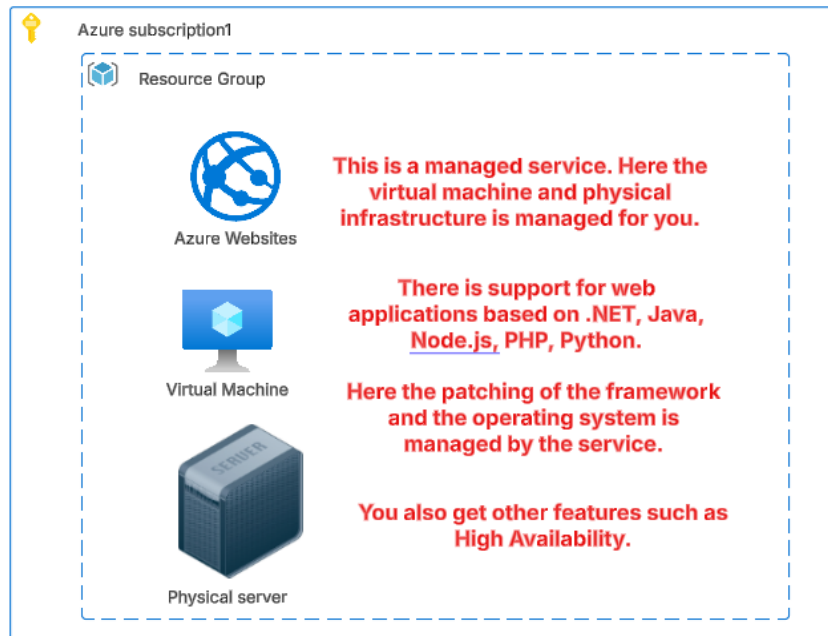


Your machines can be managed by the virtual machine scale set service. You can define rules to define different scaling conditions.

About Azure Web Apps

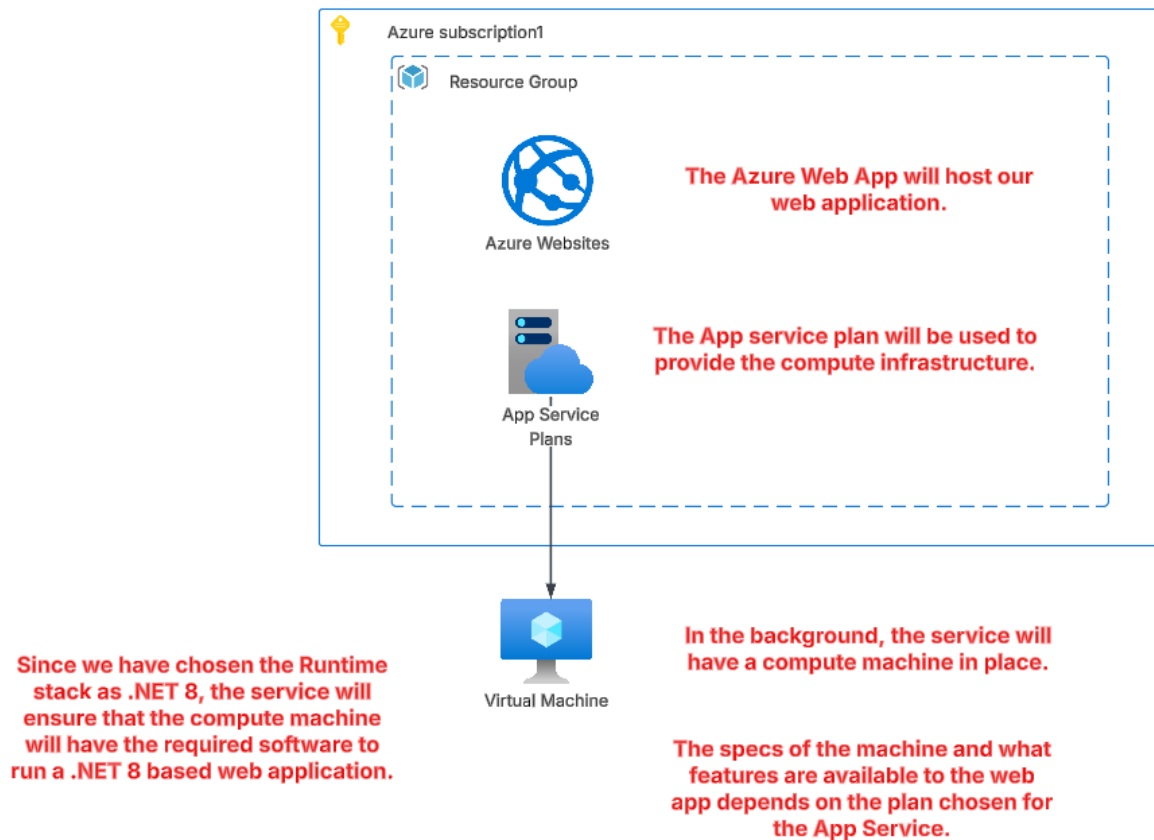
But maybe the company who wants to host the web application does not want the hassle to even manage the virtual machine infrastructure.

They can opt to use the Azure Web App service.

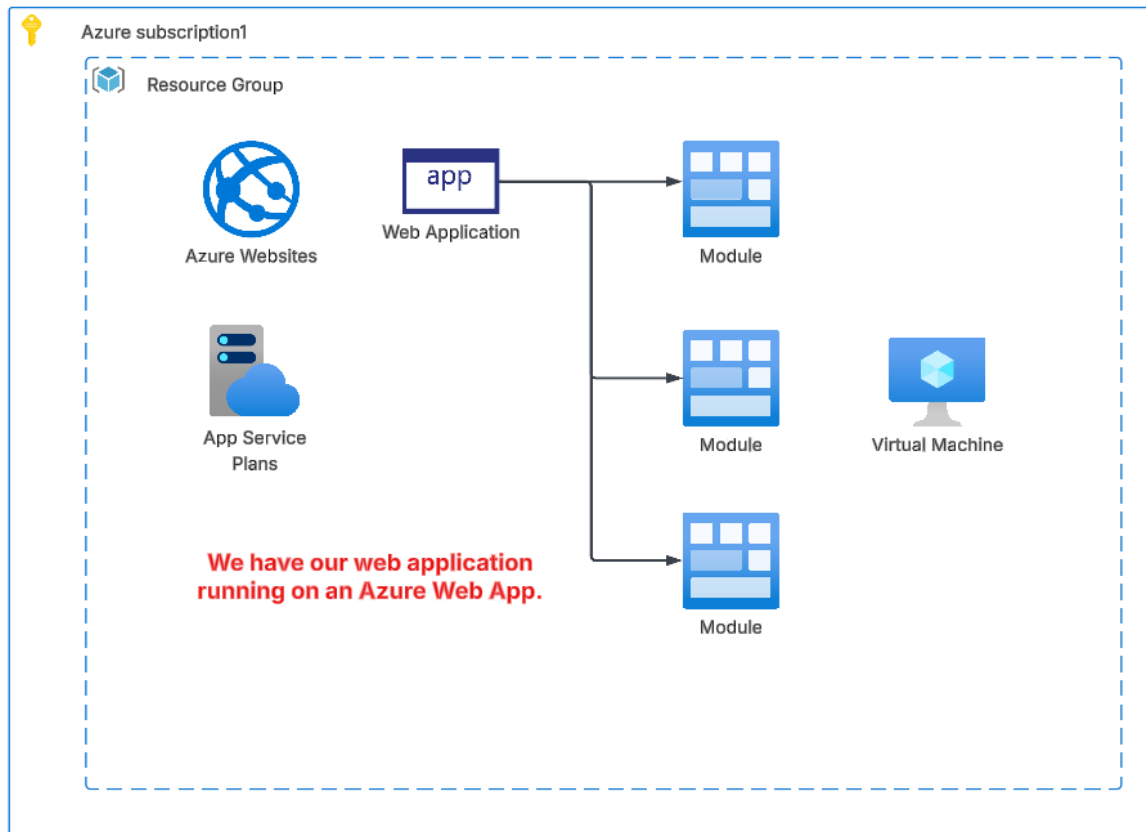


If you have a web application that fits the framework and you don't want to manage the virtual machines, then you can opt for the Azure Web App service.

But if you need to host a custom application that needs to be installed, then you would probably need to use the Azure virtual machine service.



About Azure Functions

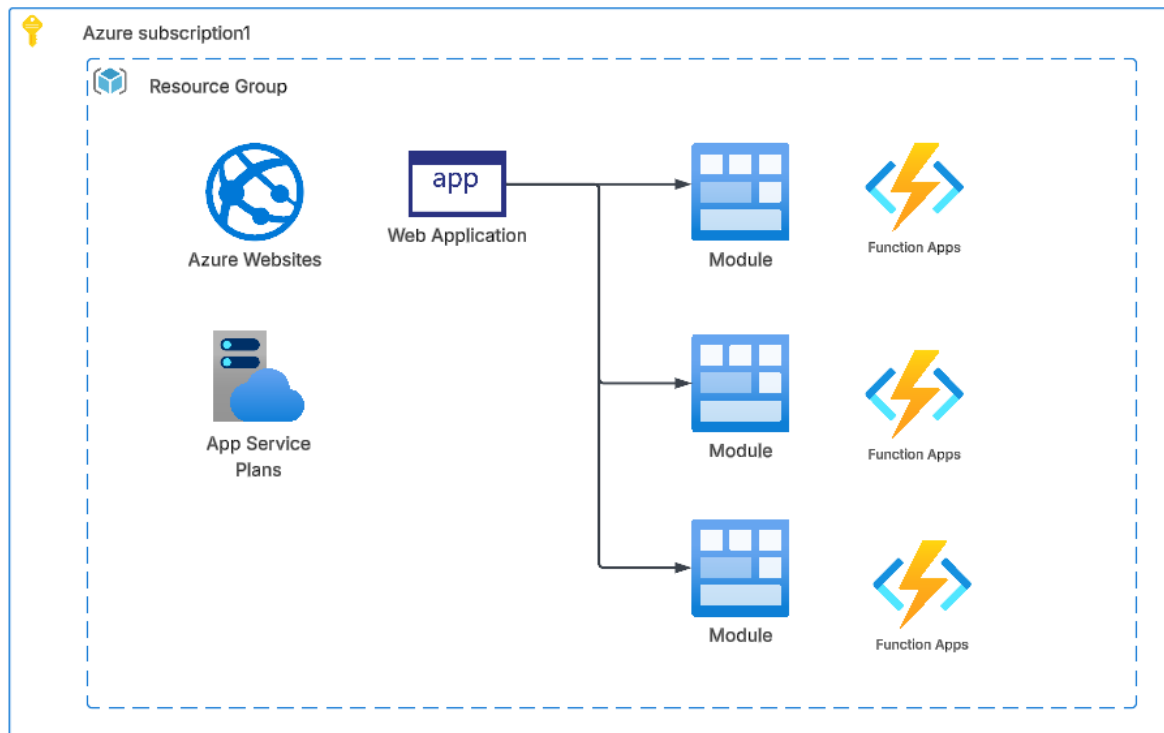


The Web Application might need to call several modules. Each module is built on code and performs a particular functionality.

These could be central modules that could be called by many web applications.

These code modules need compute to run. One option is to run these modules on Azure virtual machines.

But when it comes to these modules, they would not be running all the time. They would ideally only run when they are invoked by the web application.



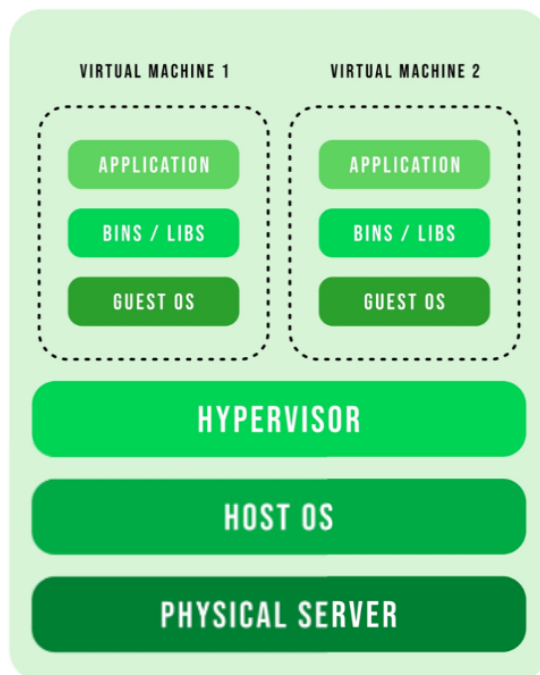
Azure provides a serverless service known as Azure Functions.

Here you don't need to maintain the underlying environment. Just publish your function code onto Azure Functions.

Your web application can invoke the Azure functions whenever required.

You only get charged based on the running time of the Azure Function.

Primer on containers



VIRTUAL MACHINE ARCHITECTURE

Virtual Machines were designed to make the most of the hardware specifications of the underlying physical machine.

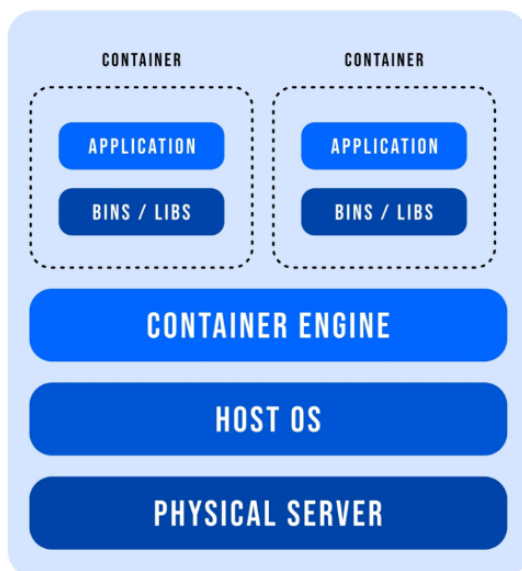
You could have multiple virtual machines hosted on the same physical server.

But then there were still some problems with it came to application deployments.

Companies would deploy multiple applications on the same virtual machine. They want to make the most of each virtual machine.

But let's say that one application requires an update at the virtual machine operating system level. This update could break the other applications running on the same virtual machine.

The other most common issue was when applications were copied from one environment to another - From development to production. The application would be perfectly working on development but would not work on production due to configuration differences.



CONTAINER ARCHITECTURE

Then came the advent of container-based tools such as Docker. These tools allowed you to package your code and all dependencies required by the application within a container.

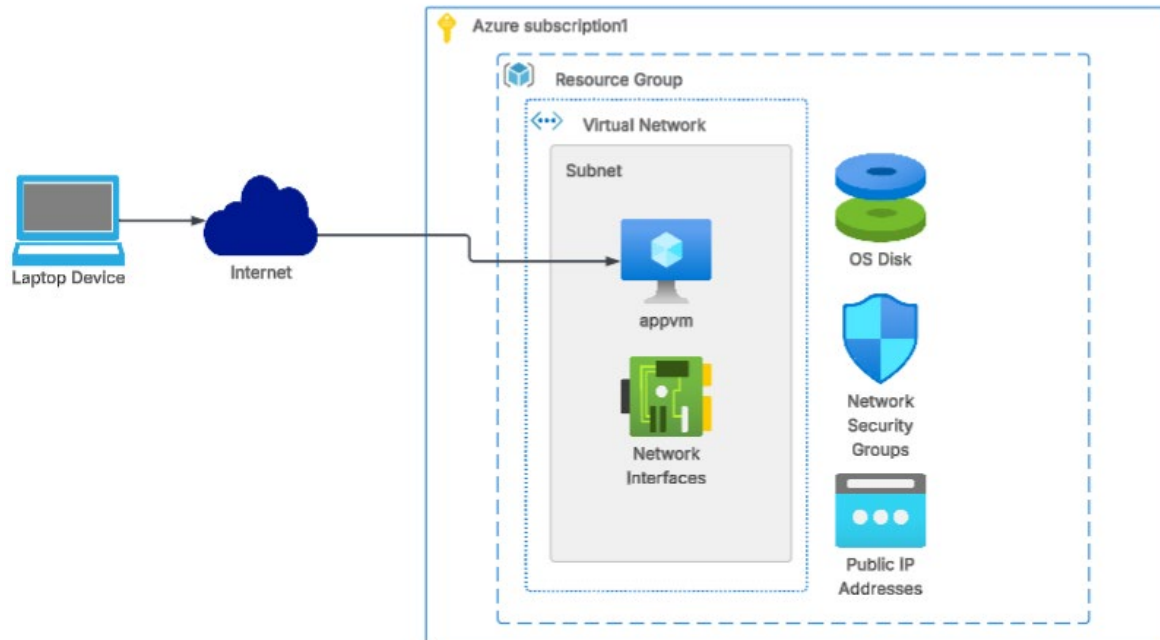
That container could the run on any machine that had the underlying supported container runtime installed such as Docker.

Each container can run independently.

You can install updates for each separte container as required by the application and it would not affect other containers running on the same host system.

Describe Azure architecture and services – Networking

Azure Virtual Network



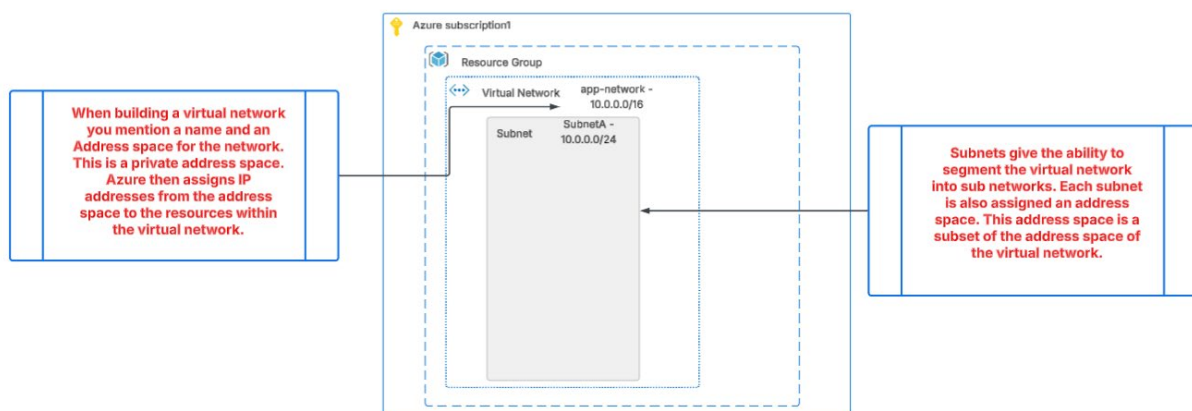
When deploying a virtual machine, it needs to be located within a virtual network.

The virtual network is an isolated network on the cloud.

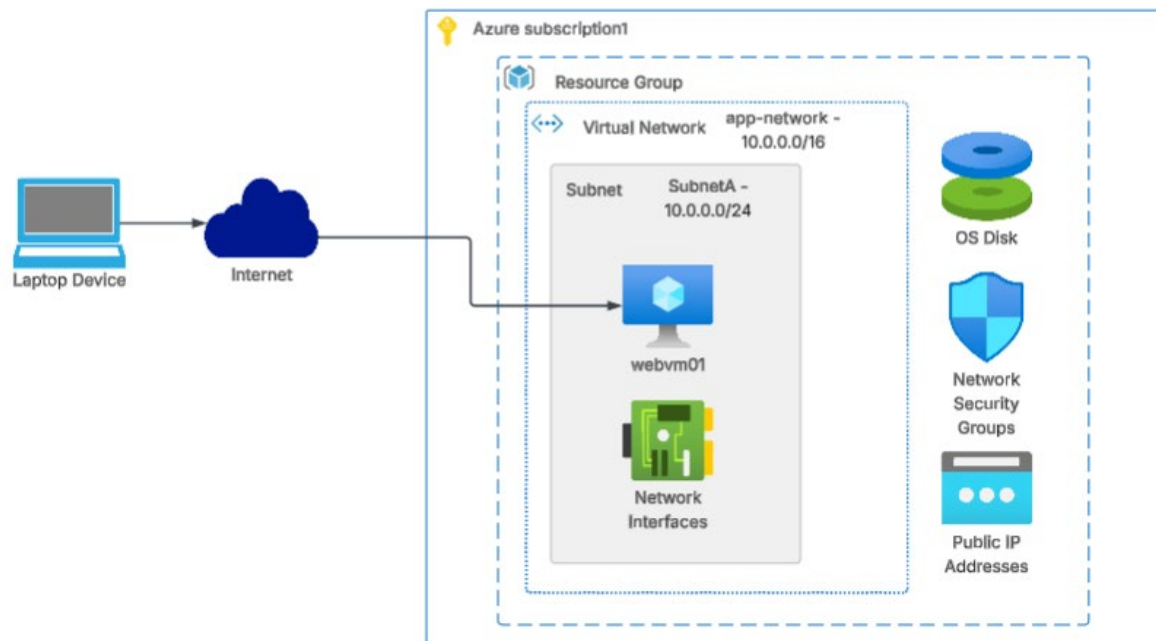
The network is similar to networks that are setup within on-premises data centers.

Once you have a network in place, you can start to deploy resources to the network.

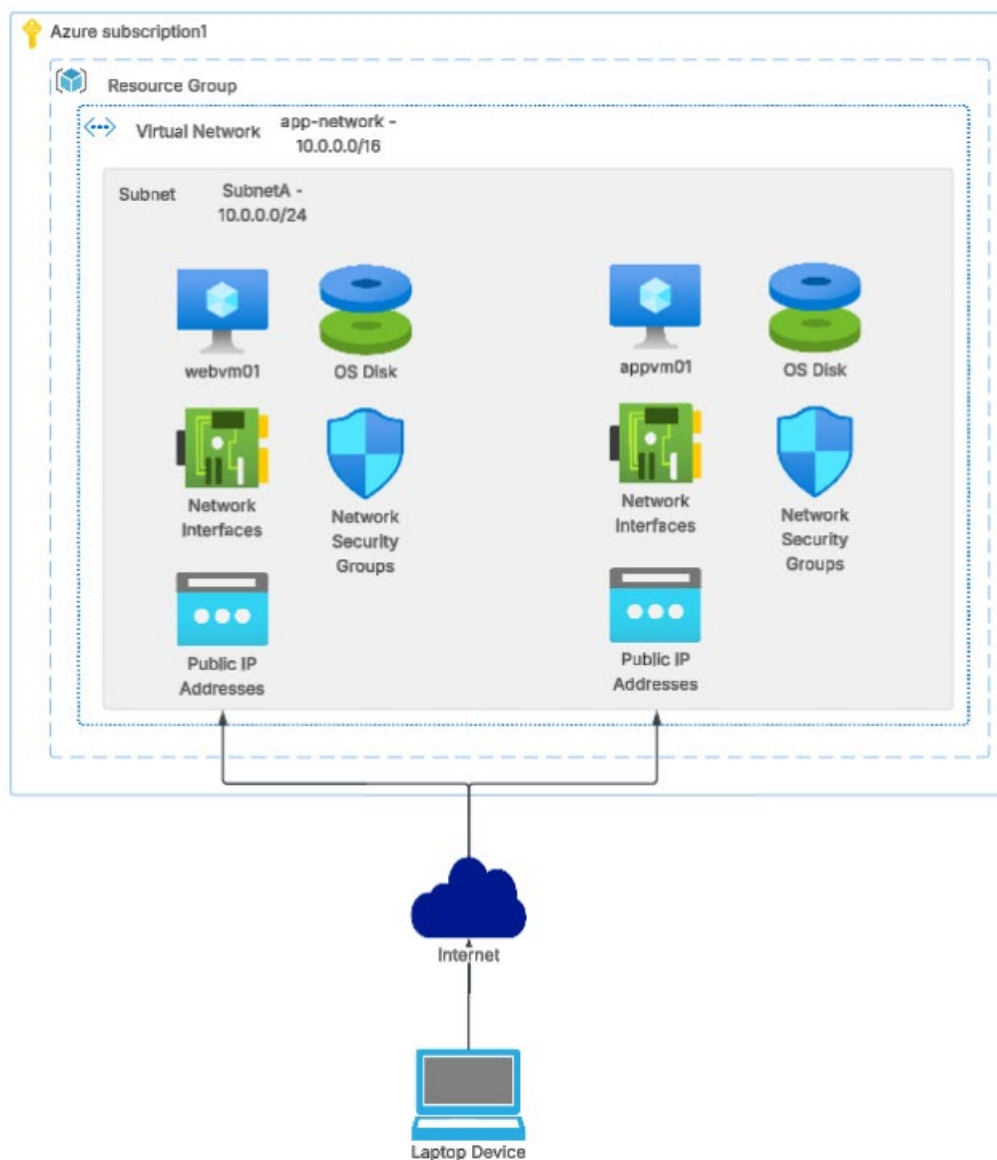
Initially let's build a virtual network



We can then deploy a virtual machine onto the virtual network



Communication across virtual machines in a virtual network – Overview



Let's deploy another machine to our virtual network

We can create another subnet in the virtual network and deploy the machine there as well.

On webvm01, let's install NGINX as the web server.

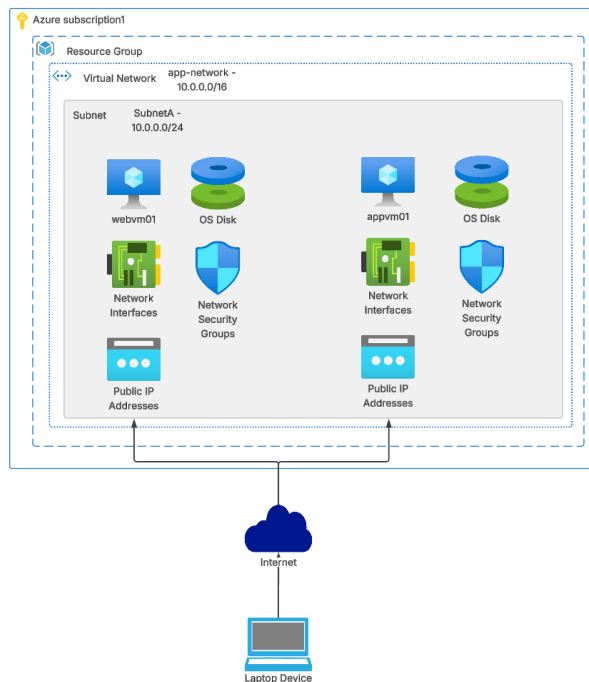
From our laptop machine, we can reach both machines via their respective Public IP addresses.

Then from appvm01, let's send an internal HTTP request via the Private IP address of webvm01.

Within the virtual network, private communication is possible via Private IP addresses.

Note that here both machines are based on a Linux distribution, but internal communication is also possible with a mix of Windows and Linux-based machines.

Network Security Groups



Network Security Groups - This is used to filter traffic between Azure resources in an Azure virtual network.

The Network security group consists of rules that allow or deny inbound and outbound traffic.

For each rule you define the following

1. A name for the rule
2. A Priority - The rules are evaluated in order of priority
3. Source or destination - This could be an IP address, A CIDR block, a service tag or an application security group.
4. A protocol - TCP, UDP, ICMP etc
5. Port Range
6. Action - Allow or Deny

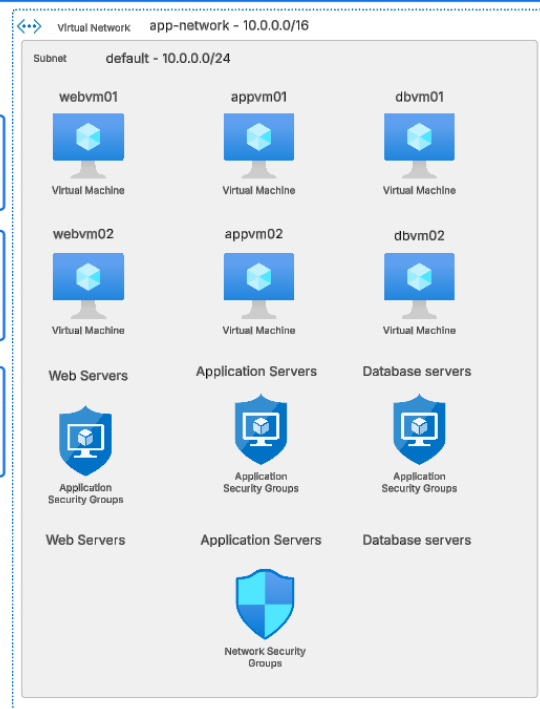
Application Security Groups

Application Security groups is just an extension onto Network Security Groups. Here you can group virtual machines when it comes to defining the required security policies.

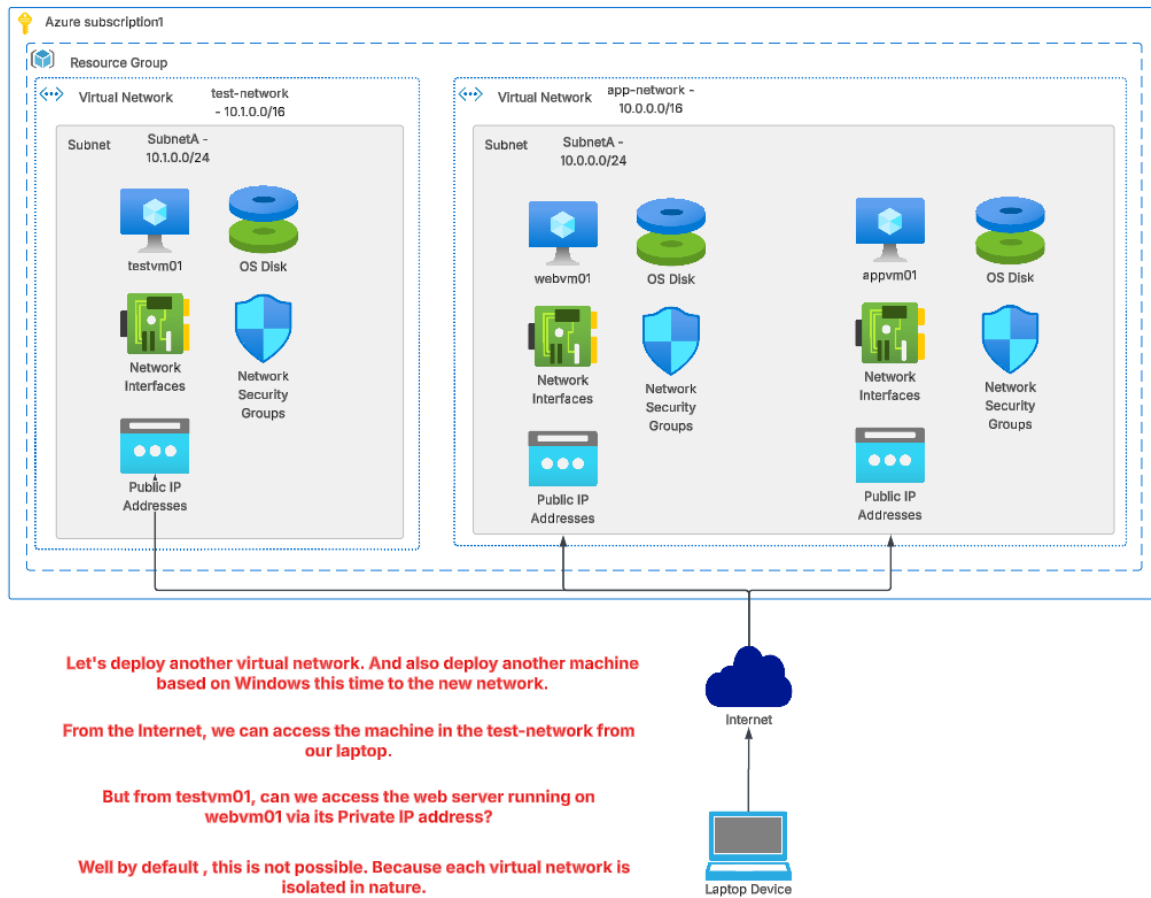
We can define three Application Security Groups, one for each set of servers.

We have one Network Security Group. We define a rule that allows traffic from the Web Server Application Security Group to the Application Server Application Security group.

We can then define another rule in the Network Security Group that allows traffic from the Application Servers Application Security Group to the Database Servers Application Security Group.



Lab - Azure virtual network peering

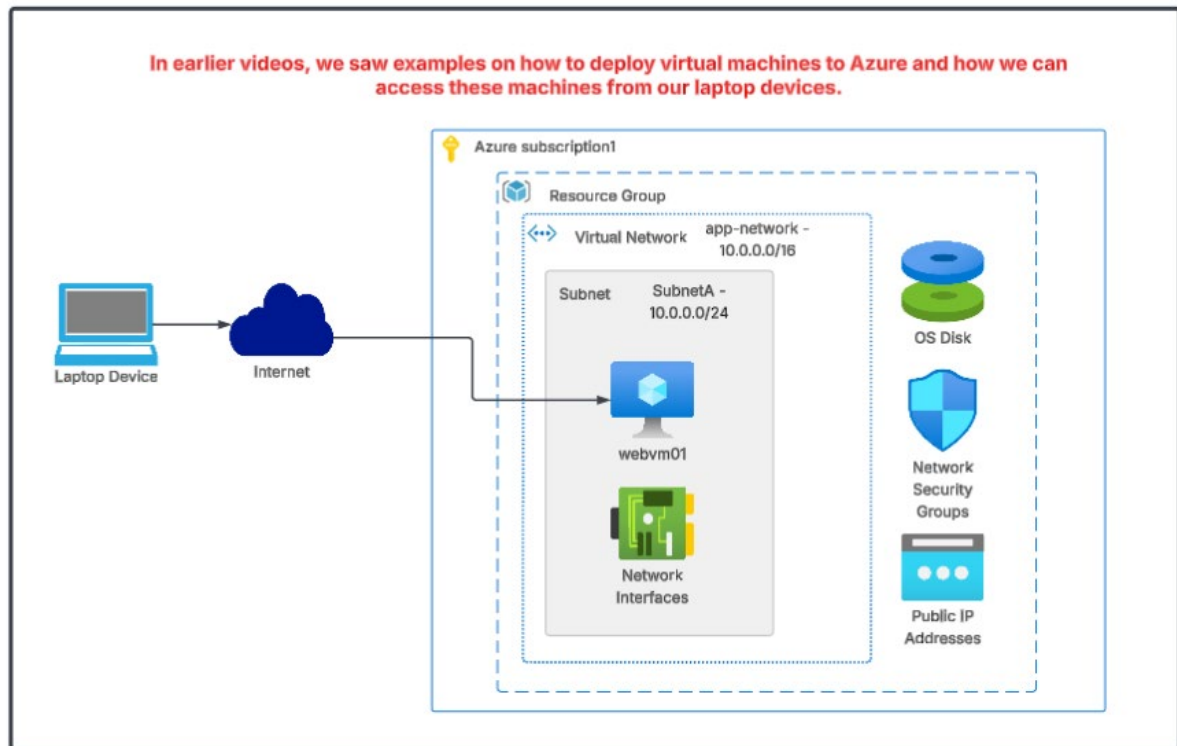


We can connect the virtual networks together using virtual network peering.

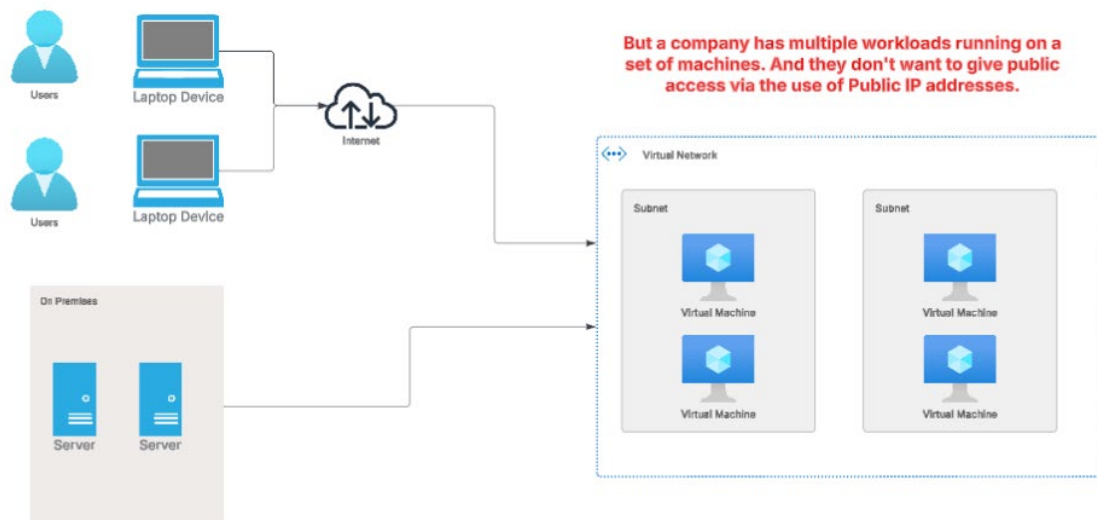
The traffic between the virtual machines across the peered virtual networks travels across the Microsoft backbone network.

You can connect virtual networks in the same region or across regions.

Overview of VPN connections to Azure



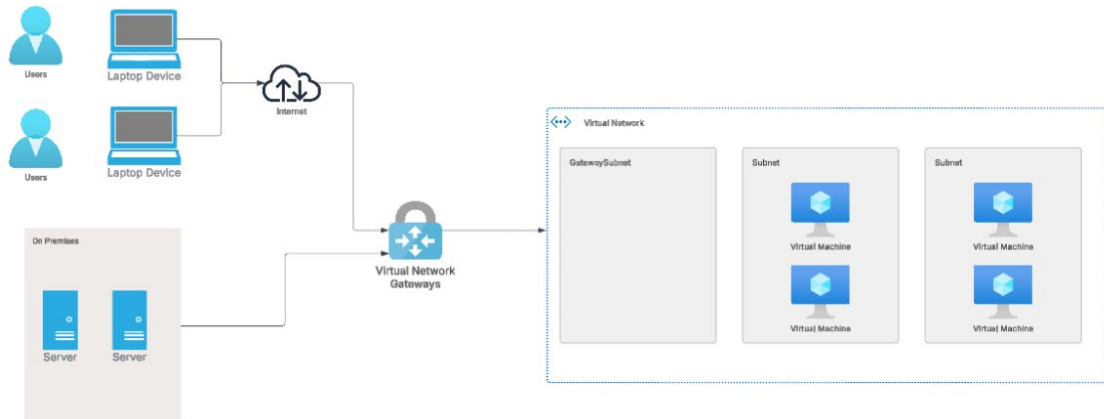
1. A set of employees need to access the workloads from their company laptops via the Internet.



2. The company wants their on-premise workloads to access the workloads in the virtual networks.

Here you create a Point-to-Site VPN connection.

1. A set of employees need to access the workloads from their company laptops via the Internet.



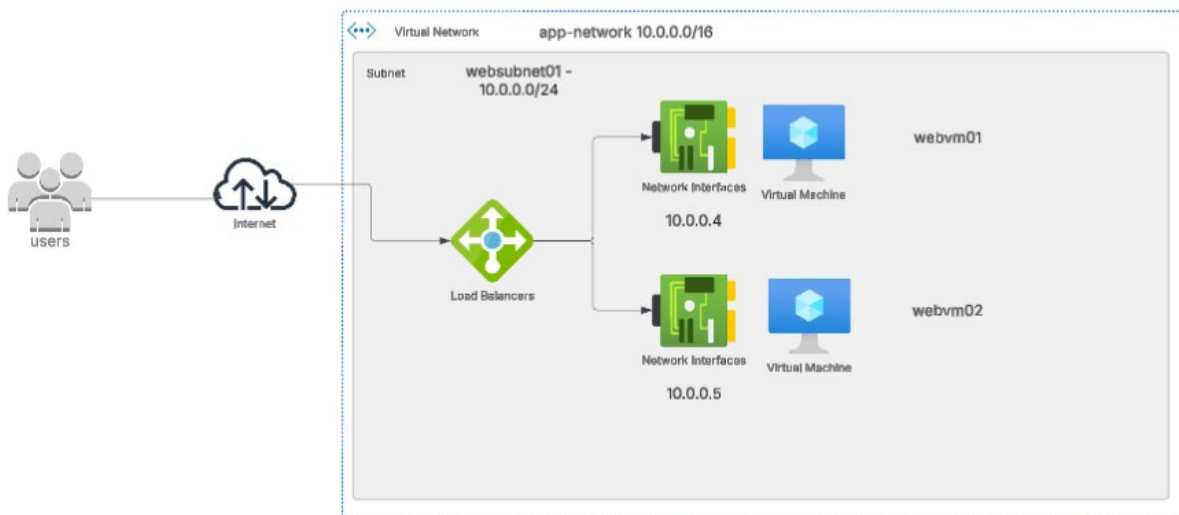
2. The company wants their on-premise workloads to access the workloads in the virtual networks.

Here you create a Site-to-Site VPN connection.

Azure Load Balancer

Azure Load Balancer

A load balancer is used to distribute the incoming network traffic across a set of backend servers.

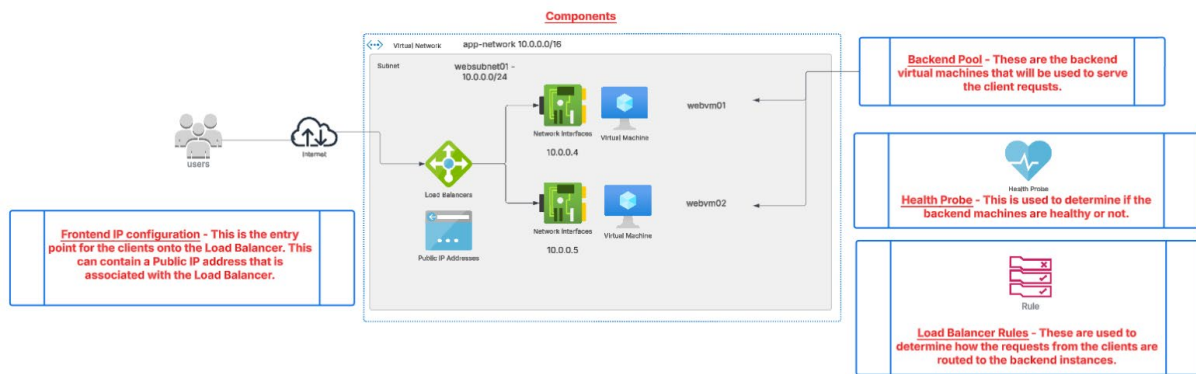


You can have an application hosted on a set of machines.

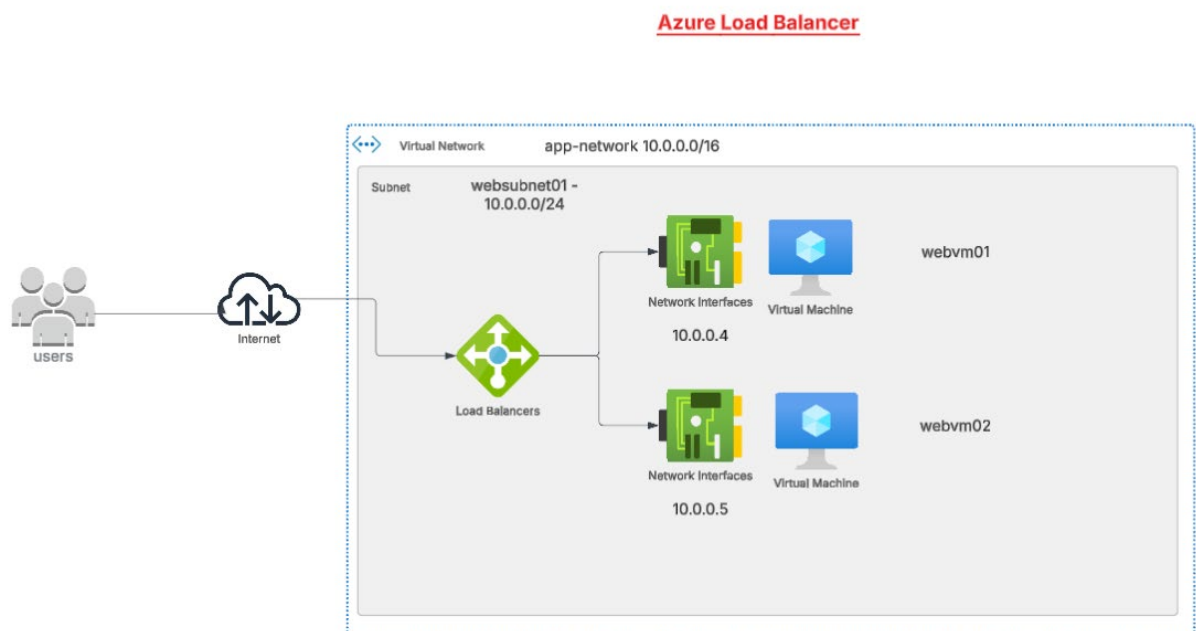
You want user traffic to be distributed equally across the machines.

For this we can make use of the Azure Load Balancer service.

Here the Azure Load Balancer can distribute traffic across the private IP addresses of the backend machines.



Lab - Azure Load Balancer – Setup



1. Let's deploy two virtual machines based on Ubuntu Linux.
2. Then install NGINX and have a [default.html](#) page on both machines.
3. We will then disassociate the Public IP addresses from both machines.
4. Then we will deploy and configure a Public-facing Azure Load Balancer.

Azure Domain name system



You have a web application located on a machine.

In order to send a request to the machine hosting the application, you need to contact the machine via its IP address.

In the browser , <https://50.47.100.20>

Users cannot easily remember IP addresses. Hence we have domain names which are user friendly to remember. They can be linked to IP addresses.

We can map a domain of cloudportalhub.com to 50.47.100.20 so users can type <https://cloudportalhub.com> in their browser.



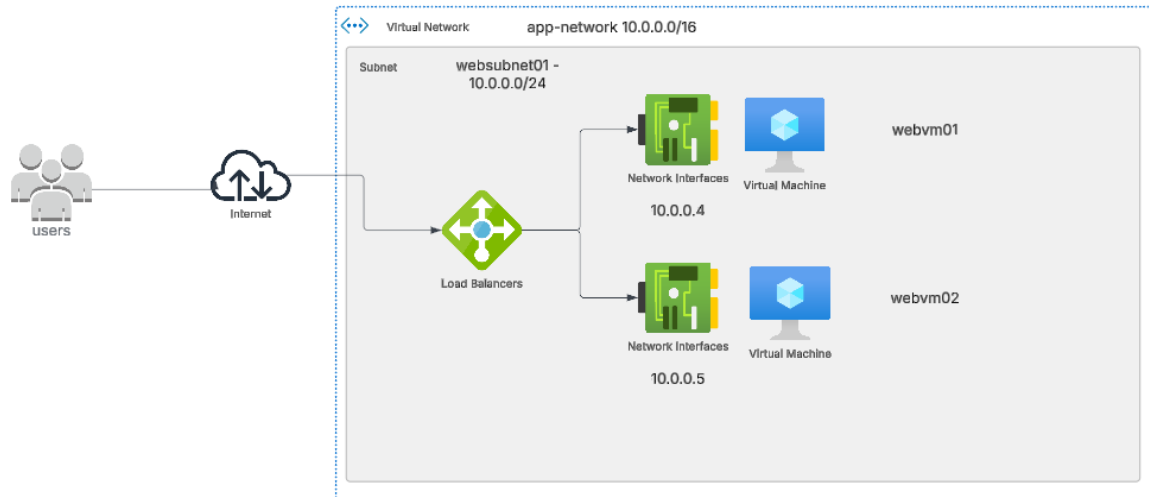
There are many DNS servers on the Internet.

Your request for cloudportalhub.com first needs to be resolved to the IP address of the machine

A request is made from your machine onto the Internet via the various DNS servers on the Internet to first find out the IP address mapped to cloudportalhub.com.

Once a response comes back , then the laptop will make a connection via the Internet onto the machine hosting the web application.





Azure provides their own in-built DNS that manages the DNS zones and records.

Azure also provides internal name resolution for virtual machines.

We can assign a DNS name for the public IP address and access the web server via the public-based DNS name.

Each virtual machine also gets an internal DNS name. Internally we can access the machines via the internal DNS name.

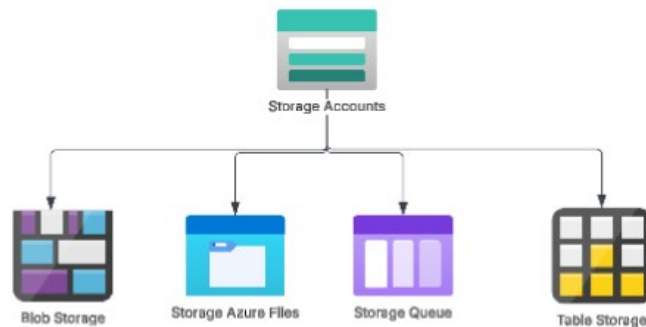


We can also create Azure DNS zones. You can create the required record sets in these zones.

Describe Azure architecture and services – Storage

Introduction to Azure Storage Accounts

Azure Storage Accounts - This is storage on the Azure cloud for your blob objects, files, queues and tables.

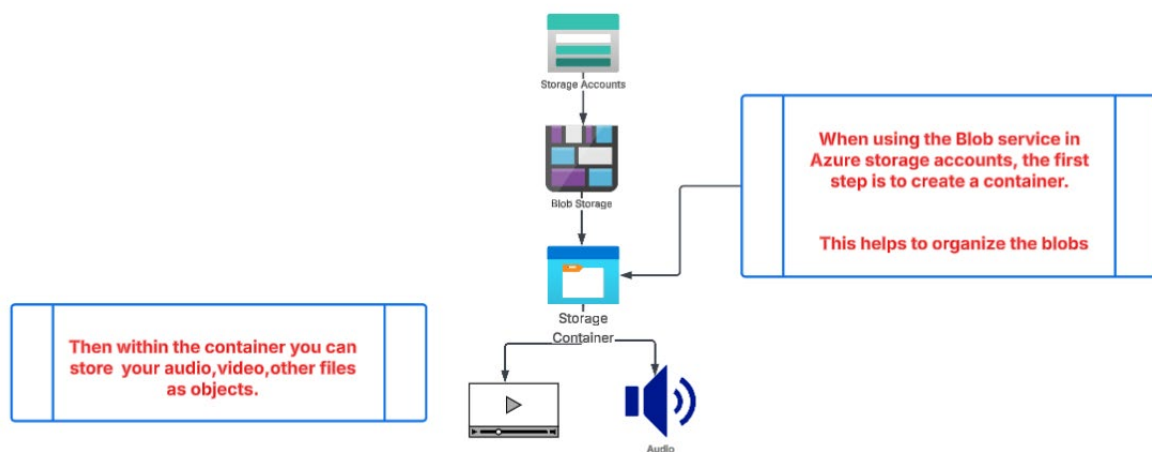


There are different types of storage accounts. We will focus on the Standard general-purpose v2 storage account. This account gives us access to the following storage services - Blob, Queue, Table and Files.

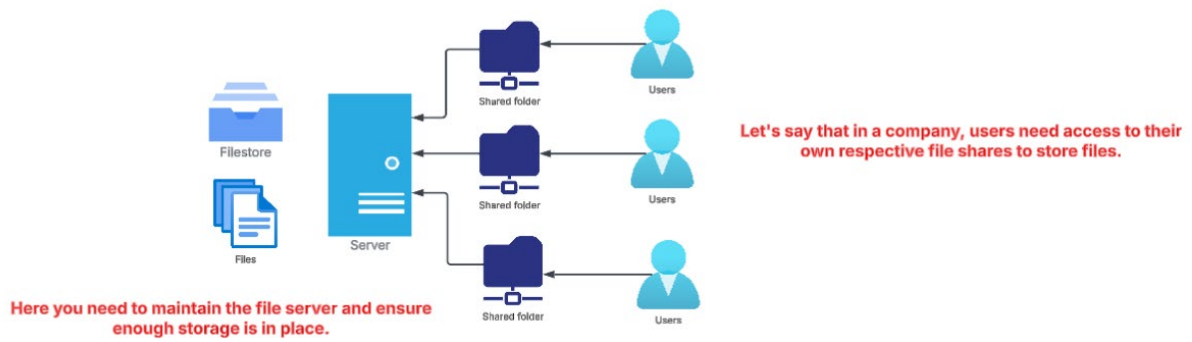
When creating a storage account, the name of the resource needs to be unique within Azure.

Blob service

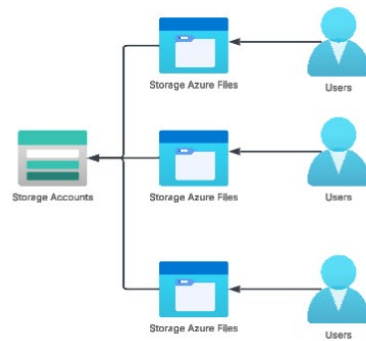
This is used for storing a large amount of unstructured data. Suitable for storing images, documents, video and audio files.



Azure Files

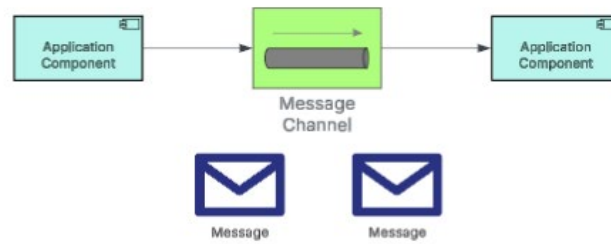


You can instead opt to use the File service in Azure Storage Accounts. This is a fully managed file share that is accessible via the Server Message Block (SMB) and Network File System (NFS) protocol.



Here the physical storage is managed for you.

If messages need to be shared across multiple application components. Here you need to have the message software and maintain it.

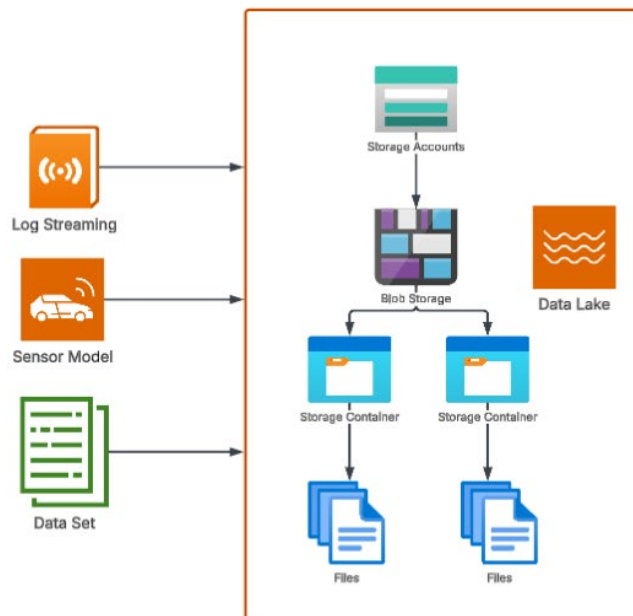


Instead we can make use of the Queue service which provides the basic messaging service.



If an application needs to store data (non-relational structured data), like let's say data about users.

Azure Storage Accounts - Access tiers



An Azure storage account has the ability to double down as a Data Lake storage account.

Companies can then stream in data from various sources onto the storage account.

Here you have data coming in at high velocities and also in large numbers.

There are various factors that govern the pricing when it comes to Azure storage. One aspect is the storage cost.

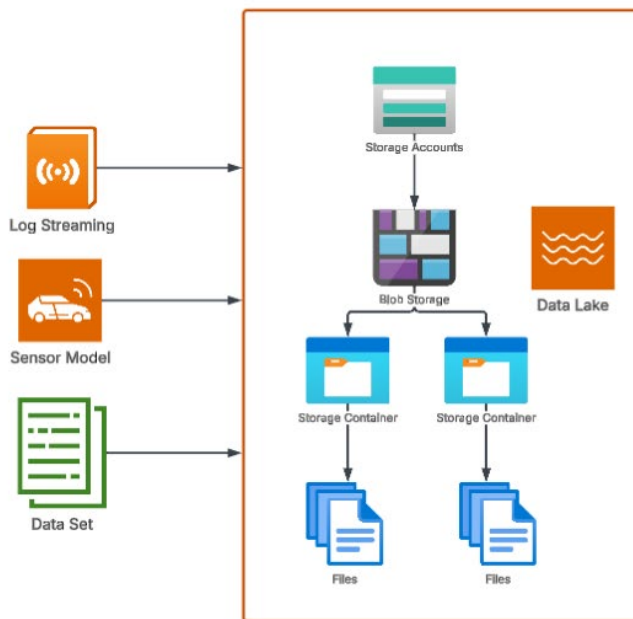
Data storage prices pay-as-you-go

All prices are per GB per month.

Data storage prices pay-as-you-go	Premium	Hot	Cool	Archive
First 50 terabyte (TB)/month	\$0.15 per GB	\$0.018 per GB	\$0.01 per GB	\$0.002 per GB
Next 450 TB/month	\$0.15 per GB	\$0.0173 per GB	\$0.01 per GB	\$0.002 per GB
Over 500 TB/month	\$0.15 per GB	\$0.0166 per GB	\$0.01 per GB	\$0.002 per GB

Reference - <https://azure.microsoft.com/en-us/pricing/details/storage/blobs/>

A company would want to monitor their storage costs.



Let's say these batch of files have been processed.

We still want to retain the files. But at the moment, they are not required for viewing or processing.

We want to see if there is any way to save on storage cost.

We have to think about saving costs because we might be receiving a large batch of files every hour.

We can change the access tier for the objects.

Hot

This is the default tier for objects. Here this is optimized for objects that are accessed frequently.

Cool

This is ideal for objects that are infrequently accessed. An object can be set to the Cool Access tier. Here the object needs to be stored for a minimum of 30 days.

Here the storage costs are lower when compared with the Hot access tier, but the access costs are higher.

Cold

This is ideal for objects that are rarely accessed or modified, but you still need access to them. An object can be set to the Cool Access tier. Here the object needs to be stored for a minimum of 90 days.

Here the storage costs are lower when compared with the Cool access tier, but the access costs are higher.

Archive

This is ideal for objects that are rarely accessed. And if you need to access them, you don't mind waiting for the data to be restored first.

Here the data needs to be stored for a minimum of 180 days.

Azure Storage Accounts - Data Redundancy

How does Azure maintain high availability of your data stored in Azure Storage Accounts.

Service Credit – hot blobs in LRS, ZRS, GRS and RA-GRS (write requests) Accounts and blobs in LRS Block Blob Storage Accounts:

Uptime Percentage	Service Credit
< 99.9%	10%
< 99%	25%

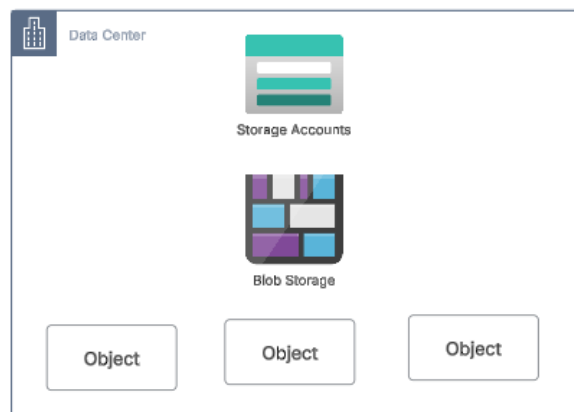
Service Credit – hot blobs in RA-GRS (read requests) Accounts:

Uptime Percentage	Service Credit
< 99.99%	10%
< 99%	25%

There are different data redundancy options in place.

Locally redundant storage

Here three copies of your data are made within a single data center.



East US

Zone redundant storage

What happens if the data center goes down. Then you don't have access to your objects.

Here the data is replicated synchronously across three Azure Availability zones in the primary region.



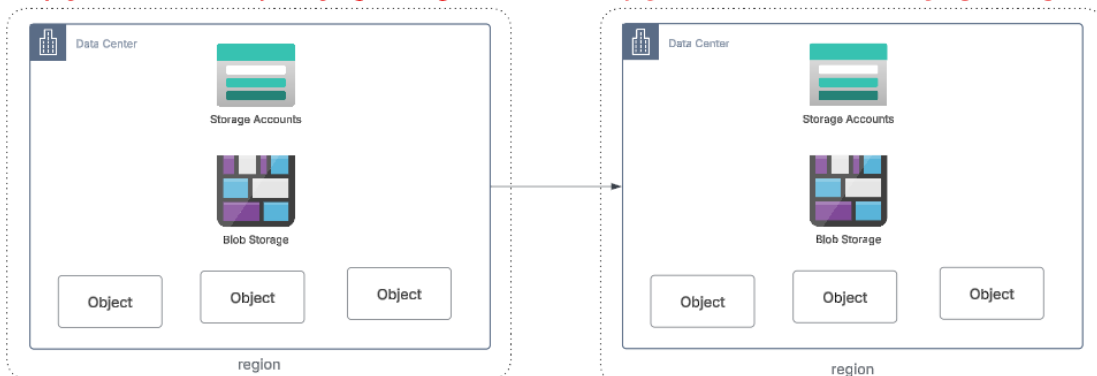
East US

But what happens if the entire region goes down. All of the Availability zones are not available.

Geo-redundant storage

Three copies of your data are made to a single physical location in the primary region using LRS

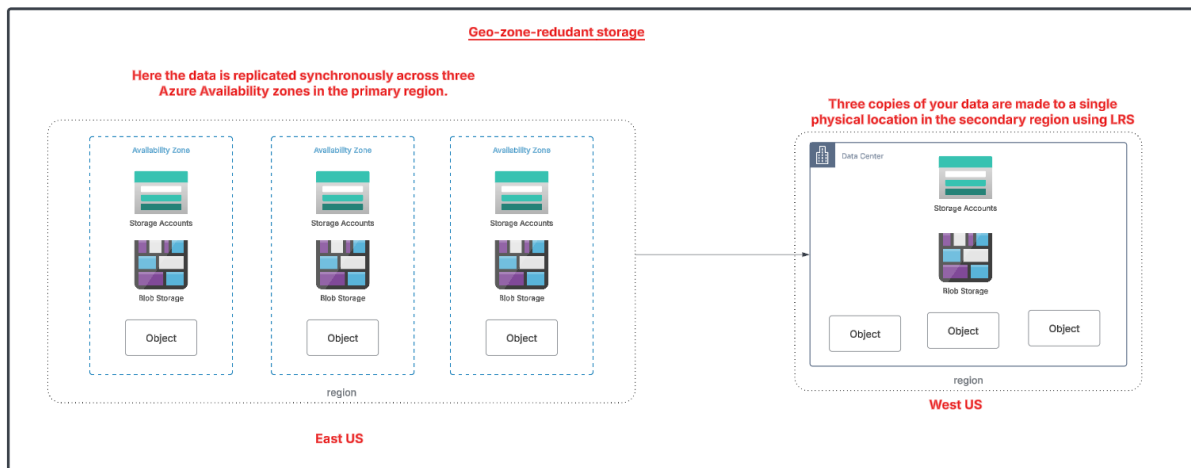
Three copies of your data are made to a single physical location in the secondary region using LRS



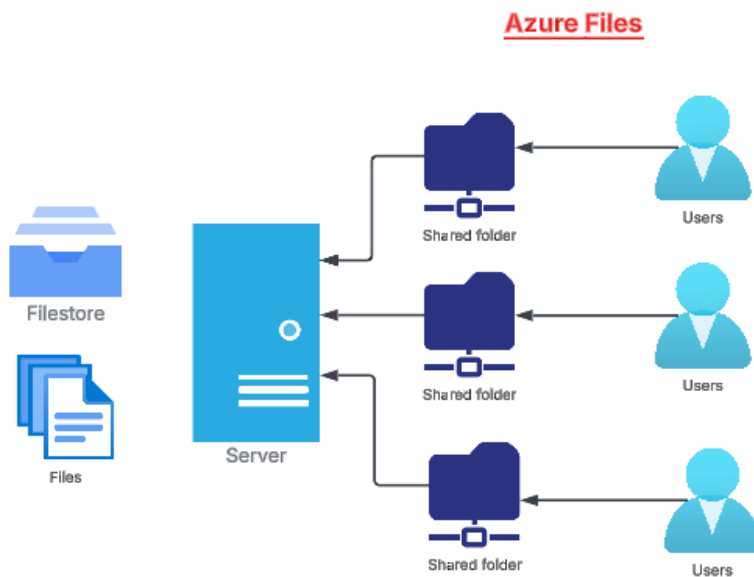
East US

West US

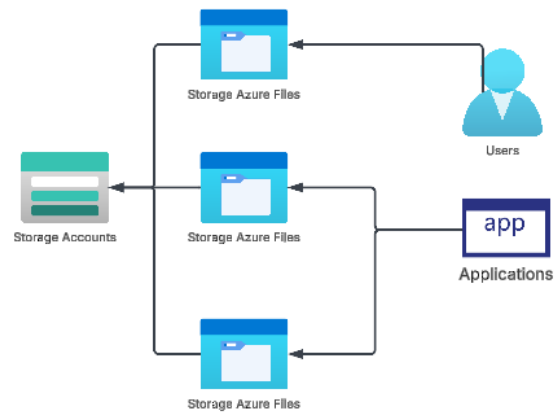
Your data is replicated to a secondary region.



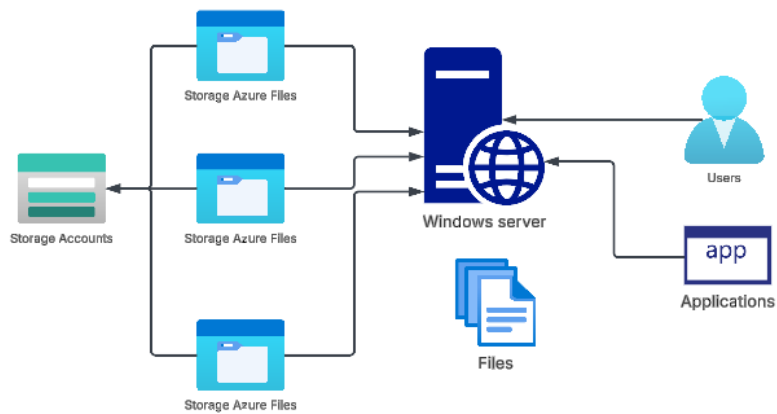
Azure File Sync service



One of the main purpose of Azure file shares is to alleviate the need of maintaining and managing the storage and systems required for the shares.



Yes, Azure file shares does provide a managed service. But sometimes users and application might need a quick retrieval of certain files. And even that fraction of latency in accessing the files might cause an issue.



With Azure File Sync, you can sync commonly accessed files onto a local Windows server. Users and applications can then fetch those files from the local Windows server.

Azure Premium Storage accounts



Storage Accounts

Here your data is stored on high performance hardware via the use of solid-state drives. These are optimized for low latency.

If your applications need fast access to data.

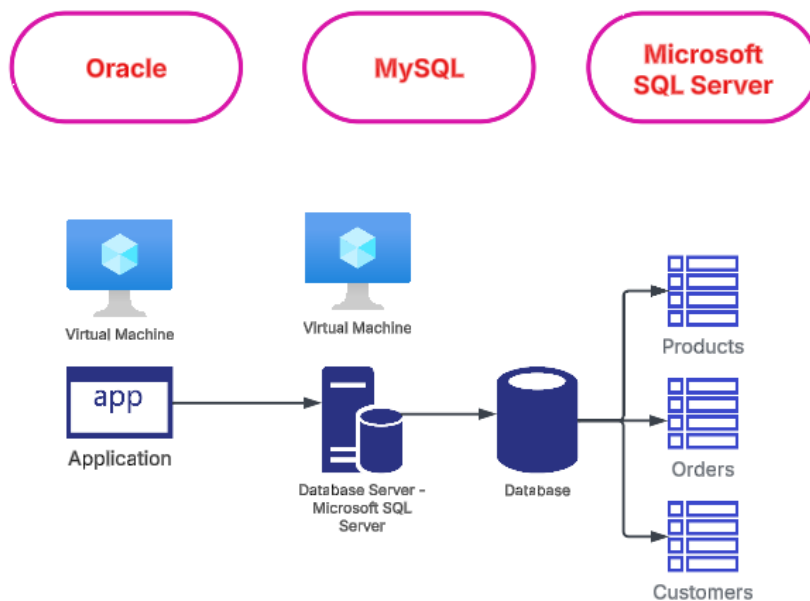
Here you have higher storage costs and lower transaction costs when compared with normal storage accounts.

These storage accounts are available for Block blobs, Page blobs and File shares.

Describe Azure architecture and services - Databases

Azure SQL Database service

You might be familiar with the concept of a relational database management system.



You could have an ecommerce application hosted on a virtual machine. The data used by the application could be hosted on a Microsoft SQL Server database.

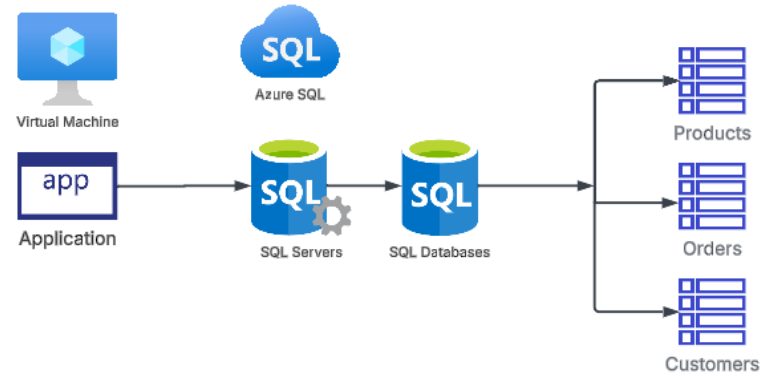
We can install the database engine on a virtual machine.

Database administrator responsibilities

1. Uptime of the database server
2. Database backups and restore
3. Patch installation at the operating system and database engine level.

Azure SQL Database

If the company does not want the burden of managing the underlying infrastructure, they can opt to use the Azure SQL Database service.



You can simply start hosting your databases. The Azure SQL database is the cloud version of Microsoft SQL Server.

Here the underlying server is managed for you. The database software will be in place. It also has features such as backup/restore and several other features.

About Azure Cosmos DB



ID	Name	Description
C01	AZ-104 Azure Adminsitrator	Azure Administration
C02	AZ-204 Azure Developer	Azure Development

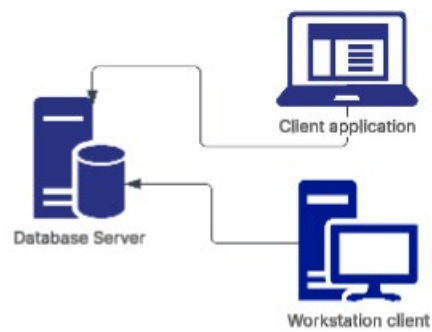
Course

ID	Name	CourseID
S01	Mark	C01
S02	James	C02

Student

What is NoSQL

Familiar with SQL-based database engines - Oracle , MySQL, Microsoft SQL Server



Applications, users connect to the database hosted on the database server.

There were issues when it came to traditional SQL database engines.

Ability to store large amounts of data - These were meant to be transactional systems and not able to manage large amounts of data.

Tables in the databases needed to have a predefined schema. But in today's world , data comes in all sizes, shapes and form.

Needed to have a more flexible way of storing data.

There were many NoSQL-based data stores developed for this purpose - MongoDB , Cassandra.

The NoSQL data stores were designed to be flexible in nature when it came to data storage.

Initially the main focus was not on building relationships across entities, but across efficient storage and retrieval of data.



Azure Cosmos DB is a fully managed NoSQL, relational and vector database.

You get fast access to your data.

Different API's

NoSQL

Data is stored in document format.

You can query for items using Structured Query Language (SQL)

MongoDB

Here documents are stored in BSON

PostgreSQL

Managed open source relational database with better performance.

Apache
Cassandra

Here data is stored in a column-oriented schema.

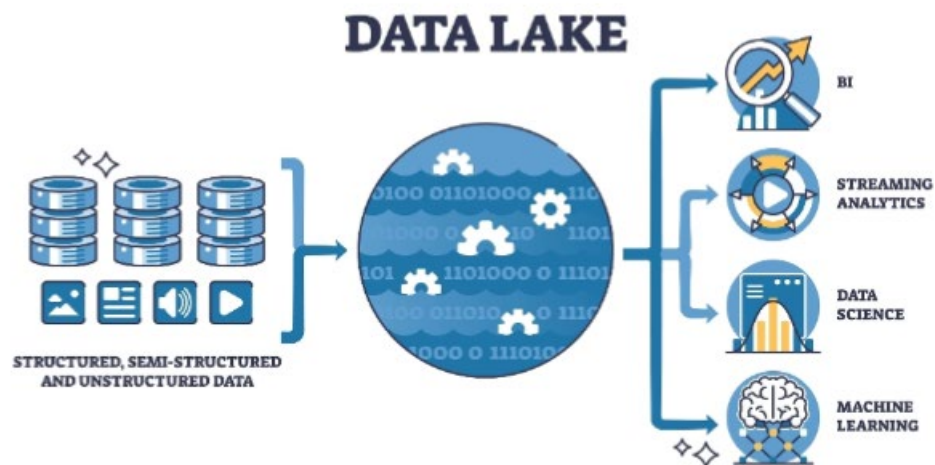
Gremlin

This allows you to store graph-based databases.

Table

Store data in the form of key/value pairs.

Enterprise Data Engineering



The data lake is a vast expanse of data. Here data is ingested at high velocities on the data lake.



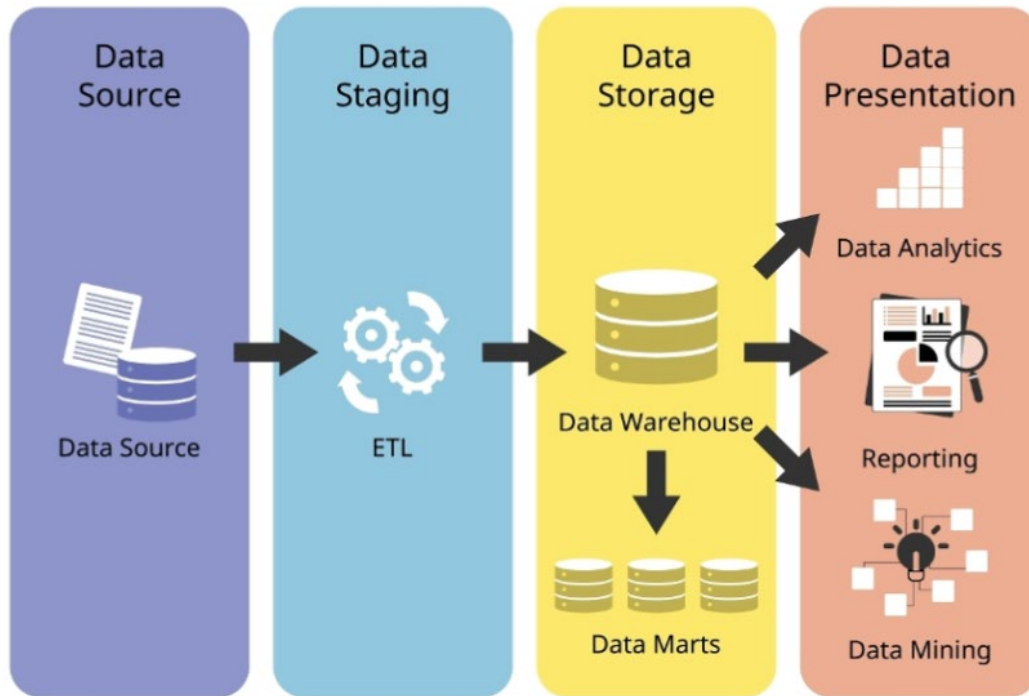
Data engineers can use tools to ingest data onto the data lake.

Initially in the data lake, the data would be in raw format, probably no structure.

The data needs to be cleansed, filtered, transformed. The data needs to be enriched to come to a form where it can add business value.

You can use tools and services to take in data, transform the data and then load it onto a destination data store.

Data Warehouse Architecture



You can then start building data warehouses. Here the data is in the form of tables.

There is a SQL engine that drives the querying of data from the data warehouse.

But here there is more of a focus on gaining insights from the data. It's more of querying for data rather than treating it as a transactional data store.

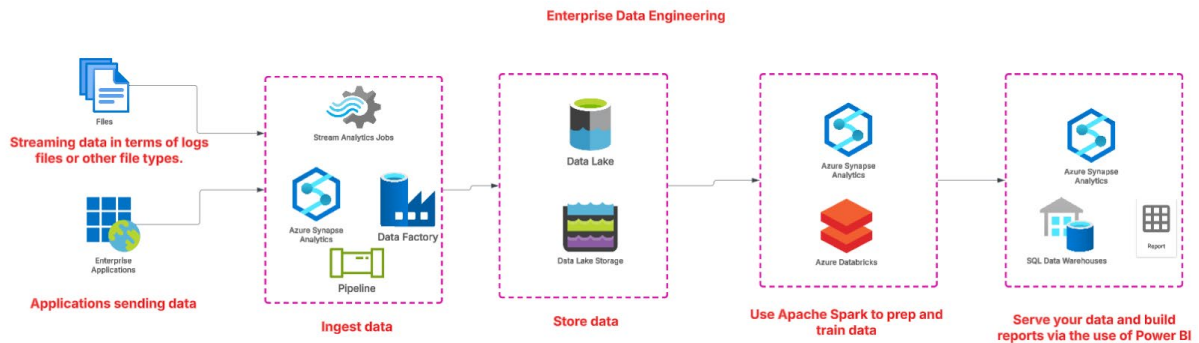
DATA LAKEHOUSE



Problems with a data lake - Governance of data because you just a lot of incoming streams of data from different data sources.

Problems with a data warehouse - It's not cheap. Its expensive to host a data warehouse. You need to make sure the warehouse is updated with data for analysis.

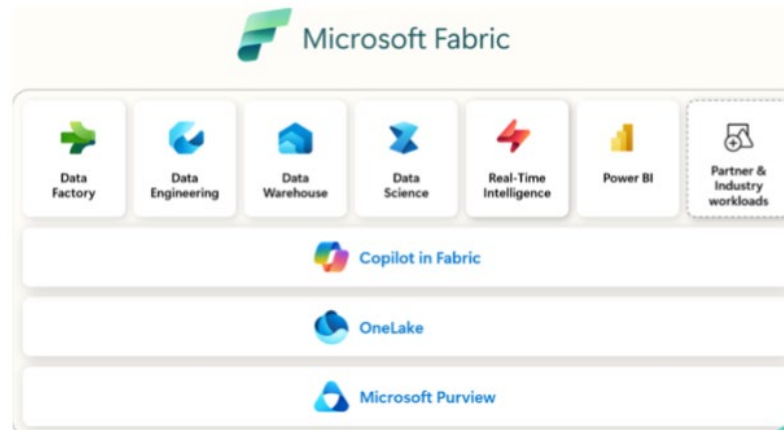
A data lakehouse is meant to the best of both worlds when it comes to a data lake and a data warehouse.



Microsoft Fabric

This is meant to be an end-to-end analytics platform. Here you can perform data movement , data processing, ingesting data, data transformation and report building.

It supports various capabilities that include Data Engineering, Data Science and Real-Time Intelligence.



Reference - <https://learn.microsoft.com/en-us/fabric/fundamentals/microsoft-fabric-overview>

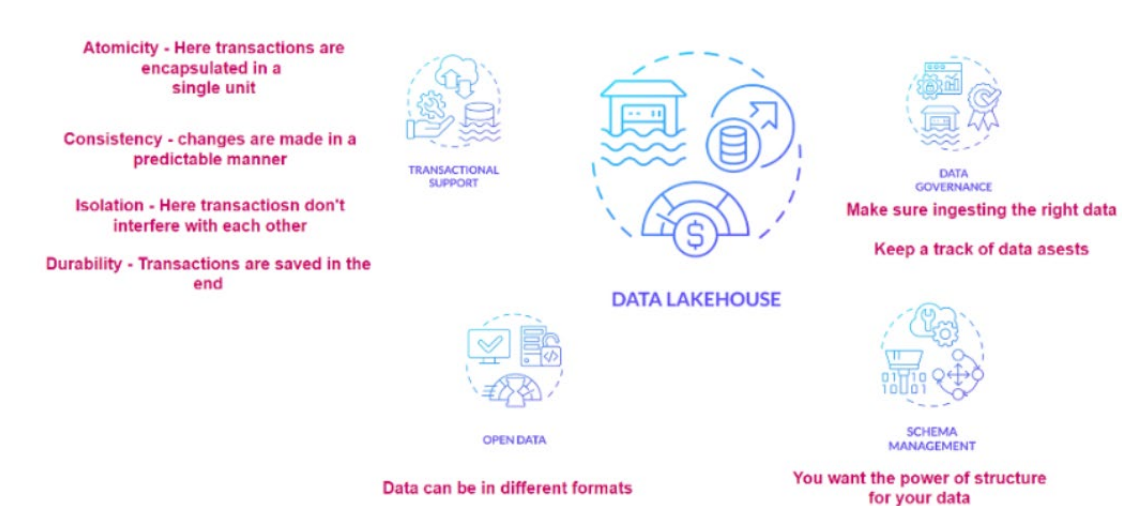
Enterprise Data Engineering - Using Azure Databricks

Databricks

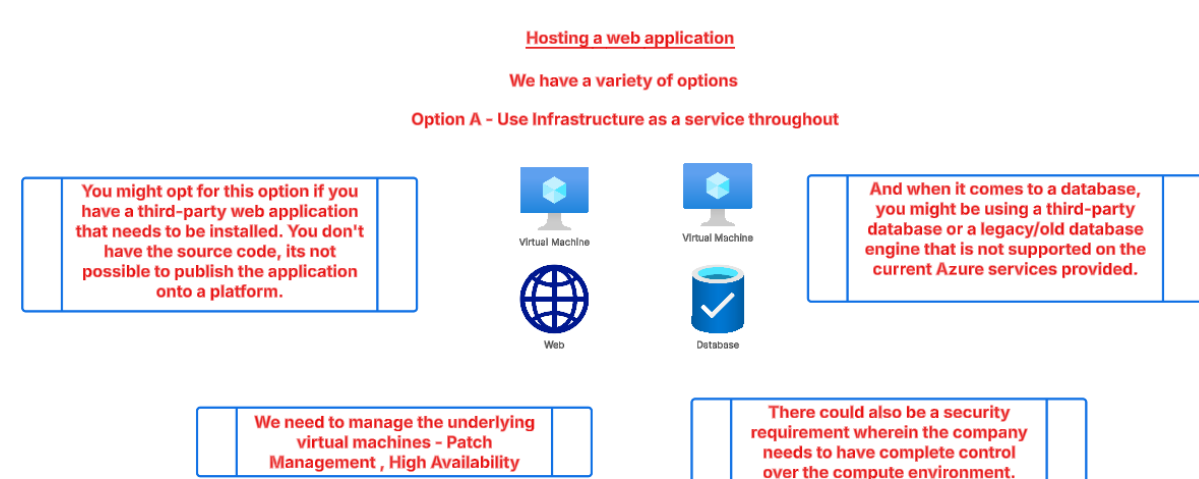
This is a company that provides a platform for developing and maintaining enterprise-grade data solutions.

The creators of Spark went ahead and created the Databricks platform





A sample architecture - use case 1



Option B - Use both Infrastructure and Platform as a service.

We use Azure Web Apps as a platform as a service. Here we don't need to worry about the underlying infrastructure.



Azure Websites



Virtual Machine



Web



Database

Option C - Complete use of Platform as a service.



Azure Websites



Azure Database MySQL Server



Web



SQL Database

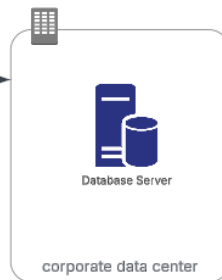
We can use Azure SQL Database or Azure Database for MySQL



Azure Websites



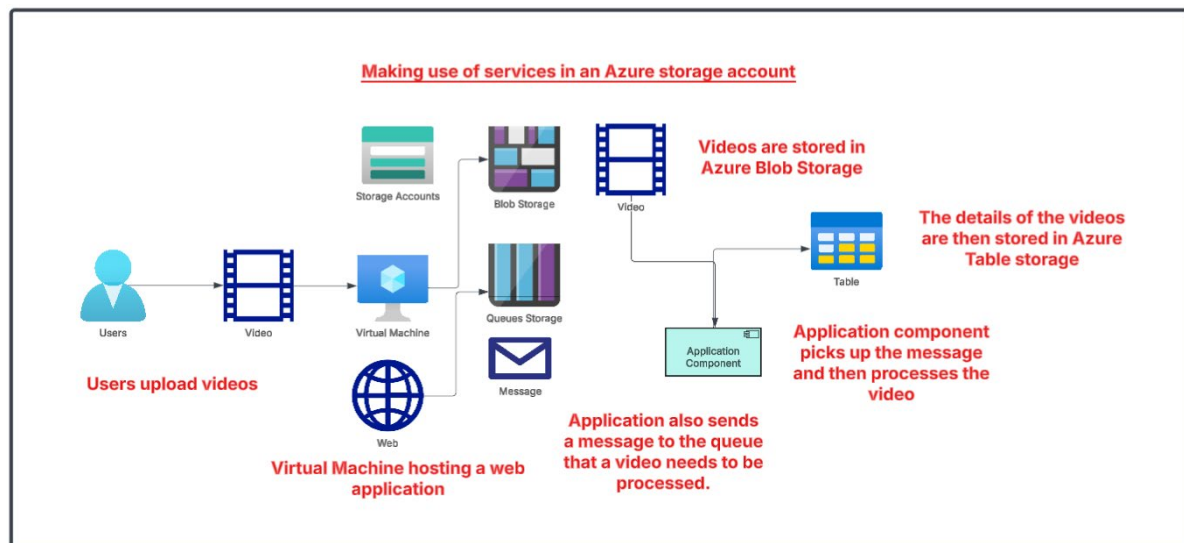
Web



corporate data center

You can also have a hybrid model where an Azure Web App might talk to a database in your on-premises environment.

A sample architecture - use case 2



Describe cloud concepts

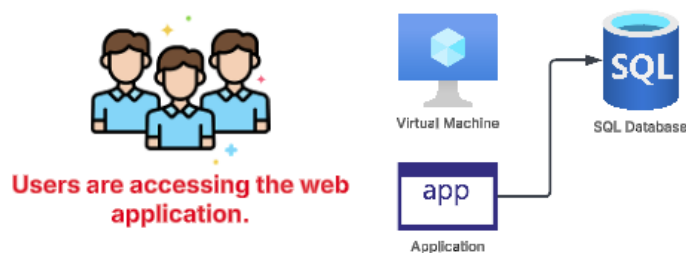
Benefits of the cloud - High Availability

High Availability

This is ensuring the application is made available with the minimum amount of downtime.

Consider the simplest use-case wherein you have a web application hosted on a virtual machine.

The application uses an Azure SQL database as the data store.



What happens if the application goes down

1. End users are not happy.
2. Operations team gets flooded with calls from users.
3. Business loses confidence on the company that is responsible for the application.

There are so many reasons as to why an application could go down.

1. An update was published to the application that caused an issue at the application level.
2. An update was published at the operating system level that caused the application to go down.
3. From the Azure side, the physical server hosting the virtual machine ran into technical issues and had to be restarted.
4. There was an issue in the overall Azure SQL database service in the region the database is hosted in.

At the infrastructure level, we could avoid failures by implementing Azure Availability sets or distributing the machines across multiple Availability zones.

At the application level, we can implement procedures to rollback to prior working application versions.

The Azure SQL database service also has a number of options available in the service itself for high availability.

Benefits of the cloud – Scalability

Scalability

This is the ability of the system to scale based on demand.

Showing 1069 VM sizes.	Subscription: Azure subscription 1	Region: East US	Current size: Standard_B1s	Image: Ubuntu Server 24.04 LTS	Learn more about VM sizes	Group by series
VM Size	Type	vCPUs	RAM (GiB)	Data disks	Max IOPS	Local storage (GiB)
▼ B-Series	Ideal for workloads that do not need continuous full CPU performance					
B1ls	General purpose	1	0.5	2	320	4 (SCSI)
B1ms	General purpose	1	2	2	640	4 (SCSI)
B1s	General purpose	1	1	2	320	4 (SCSI)
B2ms	General purpose	2	8	4	1920	16 (SCSI)
B2s	General purpose	2	4	4	1280	8 (SCSI)
B4ms	General purpose	4	16	8	2880	32 (SCSI)
B8ms	General purpose	8	32	16	4320	64 (SCSI)
> DC-Series	Designed to protect the confidentiality and integrity of code and data for general-purpose workloads					
> D-Series v6	The 6th generation D family sizes recommended for your general purpose needs					
> D-Series v4	The 4th generation D family sizes for your general purpose needs					
> D-Series v3	The 3rd generation D family sizes for your general purpose needs					
> D-Series v2	The 2nd generation D family sizes for your general purpose needs					

For a virtual machine, we know that we can scale the size of the machine at any point in time.

For an Azure Web App, we have the ability to scale the App Service Plan at any point in time

☒ Hardware view ☐ Feature view

Showing 26 App Service pricing plans

Name	ACU/vCPU	vCPU	Memory (GB)	Remote Storage (GB)	Scale (instance)	Cost per hour (instance)	Cost per month (instance)
Popular options							
Free F1	60 minutes/day co	N/A	1	1	N/A	Free	Free
Shared D1	240 minutes/day co	N/A	1	1	N/A	0.016 USD	11.899 USD
Basic B1	100	1	1.75	10	3	0.083 USD	60.59 USD
Premium v3 P0V3	195*	1	4	250	30	0.164 USD	119.72 USD
Premium v3 P1V3	195	2	8	250	30	0.328 USD	239.44 USD
Premium v3 P1mv3	195*	2	16	250	30	0.362 USD	263.968 USD
Isolated v2 I1V2	195	2	8	1000	N/A	0.56 USD	408.80 USD
Standard S1	100	1	1.75	50	10	0.11 USD	80.30 USD
Dev/Test (For less demanding workloads)							
Free F1	60 minutes/day co	N/A	1	1	N/A	Free	Free
Shared D1	240 minutes/day co	N/A	1	1	N/A	0.016 USD	11.899 USD
Basic B1	100	1	1.75	10	3	0.083 USD	60.59 USD
Basic B2	100	2	3.5	10	3	0.165 USD	120.45 USD
Basic B3	100	4	7	10	3	0.33 USD	240.90 USD



We can make use of Azure Virtual Machine scale sets to provision an identical set of machines. We can set scaling conditions to scale out and scale in based on metrics.

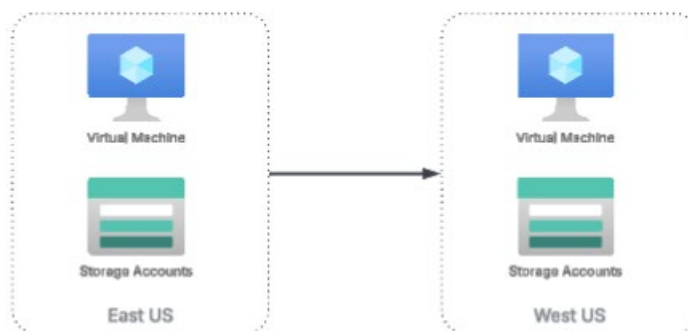
Benefits of the cloud - Disaster recovery

Disaster Recovery

How can you recover from a region failure.



Let's say that the entire East US region goes down.



We need to be able to replicate changes as they occur in our primary region. We should be able to failover to a newer region at any point in time.

For virtual machines, there is a tool known as Azure Site Recovery. And for Storage accounts, we can make use of Data redundancy.

But there is an increase in cost. Depends on how critical the application is.

Benefits of the cloud – Elasticity

Elasticity

With a cloud platform , you are able to scale based on demand.

You just get more flexibility. And with a Pay-as-you-go pricing, it makes it easier to grown and shrink on demand.

If an application needs an additional virtual machine, just spin it up. No need to buy infrastructure, set it up and then create a virtual machine.

This increases the overall agility of the application. Instead of worrying on the underlying infrastructure, you can focus more on the business and the application features.



**HIGH
ELASTICITY**

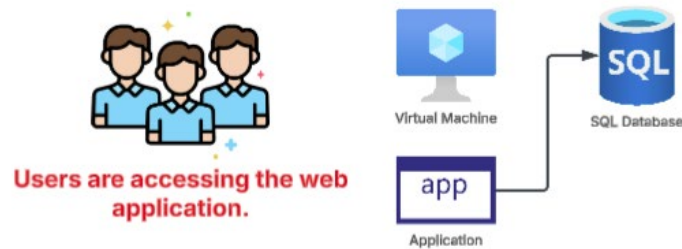
Benefits of the cloud - Fault tolerance

Fault tolerance

Can your infrastructure tolerate faults at different layers.

Consider the simplest use-case wherein you have a web application hosted on a virtual machine.

The application uses an Azure SQL database as the data store.



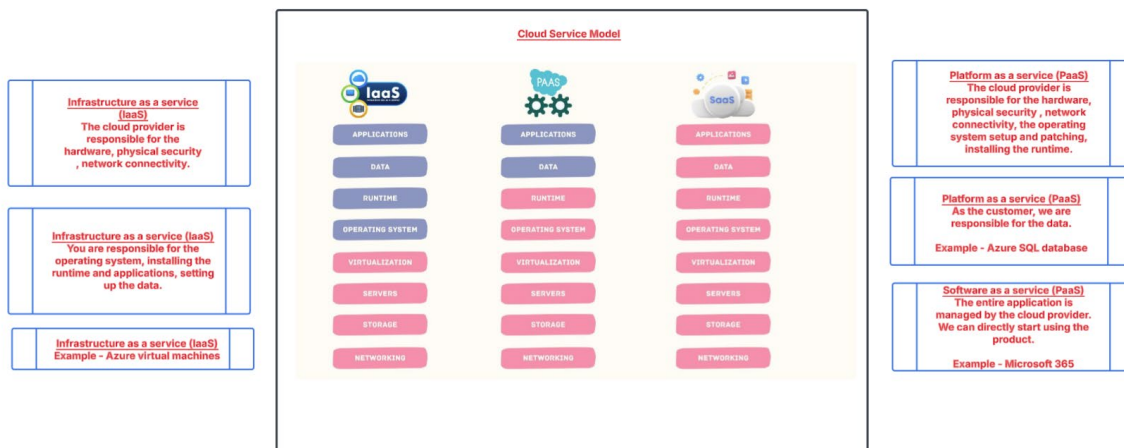
When it comes to high availability , we focus on the Infrastructure level components.

Fault tolerance applies to all aspects of the application architecture.

What happens if there is a network failure and the application components cannot talk to each other.

An update was published to the application that caused an issue at the application level.

Cloud service model



Cloud model types

Cloud Model types

Public Cloud



Azure Home



Virtual Machine



Storage Accounts



SQL Database



Virtual Machine



Azure Websites

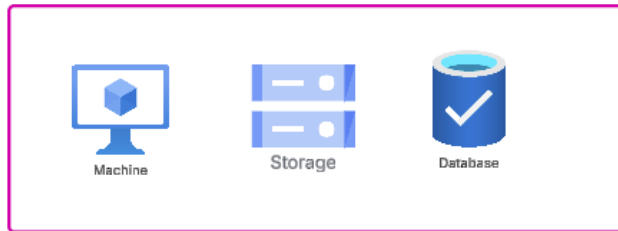
This is a platform available over the Internet.

Here many customers can use a portion of the cloud for their hosting requirements.

Key benefits

- 1. Reduction in Capital Investment**
- 2. Pay-as-you-go pricing**
- 3. Ability to scale on demand**
- 4. Make use of the latest technologies**

Private Cloud



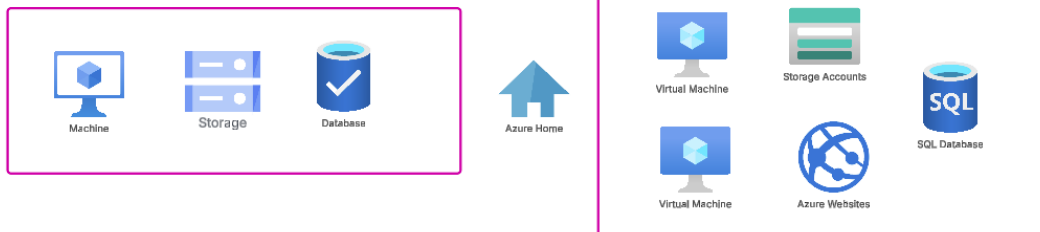
Here a company could have their own data center. And build their own sort of Azure-like platform.

They make several services available. And this is available for the company.

Benefits - A company can implement their own security protocols, they own and manage the entire infrastructure, the services are only available for the company users.

Downsides - They need to scale their infrastructure, its difficult to cover a broad aspect of services offered by cloud providers.

Hybrid Cloud



Use the best of both worlds.

Companies can host applications on their platform depending on requirements.

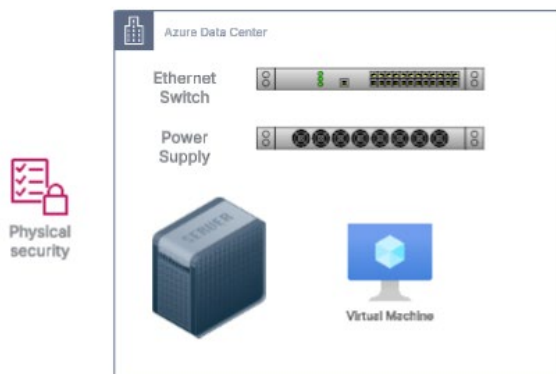
Shared Responsibility Model

Shared Responsibility Model

	Responsibility	SaaS	PaaS	IaaS	On-prem
Responsibility always retained by the customer	Information and data	Customer	Customer	Customer	Customer
	Devices (Mobile and PCs)	Customer	Customer	Customer	Customer
	Accounts and identities	Customer	Customer	Customer	Customer
Responsibility varies by type	Identity and directory infrastructure	Shared	Shared	Customer	Customer
	Applications	Shared	Shared	Customer	Customer
	Network controls	Shared	Shared	Customer	Customer
	Operating system	Shared	Shared	Customer	Customer
Responsibility transfers to cloud provider	Physical hosts	Microsoft	Microsoft	Microsoft	Customer
	Physical network	Microsoft	Microsoft	Microsoft	Customer
	Physical datacenter	Microsoft	Microsoft	Microsoft	Customer

■ Microsoft
 ■ Customer
 ■ Shared

<https://learn.microsoft.com/en-us/azure/security/fundamentals/shared-responsibility>



Infrastructure-as-a-service

Virtual Machine

Azure is responsible for the physical aspects of the data center. The physical security for the data center building.

As a customer we are responsible for the Virtual Machine.

- 1. We are responsible for operating system updates.**
- 2. Protecting the virtual machine via several controls.**
- 3. Responsible for the applications and the data on the virtual machine.**

Describe Azure architecture and services - Other services

Azure Traffic Manager

Azure Traffic Manager

This is a DNS-based traffic load balancer.

Here you have the ability to distribute traffic across public endpoints and these endpoints could be spread across different Azure regions.

Azure Traffic Manager directs client requests to the respective endpoints via a traffic-routing method.

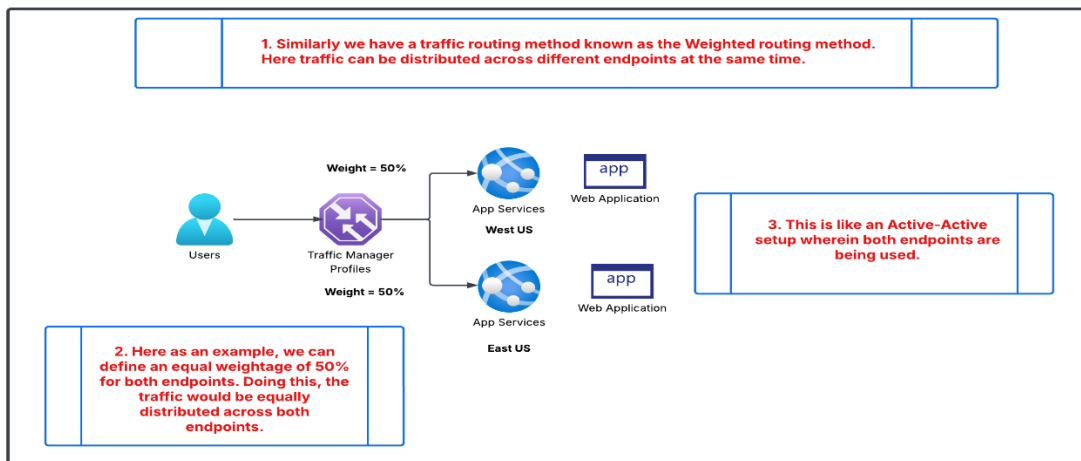
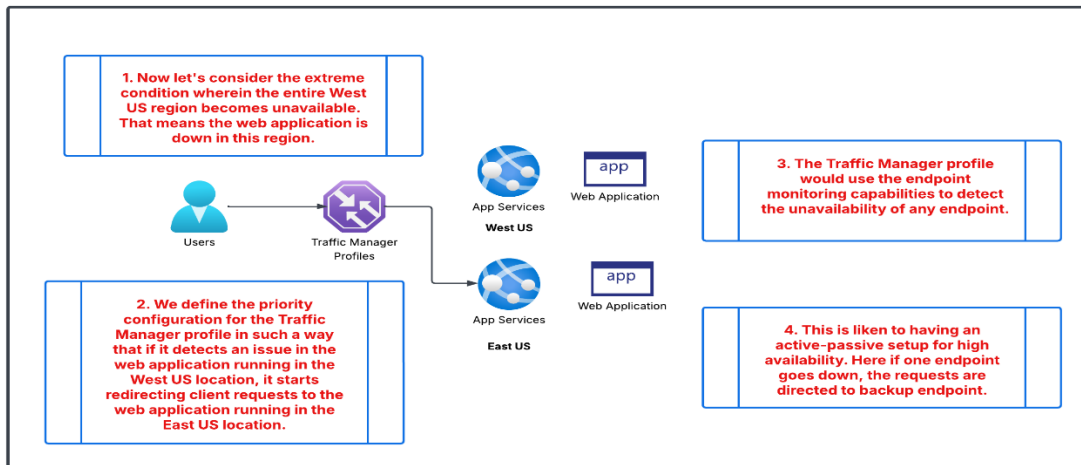
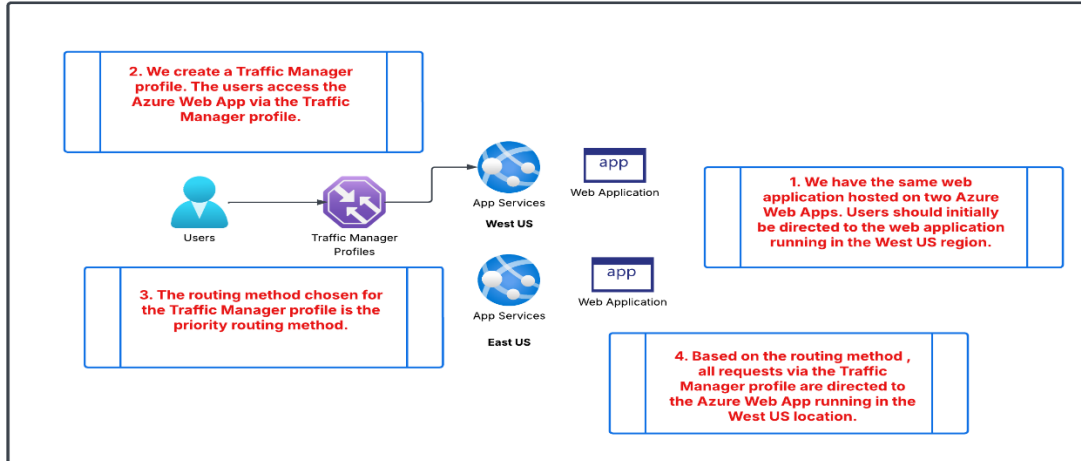
There is support for different endpoint types

Azure endpoints - These represent the services hosted in Azure.

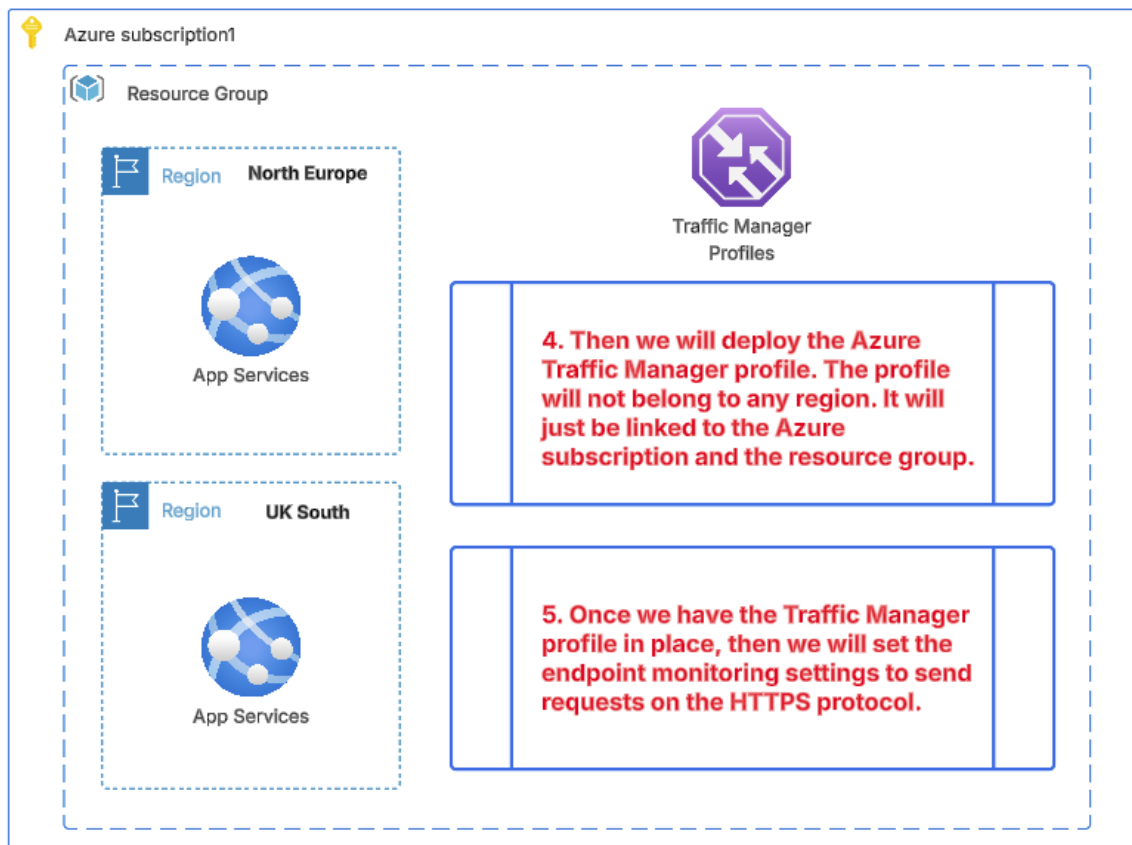
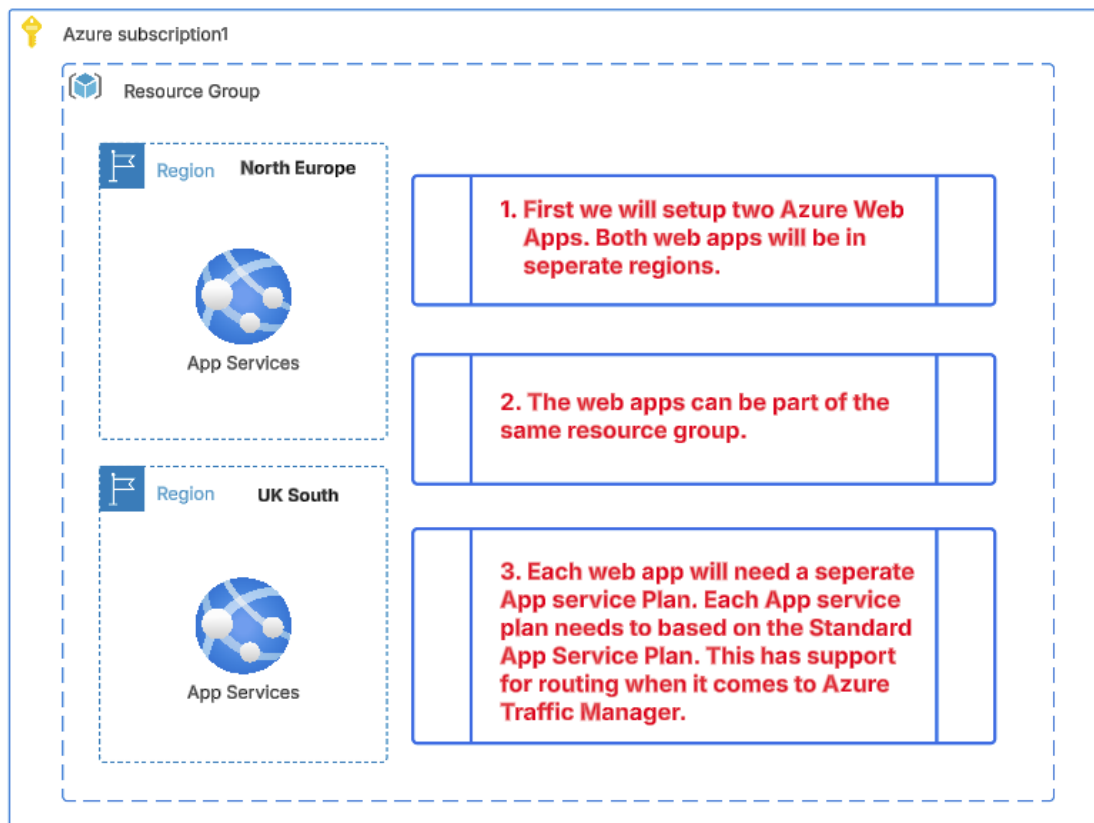
External endpoints - These represent services that are hosted outside of Azure. An example is a public-facing web application hosted on your on-premises data center.

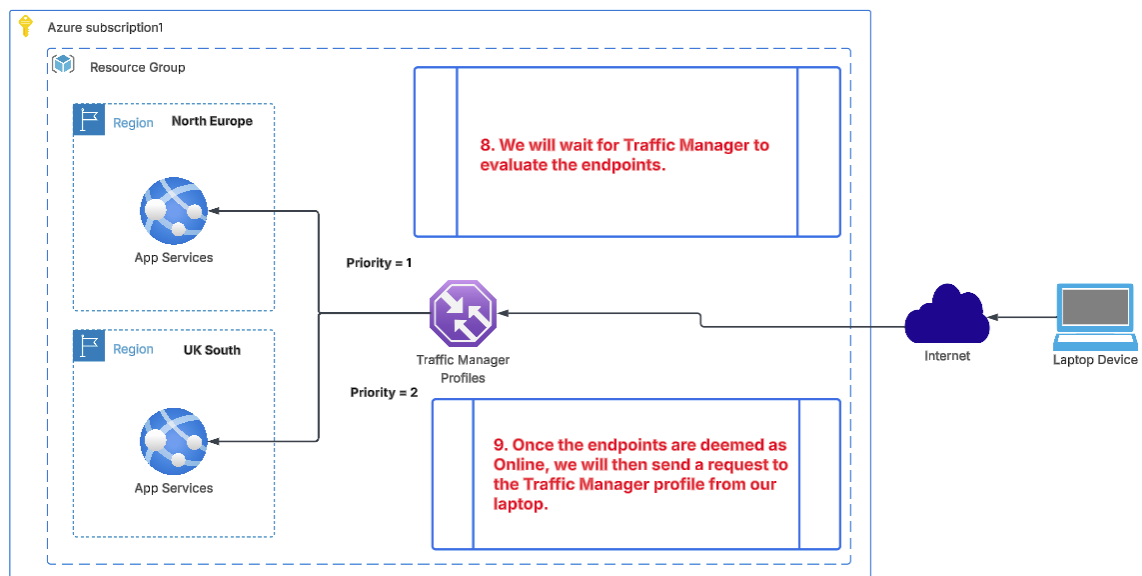
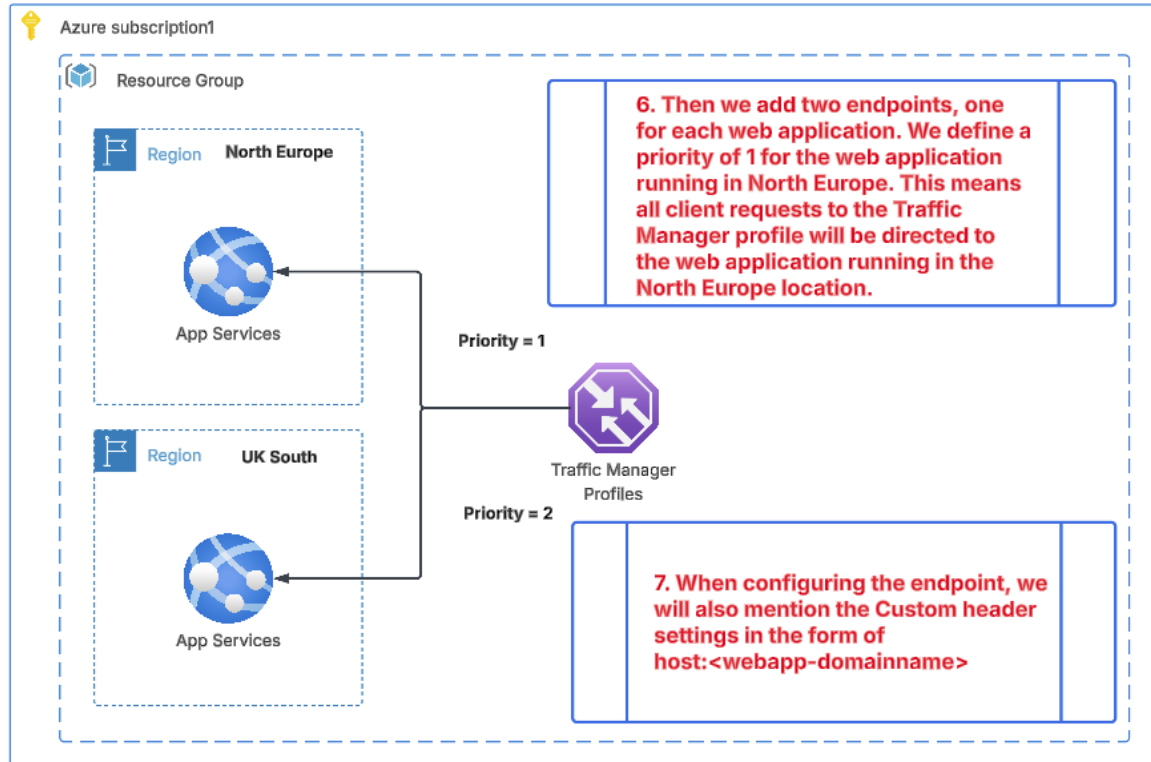
Nested endpoints - These represent the combination of Traffic Manager profiles.

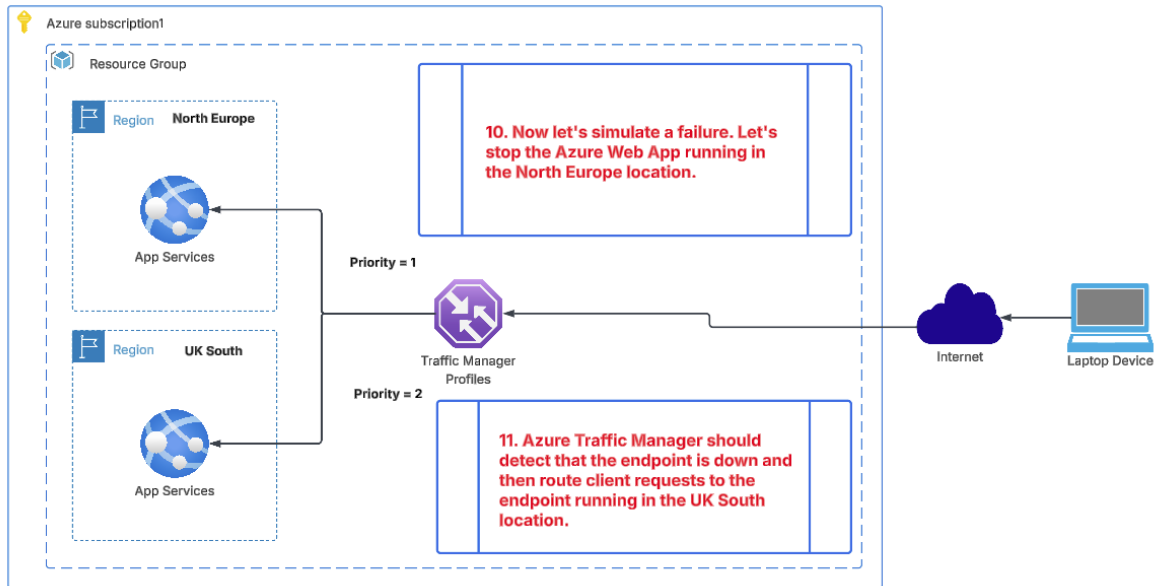
Within Traffic Manager you can also configure endpoint monitoring. Here Traffic Manager can constantly monitor the endpoint to make sure the endpoint is healthy. The Traffic Manager profile would not route the request to an unresponsive endpoint.



Lab - Azure Traffic Manager – Setup



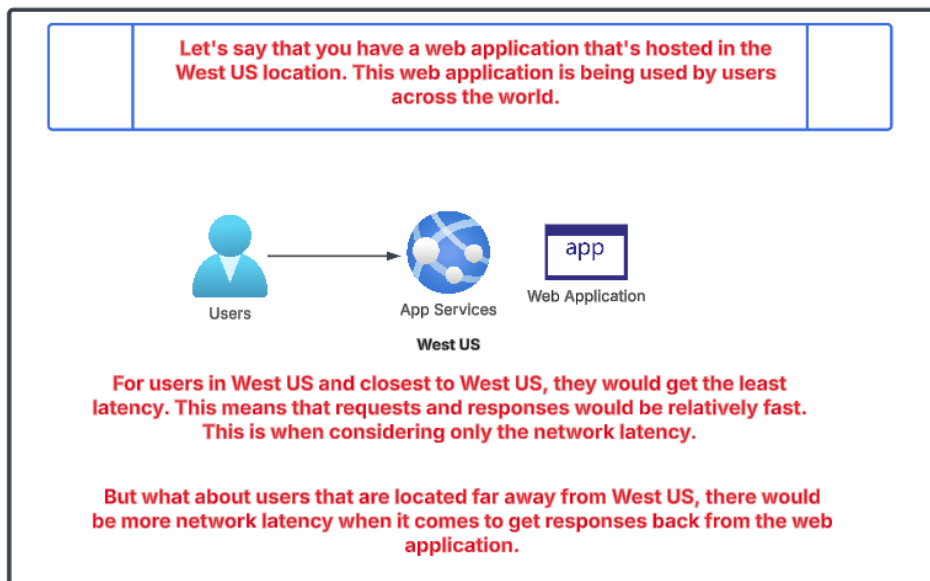




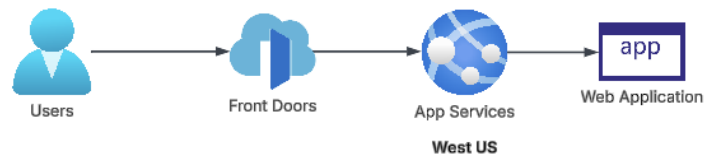
Azure Front Door

Azure Front Door

This is a Content Delivery Networking service that helps to deliver static and dynamic web content to users across the world in a fast, reliable and secure way.



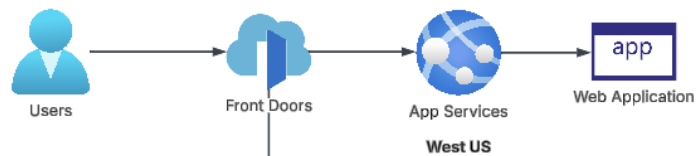
We can use Azure Front Door to provide faster access to content to users across the world.



Now when users need to access the application, they can access the application via Azure Front Door.

But what's the benefit, really what's the point of routing requests via another service.

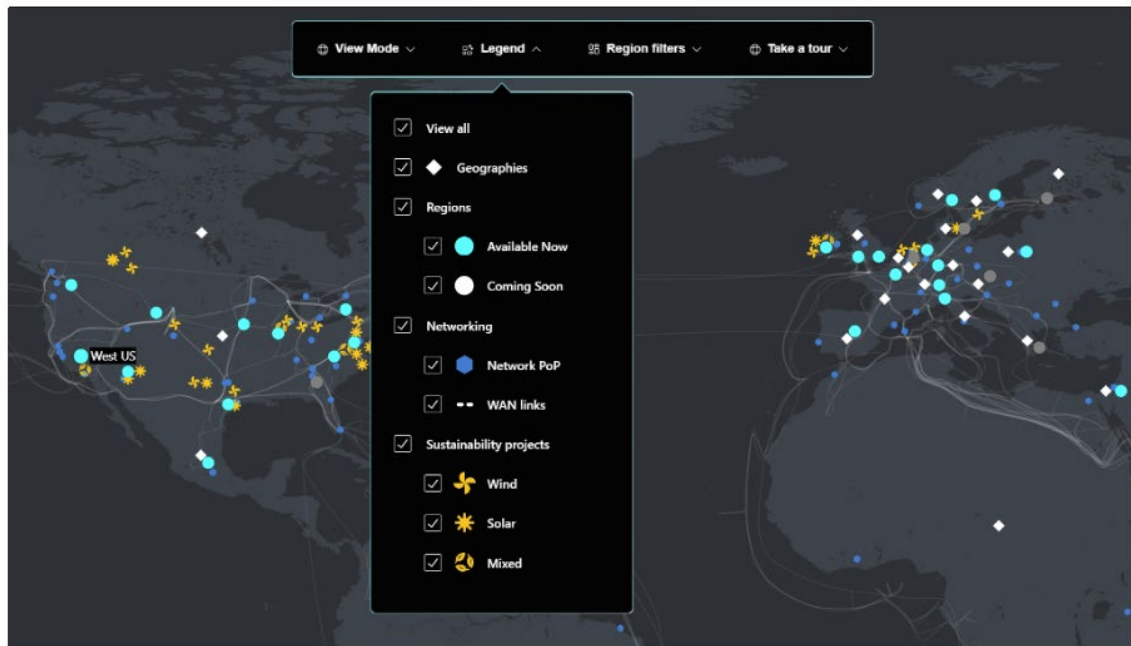
Well when Azure Front Door receives a request from a user located in another region. The response is taken from the web application and sent back to the user.



At the same time, the response is cached to a point of presence location located close to the user.

Point of Presence locations

So now when a similar user has the same request, the response is given from the PoP location instead of the request going all the way to the web application.



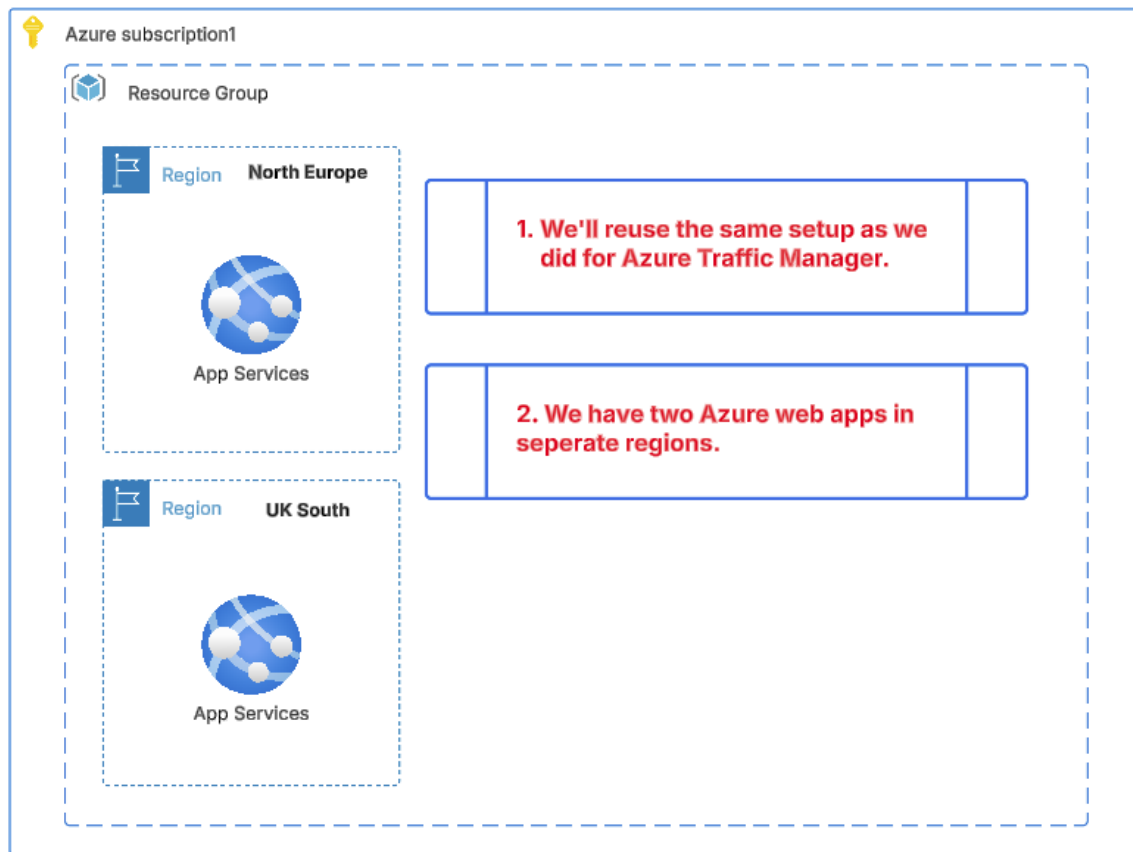
This is just one feature of Azure Front Door.

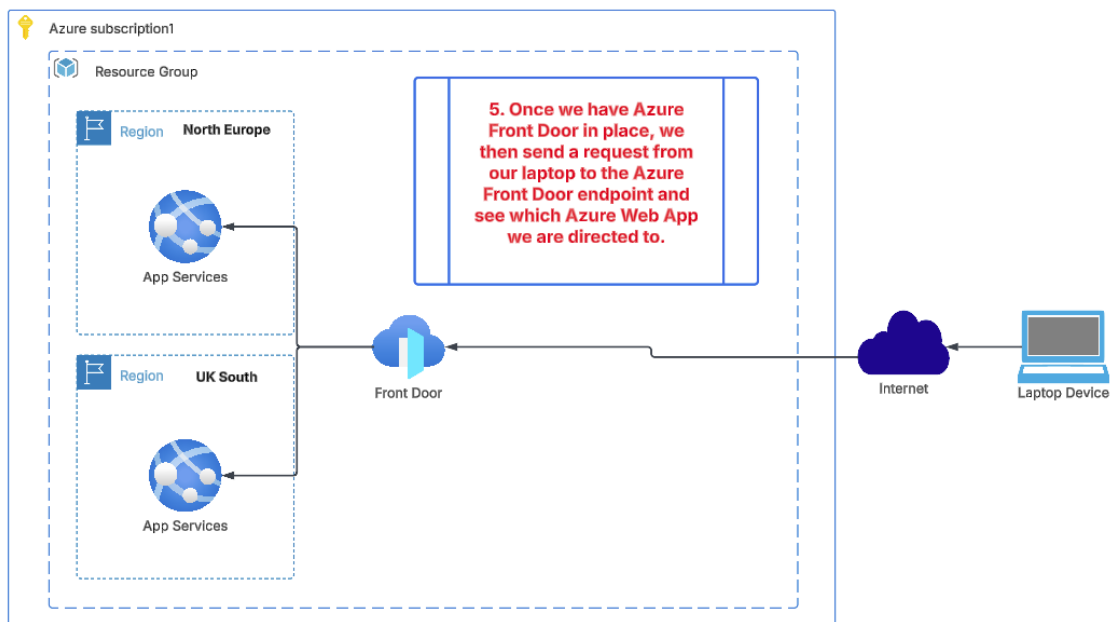
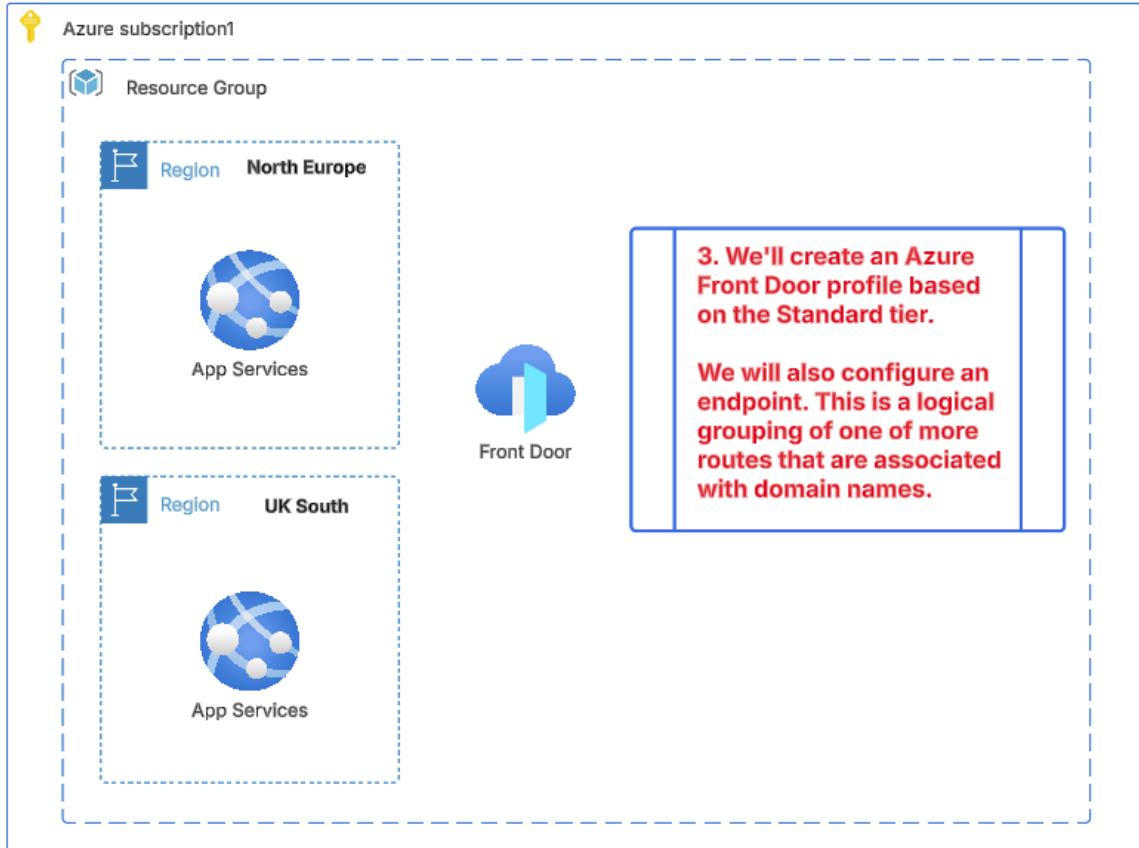
Just like Azure Traffic Manager, we can also define routing methods in Azure Front Door.

Azure Front Door can also route requests based on the performance of endpoints.

You can use another service with Azure Front Door known as the Web Application Firewall. This helps to protect web applications from known attacks.

Lab - Azure Front Door





About DevOps

What is DevOps?

This is a set of practices, principles, tools that are meant to unify the work of software development and IT operations.

Here we promote collaboration between developers, members of the QA and operations team.

We automate repetitive tasks such as code merging, testing, deployment.

We incorporate regular monitoring of applications and infrastructure.

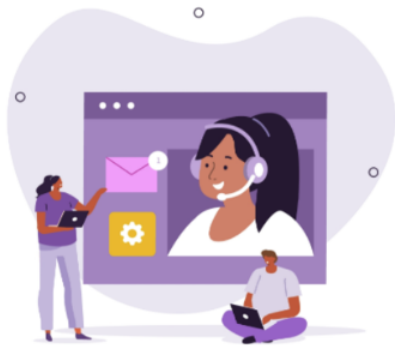


Development team is developing an application



Testing team tests the application

Application is deployed to a production environment



And then the operations team takes over

Server operations - Take cares of the servers

Database operators - Takes care of the databases

New feature

app



Sometimes developers try to apply quick fixes if they are under a time constraint.

Maybe the testing times for the applications are cut short



Operations teams are hesitant to deploy the changes.

They are in the direct line of fire. If the changes goes wrong, then it could lead to a downtime for the application.

There is always this constant tussle between development and operations.

The entire purpose of DevOps is to ensure that developers and IT operations work in collaboration.

Meetings that are held with stakeholders normally have the Developers and Operations teams.

Automation is a key aspect.

Faster delivery of application features.

Adoption of the right tools.

Big shift in the mindset and processes.



Azure DevOps Implementation

Using Azure Boards

This is a web service that allows a team to plan and track various work items during the entire development process.

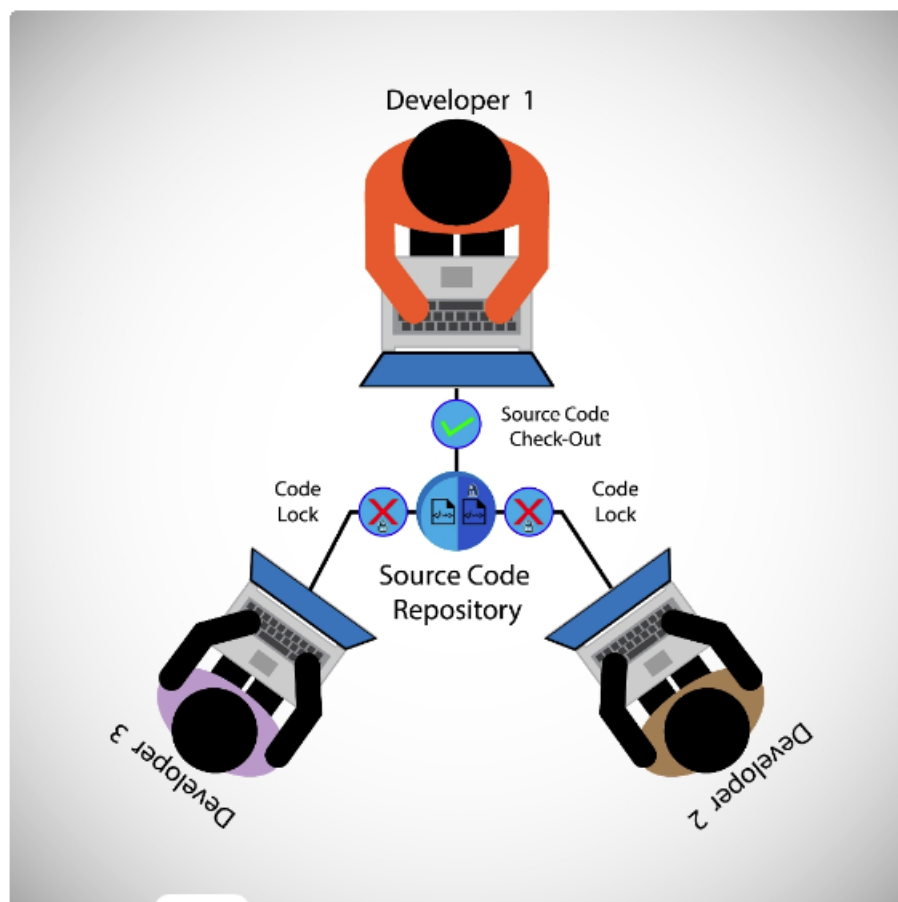
It has support for the agile methodology wherein we can create a Backlog of work items. Then we can organize the work items that need to be worked upon into Sprints

We can visualize the work in Azure Boards as cards. We can easily drag and drop the cards across various states.

Using Azure Repos

This is a version control tool that helps you to manage different code versions.

Azure Repos helps to you host and manage Git-based repositories. This is a popular version control tool.



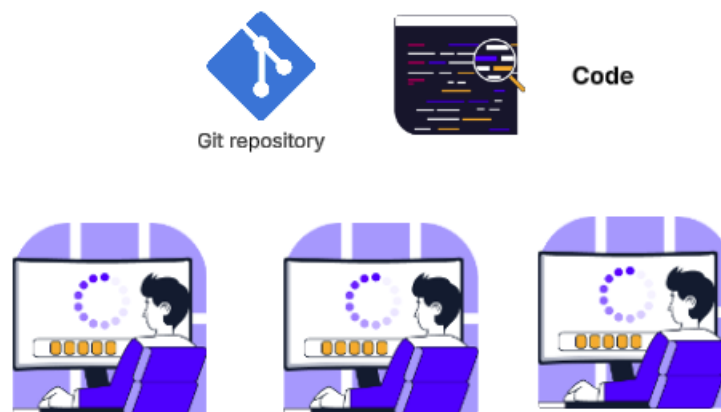
On my laptop, I will create a simple ASP.NET based application in Visual Studio Code.

On my laptop, I will install the Git client tool.

Using the Git client tool, we will create a local git repository on the laptop. Add the code files to the repository. We will then push the local Git repository to Azure Repos.

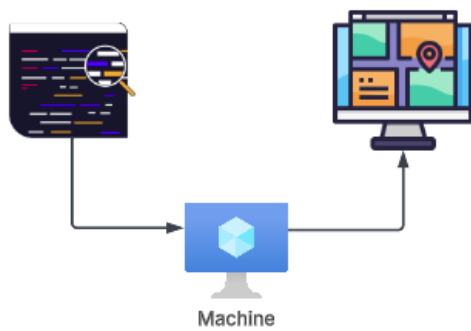
Using Azure Build Pipelines

Azure Pipelines is used for Continuous integration, continuous testing and delivery.



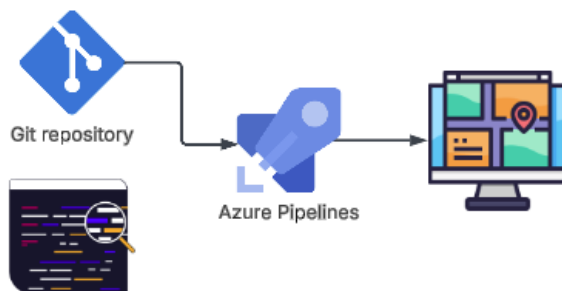
Let's consider the case wherein multiple developers are working on a code base. Based on the tasks assigned, based on the features need to be added to an application.

Now at the end of a couple of days, one set of features is ready from a development perspective. The features now need to be built and tested.



Now first the code needs to be built into an application.

To build an application, you need a machine that has the required tools and runtime required to build the code and generate the application.



Azure Pipelines has the ability to automatically detect for changes within an Azure Git repository, take the source code and build the application.

Azure Pipelines will create the compute machine with the required tools to build the application.

We will create a new Pipeline which will just take our code from Azure Repos and build the code.

Using Azure Release Pipelines

Now once our code is built, we can promote the application onto a dev environment.

The application can be tested.

It could subsequent get promoted on a production environment.

In Azure Pipelines, we can create a release pipeline that can take the build from the Build pipeline and release onto an Azure Web App.

Machine Learning

What is Machine Learning?

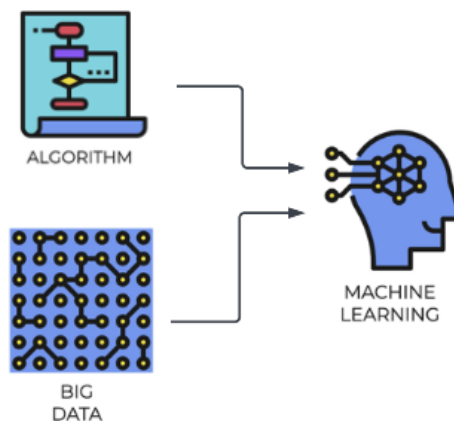


This is a subset of Artificial Intelligence that uses algorithms to analyze existing data and discover patterns in the data.

Based on the patterns you get a Machine Learning model that can be used to make predictions.

In today's world there are a ton of pre-built machine learning models. These models are trained in a variety of fields. We have pre-built models that can be used to recognize images, perform optical character recognition.

But a company might develop their own machine learning model for a particular business requirement.



To build a machine learning model, you need to use a machine learning algorithm and an input data set.

The data set can contain past and historical data that already contains information that the machine learning algorithm can use to detect patterns.

Then based on the patterns it can create a machine learning model.

The process of creating the machine learning model is known as model training.

You train a machine learning model, you test the model based on again available data.

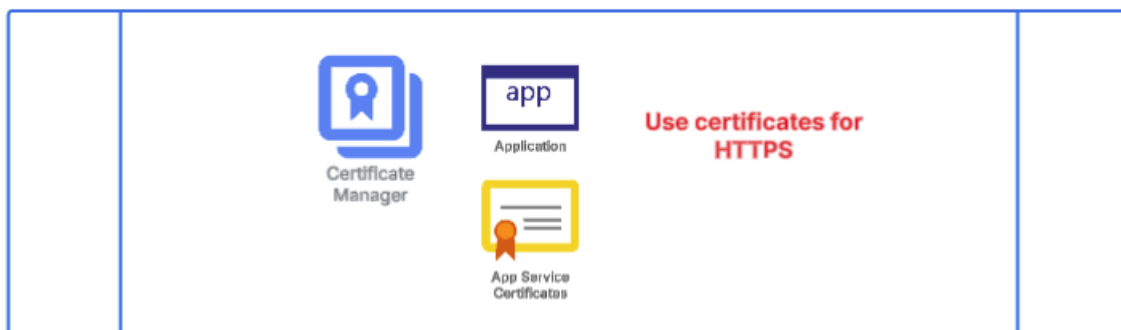
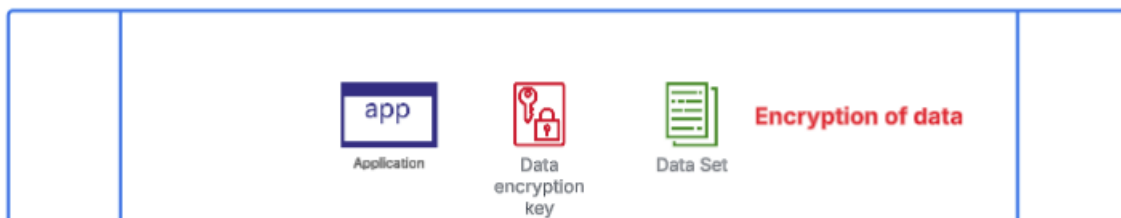
Once you have the desired machine learning model in place, you can deploy the model onto a production-based compute environment.

Lab - Azure Key Vault



Azure Key Vault

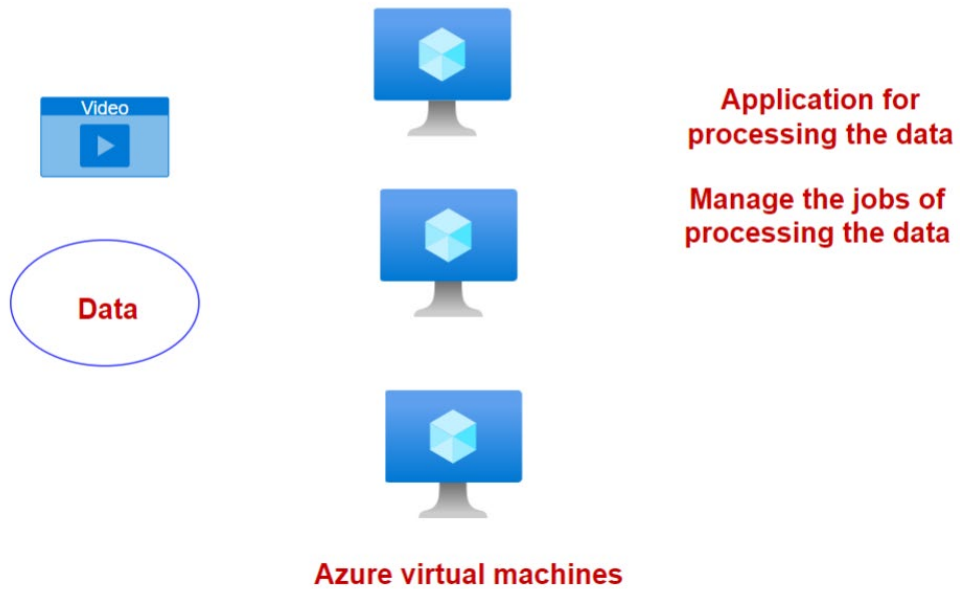
The Azure Key vault is a managed service that can be used to store secrets, encryption keys and certificates.



Azure Batch Service

Azure Batch

This is used to run large-scale parallel and high performance computing batch jobs





You can manage these batch processing jobs
with the help of the Azure Batch service



Azure Batch



Azure virtual machines

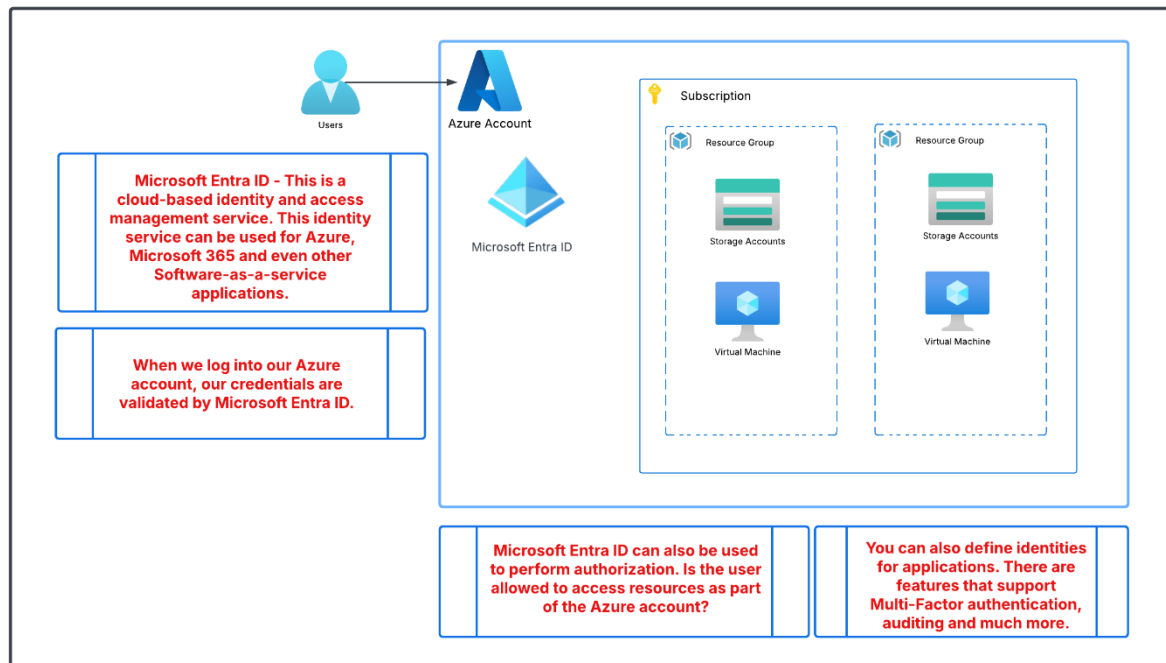
Manages the pool of virtual machines
It can ensure the application is installed on
the virtual machines

You can use Azure Storage accounts for the input files, storage of
the application and the output files.

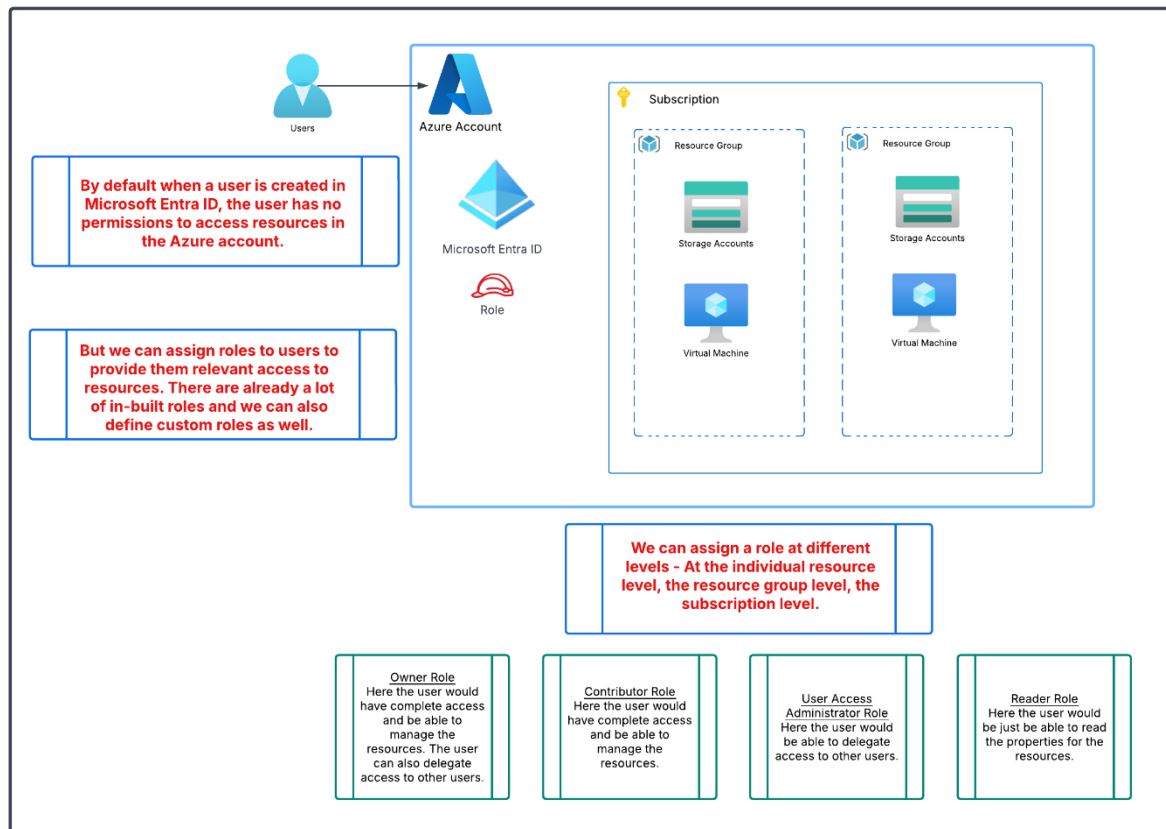
Azure Batch manages the tasks and the jobs that need to run on
the virtual machines.

Describe Azure architecture and services - Identity and Access

What is Microsoft Entra ID



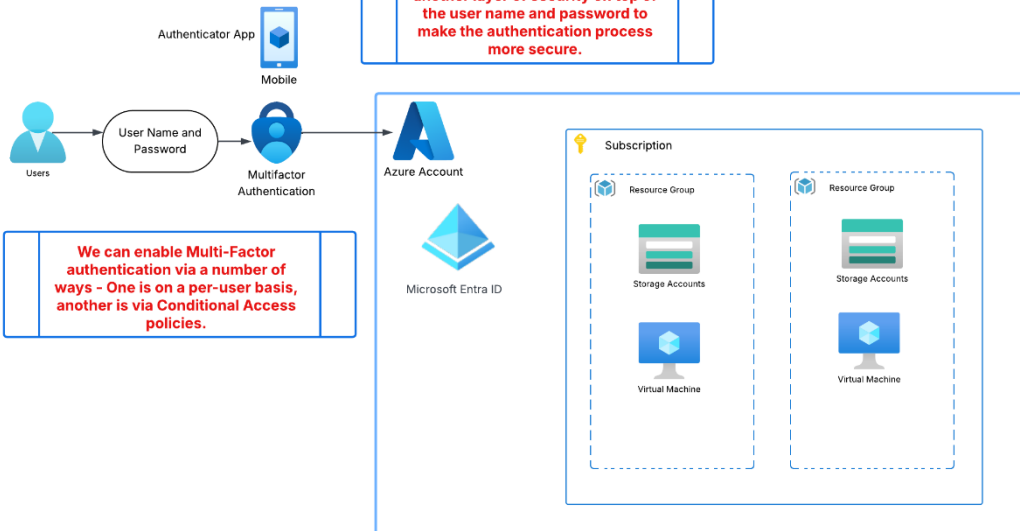
Role Based Access Control



Quick note on MFA

One additional way is to use the authenticator app on mobile devices.

Multi-Factor authentication adds another layer of security on top of the user name and password to make the authentication process more secure.



We can enable Multi-Factor authentication via a number of ways - One is on a per-user basis, another is via Conditional Access policies.

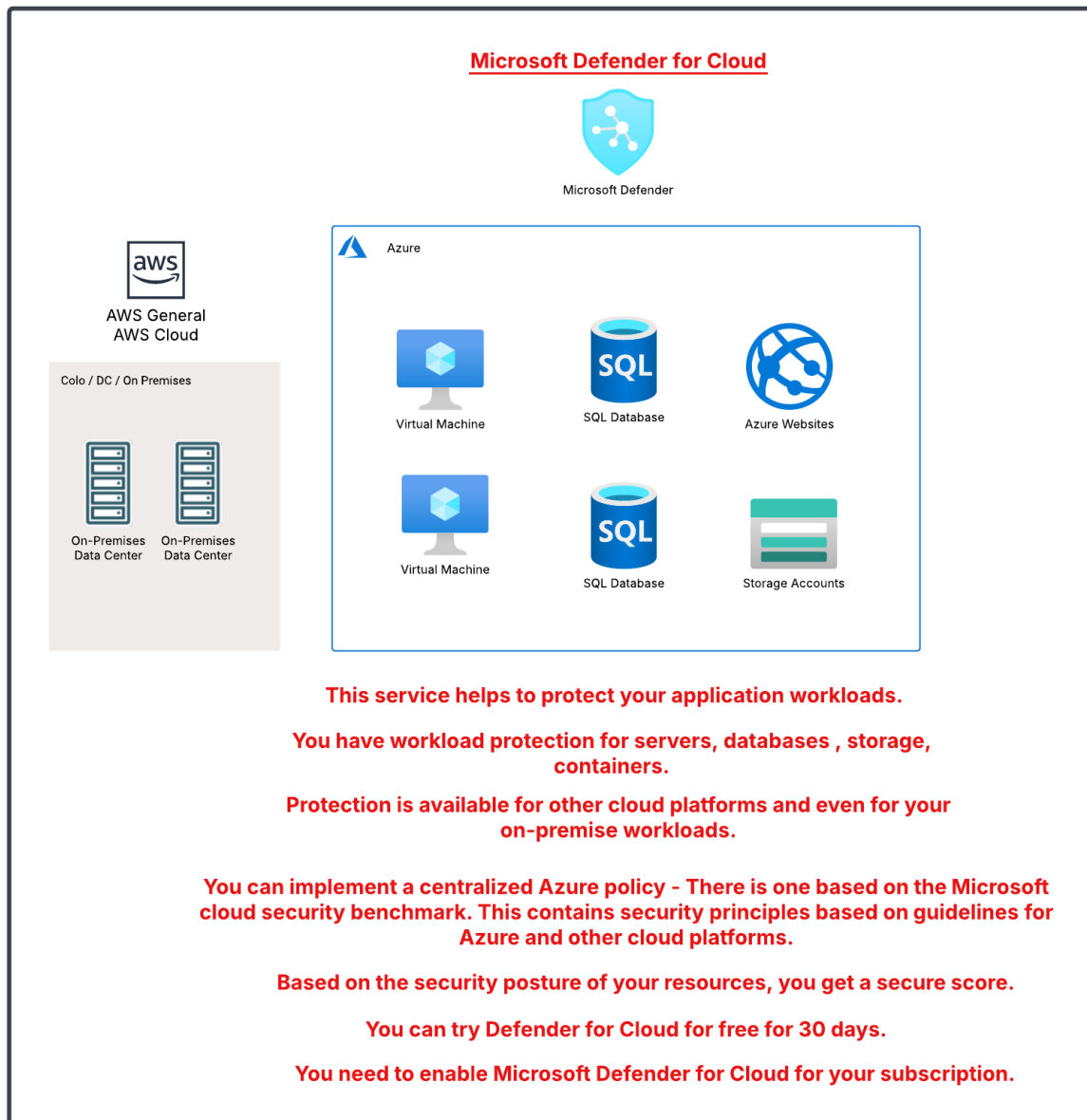
Actually in Microsoft Entra ID, there are different methods available for authentication.

Method	Primary authentication	Secondary authentication
Windows Hello for Business	Yes	MFA ¹
Microsoft Authenticator push	No	MFA and SSPR
Microsoft Authenticator passwordless	Yes	No ²
Microsoft Authenticator passkey	Yes	MFA
Authenticator Lite	No	MFA
Passkey (FIDO2)	Yes	MFA
Certificate-based authentication (CBA)	Yes	MFA
Hardware OATH tokens (preview)	No	MFA and SSPR
Software OATH tokens	No	MFA and SSPR
External authentication methods (preview)	No	MFA
Temporary Access Pass (TAP)	Yes	MFA
Text	Yes	MFA and SSPR
Voice call	No	MFA and SSPR
QR code (preview)	Yes	No
Password	Yes	No

Reference -

<https://learn.microsoft.com/en-us/entra/identity/authentication/concept-authentication-methods>

Microsoft Defender for Cloud



Quick look at Microsoft Sentinel

Microsoft Sentinel - This is a scalable , cloud-native security information and event management system (SIEM).

It also provides security orchestration, automation and response (SOAR)

You can collect data from users, devices, applications and infrastructure.



Microsoft Sentinel

Microsoft Sentinel can then be used to analyze the ingested data and detect and investigate threats.

Microsoft Sentinel also has inbuilt threat intelligence that helps to detect malicious activities.

You can use workbooks for interactive visual reports.

We can collect data via the use of data connectors.

These connectors can connect to Azure services or even to other SaaS providers.



Connectors



Microsoft Sentinel

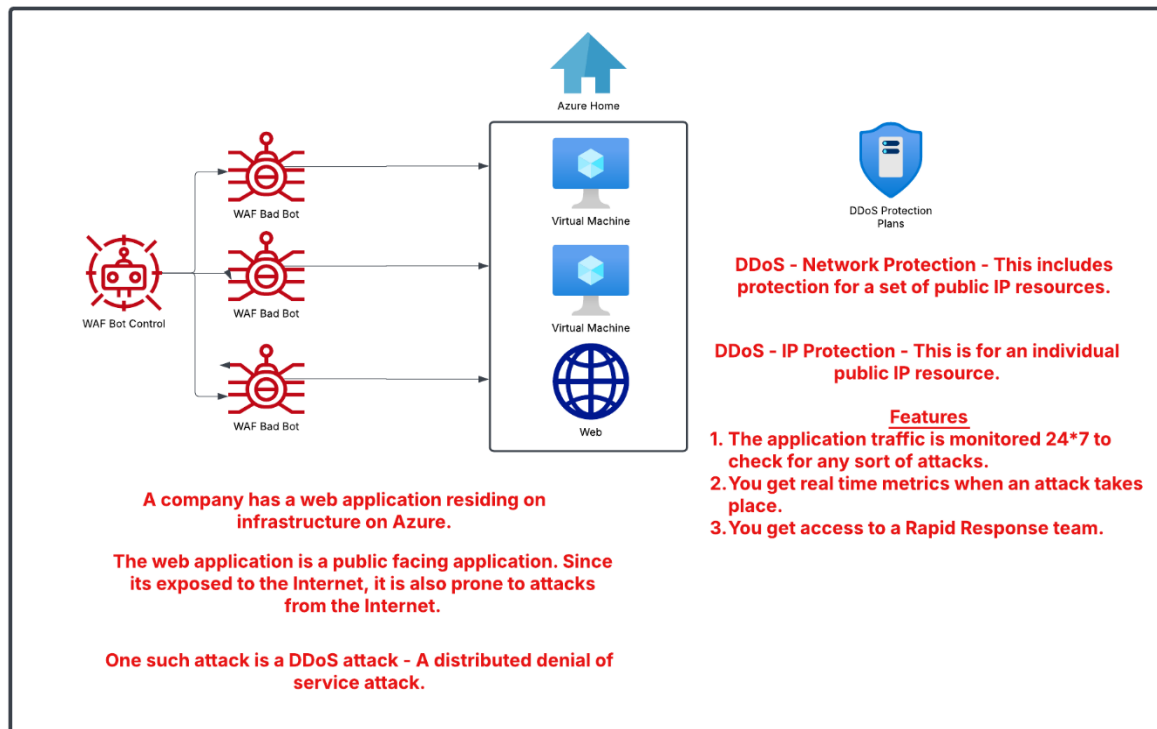


Log Analytics Workspaces

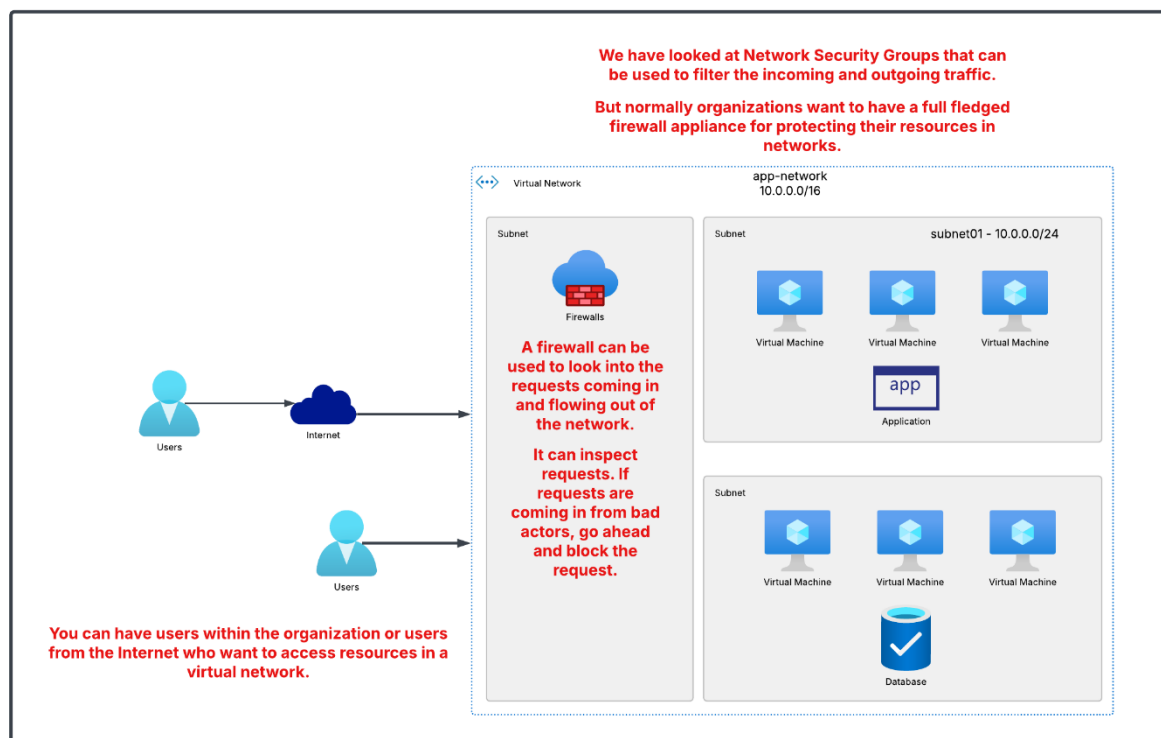
In order to collect data, we need to have a Log Analytics workspace. We can then use data connectors to stream the data onto the Log Analytics workspace.

The Microsoft Sentinel service will sit on top of the Log Analytics workspace.

Azure DDoS protection

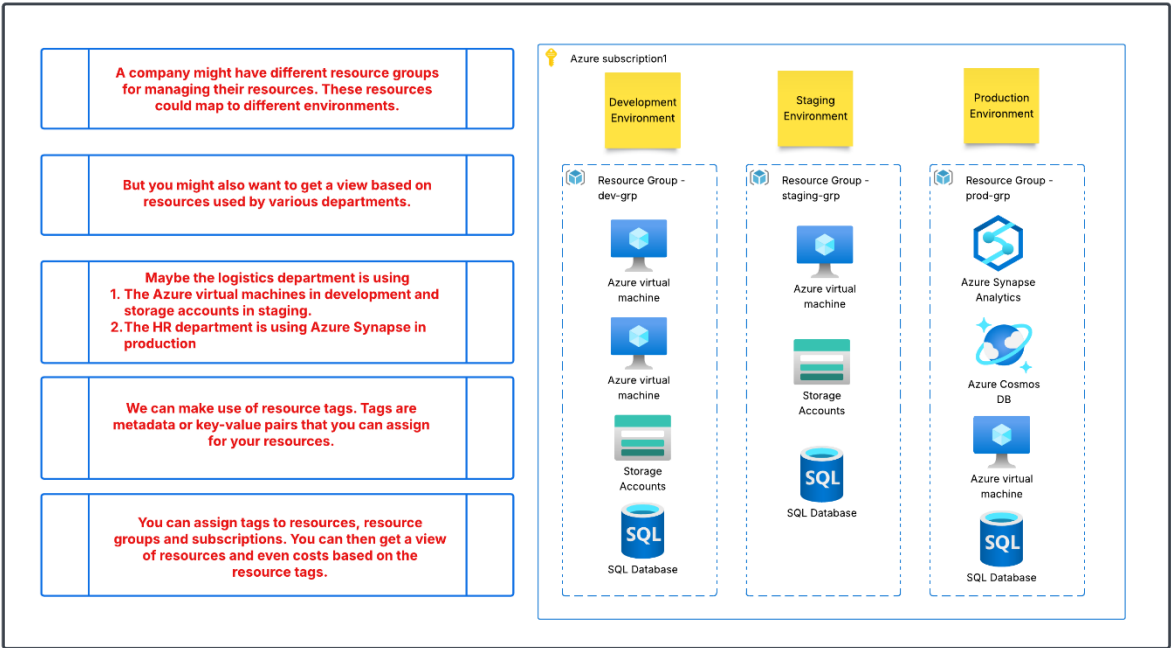


Azure Firewall

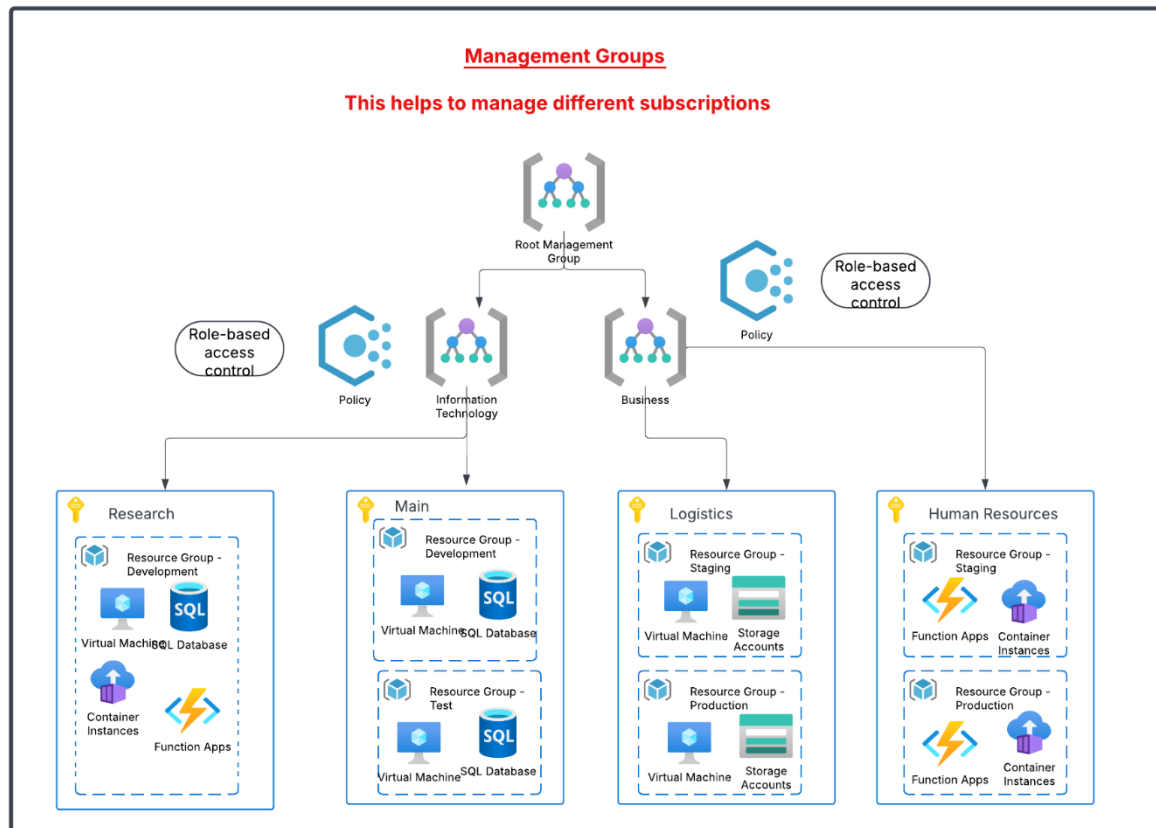


Describe Azure management and governance

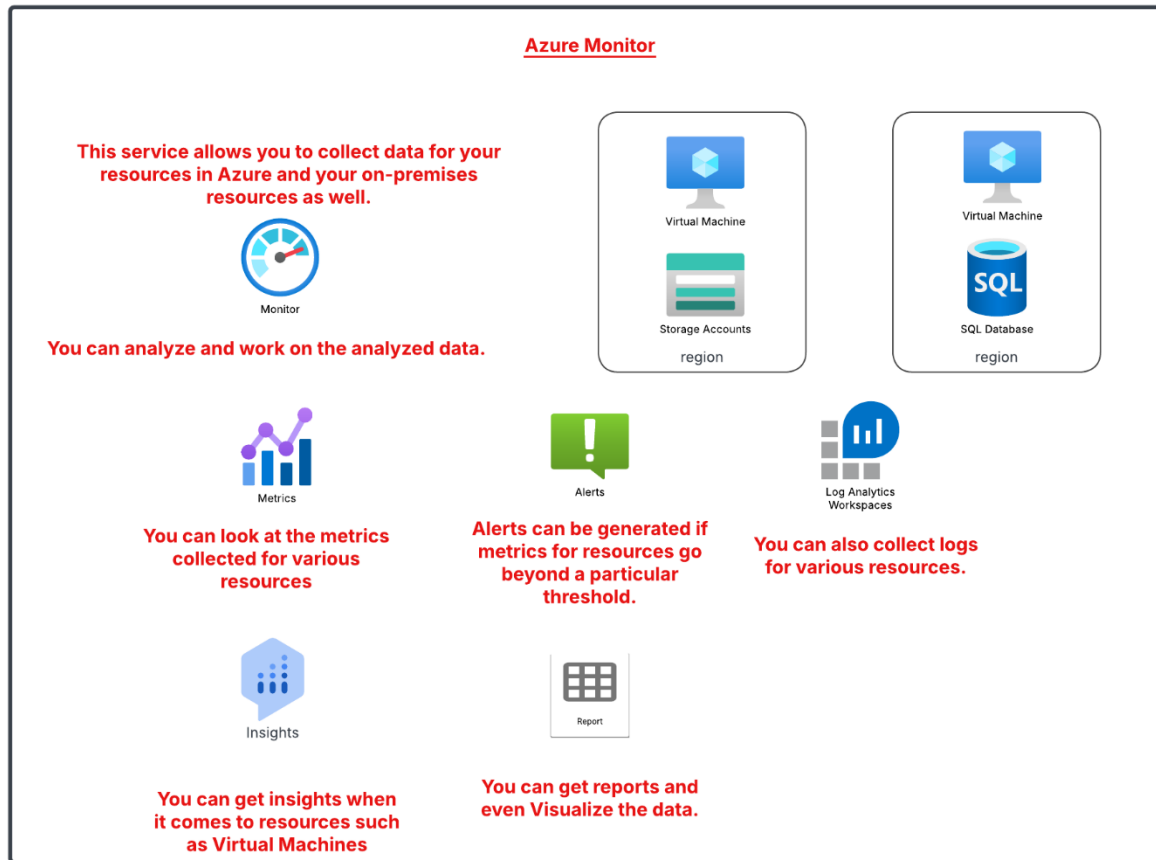
Resource tags



Management Groups



Azure Monitor service

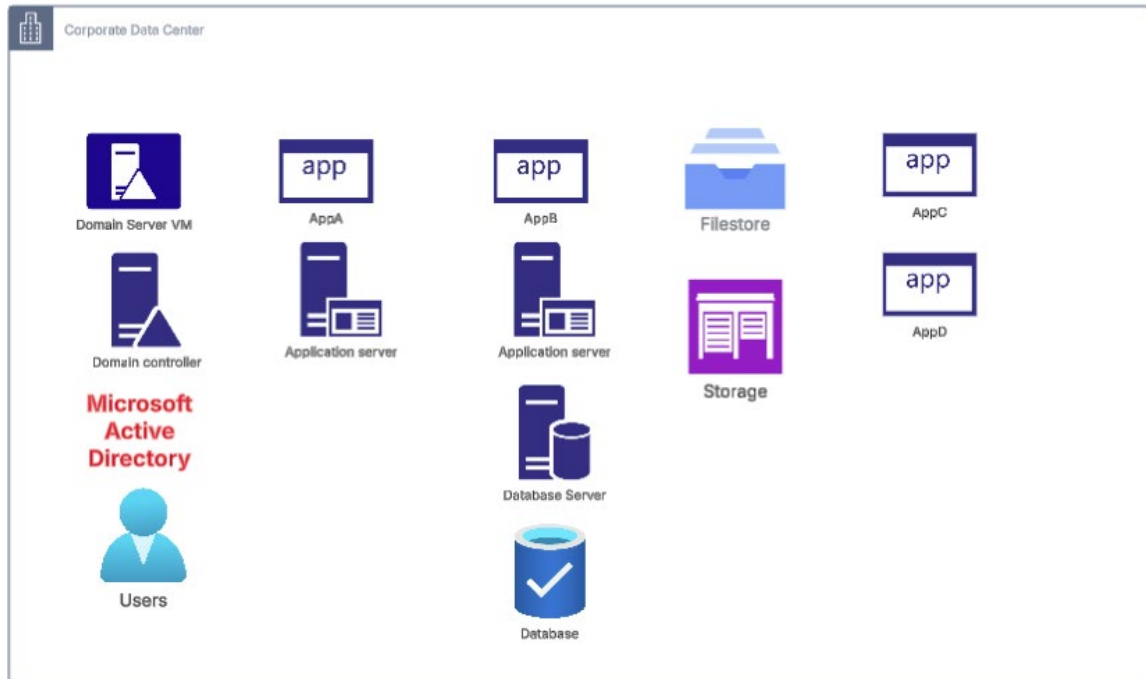


Putting everything together

Understanding your requirements

Let's consider the case where a company has a set of on-premises servers, applications, databases.

They want to migrate the workloads onto Azure.



Currently identities of users are defined using an on-premises setup of Microsoft Active Directory.

Both AppA and AppB need to be migrated onto Azure.

For AppA, the server administrators need to have complete privileges over the underlying compute infrastructure. The application needs to withstand data center level failures.

For AppB, you need to use Managed services wherever possible. The Database is currently hosted on an instance of Microsoft SQL Server.

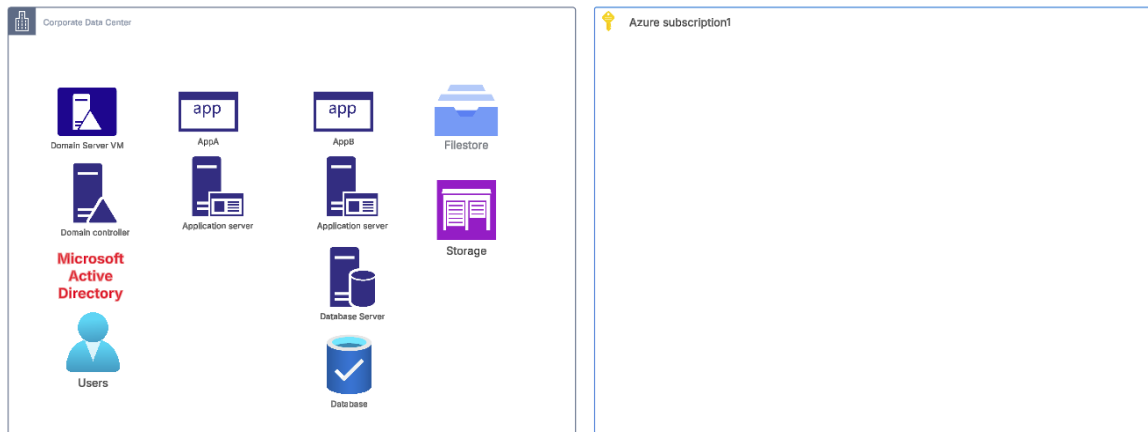
Certain applications cannot be migrated to Azure. The company needs to build a hybrid setup for applications and users on-premises to connect to resources on Azure.

The File store is a shared file share space that needs to be migrated onto Azure.

The Storage layers consists of different types of files. The files need to be migrated onto an efficient storage service on Azure. You should be able to save on storage costs for files that are rarely accessed.

We need to monitor our infrastructure using managed services on Azure.

Design and Initial implementation



What should we do first

We know that we want to migrate to Azure, and we need to decide on the services we are going to use.

But initially we can setup an Azure account and an Azure subscription. At an organization level, you might setup an Enterprise Agreement and also decide on a support plan to use.

An organization will want to ensure that they have a support plan in place. They might have a requirement wherein they need to have 24*7 support from Azure technical engineers. They want to have service reviews and advisory sessions. At the same time, they need to control costs. They can opt for the Professional Direct support plan.

Next is the costing aspect. Even though the company wants to move to the cloud. The business users would still need to approve the budget to move services to the cloud.

You need people on the ground who would setup the required infrastructure in Azure. They also need to perform the migration. And then you need to look at on-going costs.

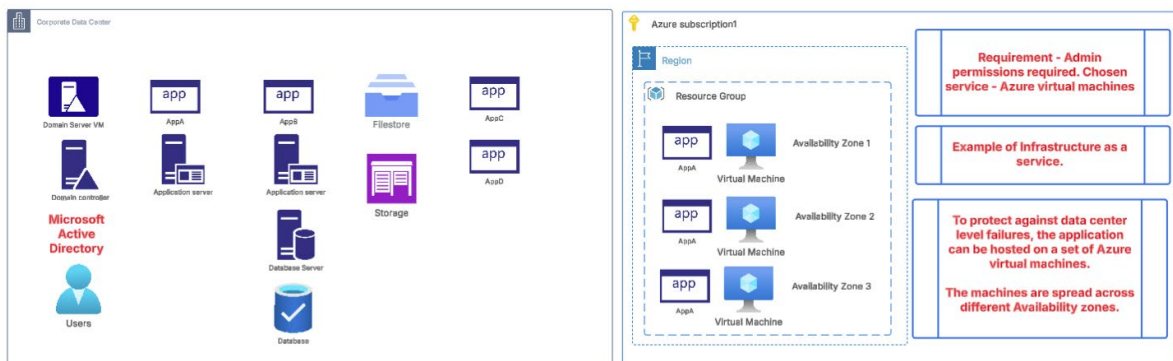
To understand the initial set of costs - You can use the Pricing Calculator. And to develop the business case for moving applications to the cloud, you can consider using the Total Cost of Ownership Calculator.

But we know that when it comes to Azure, we can have a lower entry to barrier, wherein we don't have that much of capital expenditure. And the platform provides flexibility and scalability.

Deciding on compute infrastructure

Moving Application A to Azure.

For AppA, the server administrators need to have complete privileges over the underlying compute infrastructure. The application needs to withstand data center level failures.

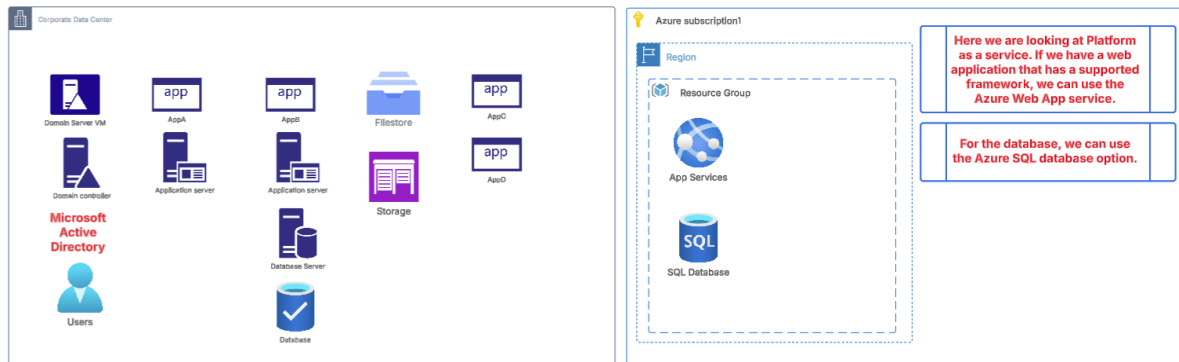


When deploying resources to Azure, for most of them, we need to deploy them to a specific region.

You decide on the region based on parameters such as proximity of users, costs etc.

Deciding on platform service usage

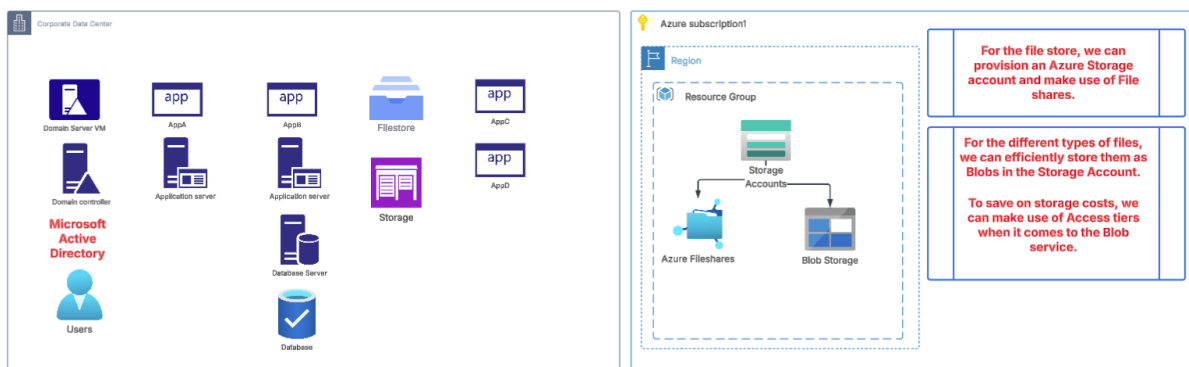
For AppB, you need to use Managed services wherever possible. The Database is currently hosted on an instance of Microsoft SQL Server.



Deciding on data storage requirements

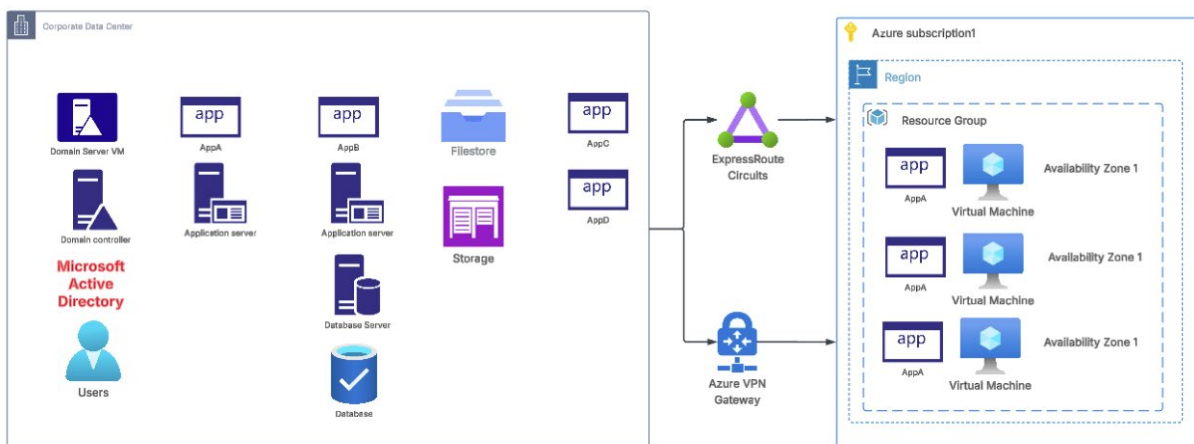
The File store is a shared file share space that needs to be migrated onto Azure.

The Storage layers consists of different types of files. The files need to be migrated onto an efficient storage service on Azure. You should be able to save on storage costs for files that are rarely accessed.



Connecting on-premises to Azure

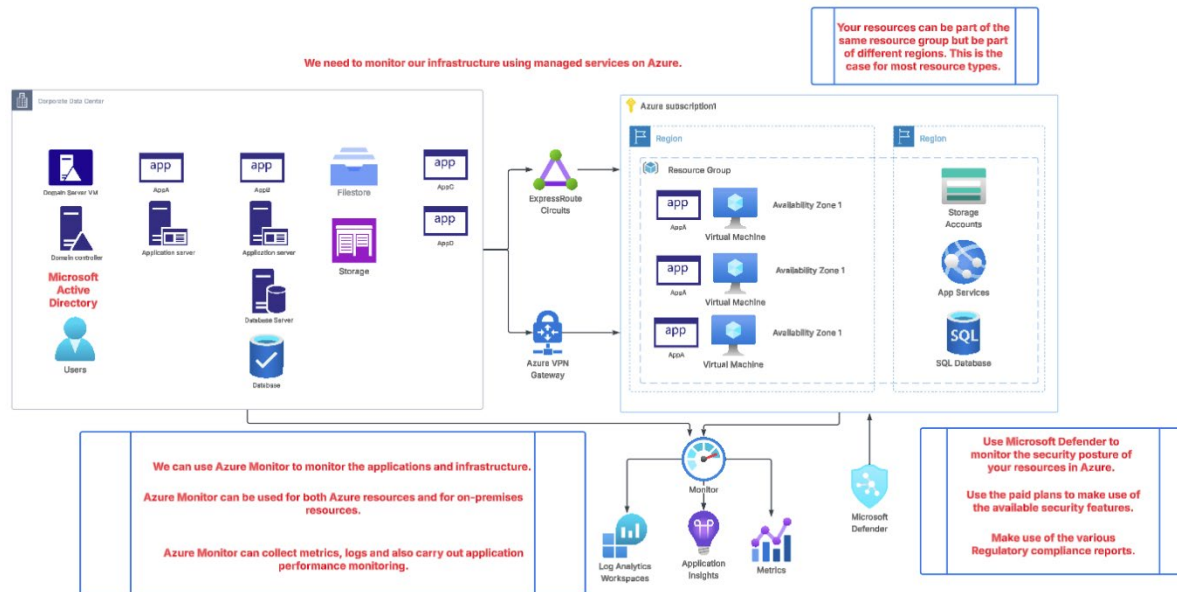
There are certain applications that cannot move to Azure at this point in time. We need to allow communication between the services on-premises and Azure.



ExpressRoute - We can create a dedicated and private connection between Azure and on-premises. Here the traffic does not traverse the Internet.

Azure VPN Gateway - We can use Azure VPN Gateways to connect on-premises to Azure. Here the traffic flows via the Internet, but its an encrypted connection.

Monitoring our infrastructure



Managing your infrastructure

